Algebra 2

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1. Order and primitive roots

Remark 1.1 (Recall). Let $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with distinct primes p_i and $\alpha_i \in \mathbb{N}$. Then $\varphi(n) = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right)$, and φ is multiplicative. If $m \in \mathbb{N}$, $a \in \mathbb{Z}$ and $(a, m) = 1$, then $a^{\varphi(m)} \equiv 1 \pmod{m}$ (Euler–Fermat).

Definition 1.2 (Order). Let $m \in \mathbb{N}$ and $a \in \mathbb{Z}$ with $(a, m) = 1$. The *order* of a modulo m, denoted $O_m(a)$, is the least $k \in \mathbb{N}$ with $a^k \equiv 1 \pmod{m}$. It exists by Remark 1.1.

Proposition 1.3. Let $m \in \mathbb{N}$, $a \in \mathbb{Z}$ with $(a, m) = 1$, and $k, s, t \in \mathbb{Z}_{\geq 0}$. Then:

1) $a^k \equiv 1 \pmod{m}$ if and only if $O_m(a) \mid k$. In particular $O_m(a) \mid \varphi(m)$.

2) $a^s \equiv a^t \pmod{m}$ if and only if $s \equiv t \pmod{O_m(a)}$.

Proof. Write $d = O_m(a)$.

1) If $d \mid k$, then $k = de$ with $e \in \mathbb{Z}_{\geq 0}$, and $a^k = \left(a^d\right)^e \equiv 1 \pmod{m}$. Conversely, let $a^k \equiv 1 \pmod{m}$ and write $k = qd + r$ with $q \in \mathbb{Z}_{\geq 0}$ and $0 \leq r < d$. Then $1 \equiv a^k = \left(a^d\right)^q a^r \equiv a^r \pmod{m}$. Since $r < d$ and d is the least positive exponent with this property, $r = 0$, so $d \mid k$. The particular case follows from Remark 1.1.

2) By symmetry assume $s \geq t$. Since $(a, m) = 1$, a^t is invertible modulo m. Hence $a^s \equiv a^t \pmod{m}$ holds if and only if $a^{s-t} \equiv 1 \pmod{m}$, which by 1) holds if and only if $d \mid s - t$, that is, $s \equiv t \pmod{d}$. ■

Example 1.4. Compute $O_{23}(5)$.

Solution. By Proposition 1.3, $O_{23}(5) \mid \varphi(23) = 22$, so $O_{23}(5) \in \{1, 2, 11, 22\}$. Here $5 \not\equiv 1$ and $5^2 = 25 \equiv 2 \not\equiv 1 \pmod{23}$. For 5^{11} use $11 = (1011)_2 = 8 + 2 + 1$ and repeated squaring: $5^2 \equiv 2$, $5^4 \equiv 4$, $5^8 \equiv 16 \pmod{23}$. Thus

$$5^{11} = 5^8 \cdot 5^2 \cdot 5 \equiv 16 \cdot 2 \cdot 5 = 160 \equiv -1 \pmod{23},$$

since $160 = 7 \cdot 23 - 1$. Hence $5^{11} \not\equiv 1$ and $O_{23}(5) = 22$. ◊

Definition 1.5 (Primitive root). Let $m \in \mathbb{N}$. An integer $a \in \mathbb{Z}$ with $(a, m) = 1$ is a *primitive root* modulo m if $O_m(a) = \varphi(m)$.

Lemma 1.6. Let $m \in \mathbb{N}$, $a \in \mathbb{Z}$ with $(a, m) = 1$, $d = O_m(a)$ and $j \in \mathbb{Z}_{\geq 0}$. Then $O_m(a^j) = d/(j, d)$.

Proof. Let $e = (j, d)$. For $k \in \mathbb{N}$, Proposition 1.3 gives $\left(a^j\right)^k \equiv 1 \pmod{m}$ if and only if $d \mid jk$, that is, $d/e \mid (j/e)k$. Since $(d/e, j/e) = 1$, this holds if and only if $d/e \mid k$. The least such k is d/e. ■

Remark 1.7 (Recall). Let p be a prime and $f \in \mathbb{Z}[x]$ a polynomial of degree $n \geq 1$ whose leading coefficient is not divisible by p. Then $f(x) \equiv 0 \pmod{p}$ has at most n pairwise incongruent solutions (Lagrange).

Theorem 1.8. Let $m \geq 2$. There exists a primitive root modulo m if and only if $m = 2$, $m = 4$, $m = p^\alpha$ or $m = 2p^\alpha$, where p is an odd prime and $\alpha \in \mathbb{N}$.

Proof. **The cases $m = 2, 4$.** $O_2(1) = 1 = \varphi(2)$. Since $3 \not\equiv 1$ and $3^2 = 9 \equiv 1 \pmod 4$, $O_4(3) = 2 = \varphi(4)$.

The case $m = p$, p an odd prime. For $d \mid p-1$ let $\psi(d)$ be the number of residues a modulo p with $p \nmid a$ and $O_p(a) = d$. By Proposition 1.3 every such order divides $p-1$, so $\sum_{d \mid p-1} \psi(d) = p-1$. Let $d \mid p-1$ with $\psi(d) \geq 1$, and let a have order d. By Proposition 1.3 the residues $1, a, \ldots, a^{d-1}$ are pairwise incongruent, and they are roots of $x^d - 1$. By Remark 1.7, $x^d - 1$ has no other roots modulo p, so every residue of order d is congruent to some a^j with $0 \leq j < d$. By Lemma 1.6, a^j has order d if and only if $(j, d) = 1$. Hence $\psi(d) = \varphi(d)$ whenever $\psi(d) \geq 1$, and so $\psi(d) \leq \varphi(d)$ for every $d \mid p-1$. Since $\sum_{d \mid p-1} \varphi(d) = p-1$ by Theorem 7.9, equality holds for every d. In particular $\psi(p-1) = \varphi(p-1) \geq 1$.

The case $m = p^\alpha$, p an odd prime, $\alpha \geq 2$. Let g be a primitive root modulo p. We may assume $g^{p-1} \not\equiv 1 \pmod{p^2}$. Indeed, $g + p$ is also a primitive root modulo p, and by the binomial theorem

$$(g+p)^{p-1} \equiv g^{p-1} + (p-1)pg^{p-2} \equiv g^{p-1} - pg^{p-2} \pmod{p^2}.$$

If $g^{p-1} \equiv 1 \pmod{p^2}$, then $(g+p)^{p-1} \equiv 1 - pg^{p-2} \not\equiv 1 \pmod{p^2}$, because $p \nmid g$; in that case replace g by $g + p$.

So $g^{p-1} = 1 + pu$ with $p \nmid u$. We claim that for every $k \in \mathbb{N}$ there is $u_k \in \mathbb{Z}$ with $p \nmid u_k$ and $g^{(p-1)p^{k-1}} = 1 + p^k u_k$. For $k = 1$ take $u_1 = u$. If the claim holds for k, then

$$g^{(p-1)p^k} = \left(1 + p^k u_k\right)^p = 1 + p^{k+1} u_k + \sum_{i=2}^{p} \binom{p}{i} p^{ik} u_k^i.$$

For $2 \leq i \leq p-1$ the prime p divides $\binom{p}{i}$, so the term is divisible by p^{2k+1}, and $2k + 1 \geq k + 2$. The term $i = p$ is divisible by p^{pk}, and $pk \geq 3k \geq k + 2$. Hence the sum equals $p^{k+2} w$ with $w \in \mathbb{Z}$, and $u_{k+1} = u_k + pw$ is not divisible by p.

Now let $d = O_{p^\alpha}(g)$. From $g^d \equiv 1 \pmod p$ and Proposition 1.3 modulo p we get $p - 1 \mid d$. Also $d \mid \varphi(p^\alpha) = (p-1)p^{\alpha-1}$. Hence $d = (p-1)p^i$ with $0 \leq i \leq \alpha - 1$. If $i \leq \alpha - 2$, then $d \mid (p-1)p^{\alpha-2}$ and so $g^{(p-1)p^{\alpha-2}} \equiv 1 \pmod{p^\alpha}$. By the claim with $k = \alpha - 1$, however, $g^{(p-1)p^{\alpha-2}} = 1 + p^{\alpha-1} u_{\alpha-1}$ with $p \nmid u_{\alpha-1}$, which is not congruent to 1 modulo p^α. Hence $d = \varphi(p^\alpha)$.

The case $m = 2p^\alpha$, p an odd prime, $\alpha \geq 1$. Let g be a primitive root modulo p^α. Replacing g by $g + p^\alpha$ if necessary, g is odd; it remains a primitive root modulo p^α, and $(g, 2p^\alpha) = 1$. If $g^d \equiv 1 \pmod{2p^\alpha}$, then $g^d \equiv 1 \pmod{p^\alpha}$, so $\varphi(p^\alpha) \mid d$. Since $\varphi(2p^\alpha) = \varphi(p^\alpha)$, Proposition 1.3 gives $O_{2p^\alpha}(g) = \varphi(2p^\alpha)$.

No primitive root otherwise. Let $m \geq 2$ be of none of the listed forms, and write $m = 2^\beta n$ with n odd. If $n = 1$, then $\beta \geq 3$. If n has two distinct prime divisors, write $n = p^\alpha n'$ with $p \nmid n'$ and $n' \geq 3$, and put $m_1 = 2^\beta p^\alpha$, $m_2 = n'$. If $n = p^\alpha$, then $\beta \geq 2$; put $m_1 = 2^\beta$, $m_2 = p^\alpha$. So either $m = 2^\beta$ with $\beta \geq 3$, or $m = m_1 m_2$ with $(m_1, m_2) = 1$ and $m_1, m_2 \geq 3$.

In the first case, $a^{2^{k-2}} \equiv 1 \pmod{2^k}$ for every odd a and every $k \geq 3$. For $k = 3$, $a^2 - 1 = (a-1)(a+1)$ is a product of two consecutive even numbers, hence divisible by 8. If $a^{2^{k-2}} = 1 + 2^k t$, then $a^{2^{k-1}} = 1 + 2^{k+1}t + 2^{2k}t^2 \equiv 1 \pmod{2^{k+1}}$. Hence $O_{2^\beta}(a) \leq 2^{\beta-2} < 2^{\beta-1} = \varphi(2^\beta)$.

In the second case, note that $\varphi(n)$ is even for $n \geq 3$: if an odd prime p divides n, then $p - 1 \mid \varphi(n)$ by Remark 1.1; otherwise $n = 2^\gamma$ with $\gamma \geq 2$ and $\varphi(n) = 2^{\gamma-1}$. Let L be the least common multiple of $\varphi(m_1)$ and $\varphi(m_2)$. Both are even, so $L \leq \varphi(m_1)\varphi(m_2)/2 = \varphi(m)/2$. For $(a, m) = 1$, Euler–Fermat gives $a^L \equiv 1$ modulo m_1 and modulo m_2, hence modulo $m = m_1 m_2$. So $O_m(a) \leq L < \varphi(m)$. ■

Remark 1.9. There is no primitive root modulo 12. Indeed, $\varphi(12) = 4$, and each of $1, 5, 7, 11$ has square congruent to 1 modulo 12, so every order is at most 2.

Theorem 1.10. Let $m \geq 2$ and suppose that g is a primitive root modulo m. Then the primitive roots modulo m are, up to congruence, exactly the powers g^j with $1 \leq j \leq \varphi(m)$ and $(j, \varphi(m)) = 1$. Hence there are $\varphi(\varphi(m))$ of them. In particular, for a prime p there are $\varphi(\varphi(p)) = \varphi(p-1)$ primitive roots modulo p.

Proof. A primitive root a satisfies $(a, m) = 1$. By Theorem 1.17 and $g^{\varphi(m)} \equiv 1 = g^0 \pmod{m}$, a is congruent to exactly one g^j with $1 \leq j \leq \varphi(m)$. By Lemma 1.6, $O_m(g^j) = \varphi(m)/(j, \varphi(m))$, which equals $\varphi(m)$ if and only if $(j, \varphi(m)) = 1$. The number of such j is $\varphi(\varphi(m))$. ■

Theorem 1.11 (cited)**.** There is a constant $c > 0$ such that $\varphi(n) > cn/\log\log n$ for every $n \geq 3$.

Procedure 1.12 (Checking primitive roots). Let p be a prime and $g \in \mathbb{Z}$ with $p \nmid g$.

1) Compute the prime divisors $q_1, \ldots, q_s$ of $p - 1$.
2) Check that $g^{(p-1)/q_i} \not\equiv 1 \pmod{p}$ for every $1 \leq i \leq s$.

Then g is a primitive root modulo p if and only if every check in step 2 succeeds.

Proof. If g is a primitive root, then $O_p(g) = p - 1$ does not divide $(p-1)/q_i < p - 1$, so $g^{(p-1)/q_i} \not\equiv 1$ by Proposition 1.3. Conversely, suppose $d = O_p(g) < p - 1$. Since $d \mid p - 1$, the quotient $(p-1)/d > 1$ has a prime divisor q_i, and then $d \mid (p-1)/q_i$, so $g^{(p-1)/q_i} \equiv 1 \pmod{p}$. ■

Theorem 1.13 (Burgess) (cited)**.** For a prime p let $n(p)$ be the least primitive root modulo p in $\mathbb{N}$. For every $\varepsilon > 0$ there is $c(\varepsilon) > 0$ such that $n(p) < c(\varepsilon) p^{1/4+\varepsilon}$ for all primes p.

Theorem 1.14 (Shoup) (cited)**.** Assume the generalized Riemann hypothesis for Dirichlet L-functions. Then there is an absolute constant $c > 0$ such that for every prime $p \geq 3$

$$n(p) \leq c\omega(p-1)^4(\log\omega(p-1) + 1)^4(\log p)^2,$$

where $\omega(k)$ is the number of distinct prime divisors of k. In particular $n(p) = O((\log p)^6)$.

Theorem 1.15 (Prime number theorem) (cited)**.** Let $\pi(x)$ be the number of primes not exceeding x. There is $c > 0$ such that

$$\pi(x) = \int_2^x \frac{dt}{\log t} + O\left(xe^{-c\sqrt{\log x}}\right) \quad (x \to \infty).$$

If the Riemann hypothesis holds, the error term is $O(\sqrt{x}\log x)$.

Definition 1.16 (Reduced residue system). Let $m \in \mathbb{N}$. Integers $a_1, ..., a_{\varphi(m)}$ form a *reduced residue system* modulo m if $(a_i, m) = 1$ for every i and $a_i \not\equiv a_j \pmod m$ for all $i \neq j$. Such a system contains exactly one element of each residue class coprime to m, since there are $\varphi(m)$ such classes.

Theorem 1.17. Let $m \in \mathbb{N}$ and $g \in \mathbb{Z}$ with $(g, m) = 1$. Then g is a primitive root modulo m if and only if $1, g, g^2, ..., g^{\varphi(m)-1}$ form a reduced residue system modulo m.

Proof. Every power of g is coprime to m. Suppose g is a primitive root and $g^i \equiv g^j \pmod m$ with $0 \leq j \leq i \leq \varphi(m) - 1$. By Proposition 1.3, $i \equiv j \pmod{\varphi(m)}$, and $0 \leq i - j < \varphi(m)$ forces $i = j$. So the $\varphi(m)$ powers are pairwise incongruent.

Conversely, if the powers form a reduced residue system, then $g^j \not\equiv g^0 = 1 \pmod m$ for $1 \leq j \leq \varphi(m) - 1$, because they are pairwise incongruent. Hence $O_m(g) \geq \varphi(m)$, and $O_m(g) \mid \varphi(m)$ by Proposition 1.3, so $O_m(g) = \varphi(m)$. ∎

2. The index

Definition 2.1 (Index, discrete logarithm). Let p be a prime, g a primitive root modulo p, and $a \in \mathbb{Z}$ with $p \nmid a$. The *index* or *discrete logarithm* of a to the base g modulo p is the unique k with $1 \leq k \leq p - 1$ and $g^k \equiv a \pmod p$. It is denoted $\operatorname{ind}_g a$. Existence and uniqueness follow from Theorem 1.17, since $g^{p-1} \equiv 1 = g^0 \pmod p$.

Proposition 2.2. Let p be a prime and g a primitive root modulo p. For $i, j \in \mathbb{Z}_{\geq 0}$, $g^i \equiv g^j \pmod p$ if and only if $i \equiv j \pmod{p-1}$. For $a, b \in \mathbb{Z}$ with $p \nmid ab$, $a \equiv b \pmod p$ if and only if $\operatorname{ind}_g a \equiv \operatorname{ind}_g b \pmod{p-1}$.

Proof. The first statement is Proposition 1.3 2) with $O_p(g) = \varphi(p) = p - 1$. For the second, $a \equiv b \pmod p$ holds if and only if $g^{\operatorname{ind}_g a} \equiv g^{\operatorname{ind}_g b} \pmod p$, and the first statement applies. ∎

Theorem 2.3. Let p be a prime, g and h primitive roots modulo p, $a, b \in \mathbb{Z}$ with $p \nmid ab$, and $n \in \mathbb{Z}_{\geq 0}$. Then, modulo $p - 1$:

1) $\operatorname{ind}_g(ab) \equiv \operatorname{ind}_g a + \operatorname{ind}_g b$,

2) $\operatorname{ind}_g(a^n) \equiv n \operatorname{ind}_g a$,

3) $\operatorname{ind}_h a \equiv \operatorname{ind}_h g \cdot \operatorname{ind}_g a$.

Proof. 1) $g^{\operatorname{ind}_g(ab)} \equiv ab$ and $g^{\operatorname{ind}_g a + \operatorname{ind}_g b} = g^{\operatorname{ind}_g a} g^{\operatorname{ind}_g b} \equiv ab \pmod p$; apply Proposition 2.2.

2) $g^{\operatorname{ind}_g(a^n)} \equiv a^n$ and $g^{n \operatorname{ind}_g a} = \left(g^{\operatorname{ind}_g a}\right)^n \equiv a^n \pmod p$; apply Proposition 2.2.

3) $h^{\operatorname{ind}_h a} \equiv a \equiv g^{\operatorname{ind}_g a} \equiv \left(h^{\operatorname{ind}_h g}\right)^{\operatorname{ind}_g a} = h^{\operatorname{ind}_h g \cdot \operatorname{ind}_g a} \pmod p$; apply Proposition 2.2 with base h. ■

Remark 2.4. Indices are computed with an index table: list g^n modulo p for $n = 1, \ldots, p-1$ and read the table in reverse.

Example 2.5. Find a primitive root g modulo 17 and compute $\operatorname{ind}_g 11$.

Solution. Here $\varphi(17) = 16$, whose only prime divisor is 2, so by Procedure 1.12 an integer g with $17 \nmid g$ is a primitive root if and only if $g^8 \not\equiv 1 \pmod{17}$. Since $2^8 = 256 = 15 \cdot 17 + 1 \equiv 1$, the integer 2 is not a primitive root. Since $3^4 = 81 \equiv -4$ and $3^8 \equiv 16 \equiv -1 \pmod{17}$, the integer 3 is a primitive root. The powers of 3 modulo 17 are:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
3^n	3	9	10	13	5	15	11	16	14	8	7	4	12	2	6	1

Reading the table in reverse, $\operatorname{ind}_3 11 = 7$. ◊

Remark 2.6 (Discrete logarithm problem). Given a prime p, a primitive root g modulo p and an integer k with $1 \le k \le p-1$, find x with $1 \le x \le p-1$ and $g^x \equiv k \pmod p$. No classical algorithm solving this problem in time polynomial in $\log p$ is known.

Remark 2.7. The index generalizes to any finite cyclic group $G = \langle g \rangle$ of order N: for $b \in G$, $\operatorname{ind}_g b$ is the unique k with $1 \le k \le N$ and $g^k = b$. The discrete logarithm problem is posed in the same way.

3. Binomial congruences and quadratic residues

Definition 3.1 (Quadratic residue). Let $p > 2$ be a prime and $a \in \mathbb{Z}$ with $(a, p) = 1$. Then a is a *quadratic residue* modulo p if $x^2 \equiv a \pmod p$ has a solution, and a *quadratic non-residue* otherwise.

Let p be a prime, $a \in \mathbb{Z}$ with $(a, p) = 1$, and $k \in \mathbb{N}$. We consider the congruence $x^k \equiv a \pmod p$.

Remark 3.2 (Recall). Let $n \in \mathbb{N}$ and $a, b \in \mathbb{Z}$. The congruence $ay \equiv b \pmod n$ is solvable if and only if $(a, n) \mid b$, and then it has exactly (a, n) pairwise incongruent solutions modulo n.

Theorem 3.3. Let p be a prime, $a \in \mathbb{Z}$ with $(a, p) = 1$, and $k \in \mathbb{N}$. The congruence $x^k \equiv a \pmod p$ is solvable if and only if

$$a^{(p-1)/(k, p-1)} \equiv 1 \pmod p.$$

In that case the number of pairwise incongruent solutions is $(k, p-1)$. If g is a primitive root modulo p, the condition is equivalent to $(k, p-1) \mid \operatorname{ind}_g a$.

Proof. Let g be a primitive root modulo p (Theorem 1.8) and $e = (k, p-1)$. Every solution x satisfies $p \nmid x$, since $p \nmid a$. By the definition of the index and Proposition 2.2, $x \mapsto \operatorname{ind}_g x$ is a bijection from the residue classes modulo p not divisible by p onto the residue classes modulo $p-1$. We seek solutions in the form $x \equiv g^{\operatorname{ind}_g x} \pmod p$. Then

$$x^k \equiv \left(g^{\operatorname{ind}_g x}\right)^k = g^{k \operatorname{ind}_g x} \quad \text{and} \quad a \equiv g^{\operatorname{ind}_g a} \pmod p,$$

so by Proposition 2.2, $x^k \equiv a \pmod p$ if and only if $k \operatorname{ind}_g x \equiv \operatorname{ind}_g a \pmod{p-1}$. Put $y = \operatorname{ind}_g x$. The solutions x modulo p correspond bijectively to the solutions y modulo $p-1$ of the linear congruence

$$ky \equiv \operatorname{ind}_g a \pmod{p-1}.$$

By Remark 3.2 it is solvable if and only if $e \mid \operatorname{ind}_g a$, and then it has exactly e solutions.

It remains to show that $e \mid \operatorname{ind}_g a$ if and only if $a^{(p-1)/e} \equiv 1 \pmod p$. We have

$$a^{(p-1)/e} \equiv \left(g^{\operatorname{ind}_g a}\right)^{(p-1)/e} = g^{\operatorname{ind}_g a \cdot (p-1)/e} \pmod p.$$

By Proposition 1.3 with $O_p(g) = p-1$, this is congruent to 1 if and only if $p-1 \mid \operatorname{ind}_g a \cdot (p-1)/e$, that is, if and only if $e \mid \operatorname{ind}_g a$. ■

Example 3.4. Solve $5x^{22} \equiv 6 \pmod{13}$.

Solution. Let $y = x^{22}$. Then $5y \equiv 6 \pmod{13}$ with $(5, 13) = 1$. Multiplying by 5 gives $25y \equiv 30 \pmod{13}$, and $25 \equiv -1$, $30 \equiv 4$, so $-y \equiv 4$ and $y \equiv -4 \equiv 9 \pmod{13}$. It remains to solve $x^{22} \equiv 9 \pmod{13}$.

Here $\varphi(13) = 12 = 2^2 \cdot 3$. Since $2^4 = 16 \equiv 3$ and $2^6 = 64 \equiv -1 \pmod{13}$, Procedure 1.12 shows that 2 is a primitive root modulo 13, that is, $O_{13}(2) = 12$. Taking indices to the base 2 and using Theorem 2.3,

$$22 \operatorname{ind}_2 x \equiv \operatorname{ind}_2 9 \pmod{12}.$$

The powers of 2 modulo 13 are:

n	1	2	3	4	5	6	7	8	9	10	11	12
2^n	2	4	8	3	6	12	11	9	5	10	7	1

Hence $\operatorname{ind}_2 9 = 8$. Let $z = \operatorname{ind}_2 x$. Then $22z \equiv 8 \pmod{12}$, and $(22, 12) = 2 \mid 8$, so there are two solutions modulo 12. Dividing by 2 gives $11z \equiv 4 \pmod 6$, that is, $-z \equiv 4$, so $z \equiv -4 \equiv 2 \pmod 6$. Thus $z \equiv 2$ or $z \equiv 8 \pmod{12}$. From $\operatorname{ind}_2 x \equiv 2$ we get $x \equiv 2^2 = 4$, and from $\operatorname{ind}_2 x \equiv 8$ we get $x \equiv 2^8 \equiv 9 \pmod{13}$. The solutions are $x \equiv 4$ and $x \equiv 9 \pmod{13}$. ◊

Corollary 3.5. Let p be an odd prime and $a \in \mathbb{Z}$ with $(a, p) = 1$. Then a is a quadratic residue modulo p if and only if $a^{(p-1)/2} \equiv 1 \pmod p$, and in that case $x^2 \equiv a \pmod p$ has exactly 2

pairwise incongruent solutions. Among the $p-1$ residue classes coprime to p there are $(p-1)/2$ quadratic residues and $(p-1)/2$ quadratic non-residues.

Proof. Since p is odd, $(2, p-1) = 2$, and the first two statements are Theorem 3.3 with $k = 2$. The map $x \mapsto x^2$ sends the $p-1$ residue classes coprime to p onto the quadratic residues, and each quadratic residue has exactly 2 preimages. Hence there are $(p-1)/2$ quadratic residues, and the remaining $(p-1)/2$ classes are non-residues. ∎

Proposition 3.6. Let p be an odd prime and $a \in \mathbb{Z}$ with $p \nmid a$. Then $a^{(p-1)/2} \equiv \pm 1 \pmod{p}$, and a is a quadratic residue modulo p if and only if $a^{(p-1)/2} \equiv 1 \pmod{p}$.

Proof. Let $b = a^{(p-1)/2}$. By Fermat's theorem $b^2 = a^{p-1} \equiv 1 \pmod{p}$, so $p \mid (b-1)(b+1)$. Since p is prime, $b \equiv 1$ or $b \equiv -1 \pmod{p}$. The criterion is Corollary 3.5. ∎

4. The Legendre symbol and quadratic reciprocity

Definition 4.1 (Legendre symbol). Let $p > 2$ be a prime and $a \in \mathbb{Z}$ with $(a, p) = 1$. The *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue modulo } p \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

Corollary 4.2 (Euler's criterion). Let $p > 2$ be a prime and $a \in \mathbb{Z}$ with $(a, p) = 1$. Then $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$.

Proof. By Proposition 3.6, $a^{(p-1)/2} \equiv 1 = \left(\frac{a}{p}\right)$ if a is a quadratic residue, and otherwise $a^{(p-1)/2} \equiv -1 = \left(\frac{a}{p}\right)$. ∎

Proposition 4.3. Let $p > 2$ be a prime and $a, b \in \mathbb{Z}$ with $(a, p) = (b, p) = 1$. Then:

1) If $a \equiv b \pmod{p}$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$.
2) $\left(\frac{a}{p}\right)\left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right)$.
3) $\left(\frac{-1}{p}\right) = 1$ if $p \equiv 1 \pmod{4}$, and $\left(\frac{-1}{p}\right) = -1$ if $p \equiv -1 \pmod{4}$.
4) $\left(\frac{2}{p}\right) = 1$ if $p \equiv \pm 1 \pmod{8}$, and $\left(\frac{2}{p}\right) = -1$ if $p \equiv \pm 3 \pmod{8}$.

Proof. 1) If $x^2 \equiv a \pmod{p}$, then $x^2 \equiv b \pmod{p}$, and conversely.

2) By Corollary 4.2, $\left(\frac{ab}{p}\right) \equiv (ab)^{(p-1)/2} = a^{(p-1)/2} b^{(p-1)/2} \equiv \left(\frac{a}{p}\right)\left(\frac{b}{p}\right) \pmod{p}$. Both sides lie in $\{1, -1\}$ and $p > 2$, so the congruence is an equality.

3) By Corollary 4.2, $\left(\frac{-1}{p}\right) \equiv (-1)^{(p-1)/2} \pmod{p}$, and as in 2) this is an equality. The exponent $(p-1)/2$ is even if and only if $p \equiv 1 \pmod{4}$.

4) is proved after Lemma 4.4. ∎

Lemma 4.4 (Gauss). Let p be an odd prime, $a \in \mathbb{Z}$ with $p \nmid a$, and $S = \{1, 2, \ldots, (p-1)/2\}$. For $j \in S$ let t_j be the least positive residue of ja modulo p, and let μ be the number of $j \in S$ with $t_j > p/2$. Then $\left(\frac{a}{p}\right) = (-1)^\mu$.

Proof. For $j \in S$ put $s_j = t_j$ if $t_j < p/2$ and $s_j = p - t_j$ otherwise. Then $s_j \in S$ and $s_j \equiv \pm ja \pmod p$, with the minus sign for exactly μ values of j. If $s_i = s_j$, then $ia \equiv \pm ja$, so $i \equiv \pm j \pmod p$ because $p \nmid a$. Since $2 \le i + j \le p - 1$, the minus sign is impossible, so $i = j$. Hence $j \mapsto s_j$ is a permutation of S, and

$$\prod_{j \in S} ja \equiv (-1)^\mu \prod_{j \in S} s_j = (-1)^\mu \left(\frac{p-1}{2}\right)! \pmod p.$$

The left side is $a^{(p-1)/2}((p-1)/2)!$. Cancelling the unit $((p-1)/2)!$ gives $a^{(p-1)/2} \equiv (-1)^\mu$, and Corollary 4.2 gives $\left(\frac{a}{p}\right) \equiv (-1)^\mu \pmod p$. Both sides lie in $\{1, -1\}$ and $p > 2$, so they are equal. ■

Proof of Proposition 4.3 4). Take $a = 2$ in Lemma 4.4. For $j \in S$ we have $2j \le p - 1$, so $t_j = 2j$, and μ is the number of $j \in S$ with $j > p/4$, namely $\mu = (p-1)/2 - \lfloor p/4 \rfloor$. Writing $p = 8l + r$ with $r \in \{1, 3, 5, 7\}$:

$$r = 1 : \mu = 4l - 2l = 2l, \quad r = 3 : \mu = (4l + 1) - 2l = 2l + 1,$$
$$r = 5 : \mu = (4l + 2) - (2l + 1) = 2l + 1, \quad r = 7 : \mu = (4l + 3) - (2l + 1) = 2l + 2.$$

So μ is even for $p \equiv \pm 1 \pmod 8$ and odd for $p \equiv \pm 3 \pmod 8$. ■

Lemma 4.5. Let p be an odd prime, a an odd integer with $p \nmid a$, and μ as in Lemma 4.4. Then

$$\mu \equiv \sum_{j=1}^{(p-1)/2} \left\lfloor \frac{ja}{p} \right\rfloor \pmod 2.$$

Proof. With the notation of Lemma 4.4, $ja = p\lfloor ja/p \rfloor + t_j$. Let $u_1, \ldots, u_\lambda$ be the t_j below $p/2$ and $v_1, \ldots, v_\mu$ those above. By the proof of Lemma 4.4, the numbers $u_1, \ldots, u_\lambda, p - v_1, \ldots, p - v_\mu$ are $1, \ldots, (p-1)/2$ in some order. Summing over $j \in S$,

$$a \sum_{j \in S} j = p \sum_{j \in S} \left\lfloor \frac{ja}{p} \right\rfloor + \sum_i u_i + \sum_i v_i, \quad \sum_{j \in S} j = \sum_i u_i + \mu p - \sum_i v_i.$$

Subtracting, $(a-1)\sum_{j \in S} j = p\left(\sum_{j \in S} \lfloor ja/p \rfloor - \mu\right) + 2\sum_i v_i$. The left side is even because a is odd, and p is odd, so $\sum_{j \in S} \lfloor ja/p \rfloor - \mu$ is even. ■

Theorem 4.6 (Quadratic reciprocity law). Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right) = \begin{cases} -\left(\frac{q}{p}\right) & \text{if } p \equiv q \equiv -1 \pmod 4 \\ \left(\frac{q}{p}\right) & \text{otherwise.} \end{cases}$$

Equivalently, $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$.

Proof. The two forms are equivalent: both symbols are ± 1, and $\frac{p-1}{2} \cdot \frac{q-1}{2}$ is odd if and only if both factors are odd, that is, if and only if $p \equiv q \equiv -1 \pmod 4$.

By Lemma 4.4 and Lemma 4.5, $\left(\frac{q}{p}\right) = (-1)^A$ and $\left(\frac{p}{q}\right) = (-1)^B$ with

$$A = \sum_{j=1}^{(p-1)/2} \left\lfloor \frac{jq}{p} \right\rfloor, \quad B = \sum_{k=1}^{(q-1)/2} \left\lfloor \frac{kp}{q} \right\rfloor.$$

Consider the lattice points (j,k) with $1 \le j \le (p-1)/2$ and $1 \le k \le (q-1)/2$; there are $\frac{p-1}{2} \cdot \frac{q-1}{2}$ of them. None satisfies $qj = pk$, since $p \nmid qj$. For fixed j, the points with $k < qj/p$ are those with $1 \le k \le \lfloor qj/p \rfloor$, and $\lfloor qj/p \rfloor \le (q-1)/2$ because $qj/p < q/2$. So A counts the points with $qj > pk$, and symmetrically B counts those with $qj < pk$. Hence $A + B = \frac{p-1}{2} \cdot \frac{q-1}{2}$. ■

Example 4.7. Compute $\left(\frac{12}{31}\right)$.

Solution.

$$\left(\frac{12}{31}\right) = \left(\frac{2^2 \cdot 3}{31}\right) = \left(\frac{2}{31}\right)^2 \left(\frac{3}{31}\right) = \left(\frac{3}{31}\right) = -\left(\frac{31}{3}\right) = -\left(\frac{1}{3}\right) = -1.$$

The fourth equality is Theorem 4.6 with $3 \equiv 31 \equiv -1 \pmod 4$, and the fifth uses $31 \equiv 1 \pmod 3$. Hence $x^2 \equiv 12 \pmod{31}$ has no solution. ◊

Example 4.8. Compute $\left(\frac{1783}{7523}\right)$, where 1783 and 7523 are primes.

Solution.

$$\left(\frac{1783}{7523}\right) = -\left(\frac{7523}{1783}\right) = -\left(\frac{391}{1783}\right) = \left(\frac{1783}{391}\right) = \left(\frac{219}{391}\right) = \left(\frac{3}{391}\right)\left(\frac{73}{391}\right).$$

The first equality is Theorem 4.6 with $1783 \equiv 7523 \equiv -1 \pmod 4$, and the second uses $7523 = 4 \cdot 1783 + 391$. From here on $391 = 17 \cdot 23$ appears as a denominator, so the symbols with denominator 391 are Jacobi symbols. The third equality is Theorem 5.4 with $391 \equiv 1783 \equiv -1 \pmod 4$, the fourth is Proposition 5.3 1) with $1783 = 4 \cdot 391 + 219$, and the fifth is Proposition 5.3 2) with $219 = 3 \cdot 73$.

$$\left(\frac{3}{391}\right)\left(\frac{73}{391}\right) = -\left(\frac{391}{3}\right)\left(\frac{391}{73}\right) = -\left(\frac{1}{3}\right)\left(\frac{26}{73}\right) = -\left(\frac{2}{73}\right)\left(\frac{13}{73}\right) = -\left(\frac{13}{73}\right)$$
$$= -\left(\frac{73}{13}\right) = -\left(\frac{8}{13}\right) = -\left(\frac{2}{13}\right)^3 = -(-1)^3 = 1.$$

The first equality is Theorem 5.4 with $3 \equiv 391 \equiv -1 \pmod 4$ and $73 \equiv 1 \pmod 4$. The remaining symbols have prime denominators. We used $391 = 130 \cdot 3 + 1$, $391 = 5 \cdot 73 + 26$, $\left(\frac{2}{73}\right) = 1$ since $73 \equiv 1 \pmod 8$, Theorem 4.6 with $13 \equiv 1 \pmod 4$, $73 = 5 \cdot 13 + 8$, and $\left(\frac{2}{13}\right) = -1$ since $13 \equiv -3 \pmod 8$. ◊

5. The Jacobi symbol

Definition 5.1 (Jacobi symbol). Let $n \in \mathbb{N}$ be odd with $n > 1$, let $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with distinct primes p_i, and let $a \in \mathbb{Z}$ with $(a, n) = 1$. The *Jacobi symbol* is

$$\left(\frac{a}{n}\right) = \left(\frac{a}{p_1}\right)^{\alpha_1} \cdots \left(\frac{a}{p_r}\right)^{\alpha_r},$$

where the factors are Legendre symbols. For n prime it coincides with the Legendre symbol, which justifies the notation.

Lemma 5.2. For odd integers u and v,

$$\frac{uv-1}{2} \equiv \frac{u-1}{2} + \frac{v-1}{2} \pmod 2, \qquad \frac{u^2v^2-1}{8} \equiv \frac{u^2-1}{8} + \frac{v^2-1}{8} \pmod 2.$$

Proof. The differences are $(u-1)(v-1)/2$ and $(u^2-1)(v^2-1)/8$. In the first, $4 \mid (u-1)(v-1)$. In the second, 8 divides each of u^2-1 and v^2-1, since each is a product of two consecutive even numbers, so $64 \mid (u^2-1)(v^2-1)$. ∎

Proposition 5.3. Let $m, n \in \mathbb{N}$ be odd with $m, n > 1$, and let $a, b \in \mathbb{Z}$.

1) If $(a, n) = 1$ and $a \equiv b \pmod n$, then $\left(\frac{a}{n}\right) = \left(\frac{b}{n}\right)$.
2) If $(ab, n) = 1$, then $\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right)\left(\frac{b}{n}\right)$.
3) $\left(\frac{-1}{n}\right) = 1$ if $n \equiv 1 \pmod 4$, and $\left(\frac{-1}{n}\right) = -1$ if $n \equiv -1 \pmod 4$.
4) $\left(\frac{2}{n}\right) = 1$ if $n \equiv \pm 1 \pmod 8$, and $\left(\frac{2}{n}\right) = -1$ if $n \equiv \pm 3 \pmod 8$.
5) If $(a, mn) = 1$, then $\left(\frac{a}{mn}\right) = \left(\frac{a}{m}\right)\left(\frac{a}{n}\right)$.

Proof. 1) and 2) follow factor by factor from Proposition 4.3 1) and 2), since $a \equiv b \pmod n$ implies $a \equiv b \pmod{p_i}$. 5) holds because the prime factorization of mn is obtained by adding the exponents in those of m and n.

For 3) and 4), Proposition 4.3 gives $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$ and $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$ for odd primes p; for the second, $p \equiv \pm 1 \pmod 8$ gives $p^2 \equiv 1 \pmod{16}$ and $p \equiv \pm 3 \pmod 8$ gives $p^2 \equiv 9 \pmod{16}$. By Lemma 5.2, $n \mapsto (-1)^{(n-1)/2}$ and $n \mapsto (-1)^{(n^2-1)/8}$ are multiplicative on odd integers. Hence $\left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}$ and $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$, and the same computations modulo 4 and 16 give the stated cases. ∎

Theorem 5.4 (Reciprocity for the Jacobi symbol). Let $m, n \in \mathbb{N}$ be odd with $m, n > 1$ and $(m, n) = 1$. Then

$$\left(\frac{m}{n}\right) = \begin{cases} -\left(\frac{n}{m}\right) & \text{if } n \equiv m \equiv -1 \pmod 4 \\ \left(\frac{n}{m}\right) & \text{otherwise.} \end{cases}$$

Proof. As in Theorem 4.6, the claim is equivalent to $\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}}$. Write $m = p_1 \cdots p_s$ and $n = q_1 \cdots q_t$ as products of not necessarily distinct primes; $p_i \neq q_j$ since $(m,n) = 1$. By Proposition 5.3 2) and 5),

$$\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = \prod_{i,j}\left(\frac{p_i}{q_j}\right)\left(\frac{q_j}{p_i}\right) = \prod_{i,j}(-1)^{P_i Q_j} = (-1)^{\left(\sum_i P_i\right)\left(\sum_j Q_j\right)},$$

where $P_i = (p_i - 1)/2$, $Q_j = (q_j - 1)/2$, and the middle equality is Theorem 4.6. By Lemma 5.2, $\sum_i P_i \equiv (m-1)/2$ and $\sum_j Q_j \equiv (n-1)/2 \pmod 2$. ∎

Example 5.5. Compute $\left(\frac{31}{95}\right)$.

Solution. $\left(\frac{31}{95}\right) = -\left(\frac{95}{31}\right) = -\left(\frac{2}{31}\right) = -1$, by Theorem 5.4 with $31 \equiv 95 \equiv -1 \pmod 4$, then $95 = 3 \cdot 31 + 2$, and $\left(\frac{2}{31}\right) = 1$ since $31 \equiv -1 \pmod 8$. ◊

Example 5.6. Compute $\left(\frac{60}{101}\right)$.

Solution.

$$\left(\frac{60}{101}\right) = \left(\frac{2^2 \cdot 15}{101}\right) = \left(\frac{2}{101}\right)^2\left(\frac{15}{101}\right) = \left(\frac{15}{101}\right) = \left(\frac{101}{15}\right)$$
$$= \left(\frac{11}{15}\right) = -\left(\frac{15}{11}\right) = -\left(\frac{4}{11}\right) = -\left(\frac{2}{11}\right)^2 = -1.$$

The fourth equality is Theorem 5.4 with $101 \equiv 1 \pmod 4$, the fifth uses $101 = 6 \cdot 15 + 11$, and the sixth is Theorem 5.4 with $11 \equiv 15 \equiv -1 \pmod 4$. ◊

Remark 5.7. If $x^2 \equiv a \pmod n$ is solvable, then a is a quadratic residue modulo every prime divisor of n, so $\left(\frac{a}{n}\right) = 1$. Hence $\left(\frac{a}{n}\right) = -1$ implies that $x^2 \equiv a \pmod n$ has no solution. The converse fails: $\left(\frac{2}{15}\right) = \left(\frac{2}{3}\right)\left(\frac{2}{5}\right) = (-1)(-1) = 1$, but $x^2 \equiv 2 \pmod 3$ has no solution, so neither has $x^2 \equiv 2 \pmod{15}$.

6. Polynomial congruences and Hensel's lemma

Definition 6.1 (Polynomial congruence). Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_i \in \mathbb{Z}$ and $a_n \neq 0$, and let $m \in \mathbb{N}$. A *polynomial congruence* is a congruence $f(x) \equiv 0 \pmod m$.

Let p be a prime. Starting from a solution of $f(x) \equiv 0 \pmod p$, we lift it to solutions modulo p^k. Here f' denotes the formal derivative of f, which again lies in $\mathbb{Z}[x]$.

Theorem 6.2 (Hensel). Let $f \in \mathbb{Z}[x]$, p a prime, and $c \in \mathbb{Z}$ with $f(c) \equiv 0 \pmod p$ and $f'(c) \not\equiv 0 \pmod p$. Then for every $k \in \mathbb{N}$ the congruence $f(x) \equiv 0 \pmod{p^k}$ has exactly one solution $x \equiv c_k \pmod{p^k}$ with $c_k \equiv c \pmod p$.

Lemma 6.3. Let $f \in \mathbb{Z}[x]$, p a prime, $a, t \in \mathbb{Z}$ and $j \in \mathbb{N}$. Then

$$f(a+tp^j) \equiv f(a) + tp^j f'(a) \pmod{p^{j+1}}.$$

Proof. For a polynomial of degree n the Taylor expansion is exact:

$$f(a+tp^j) = f(a) + tp^j f'(a) + t^2 p^{2j}\frac{f''(a)}{2!} + \cdots + t^n p^{nj}\frac{f^{(n)}(a)}{n!}.$$

Each $f^{(r)}(a)/r!$ is an integer, because $(x^s)^{(r)} = s(s-1)\cdots(s-r+1)x^{s-r}$ and $s(s-1)\cdots(s-r+1)/r! = \binom{s}{r}$. Every term with $r \geq 2$ is divisible by p^{2j}, and $2j \geq j+1$ since $j \geq 1$. ∎

Proof of Theorem 6.2. We proceed by induction on k. For $k=1$, $c_1 = c$ is a solution, and every solution congruent to c modulo p is congruent to c_1.

Let $k \geq 2$ and assume that $f(x) \equiv 0 \pmod{p^{k-1}}$ has exactly one solution $x \equiv c_{k-1} \pmod{p^{k-1}}$ with $c_{k-1} \equiv c \pmod{p}$. Let x be a solution of $f(x) \equiv 0 \pmod{p^k}$ with $x \equiv c \pmod{p}$. Then $f(x) \equiv 0 \pmod{p^{k-1}}$, so $x \equiv c_{k-1} \pmod{p^{k-1}}$ by the uniqueness in the induction hypothesis, that is, $x = c_{k-1} + tp^{k-1}$ with $t \in \mathbb{Z}$. By Lemma 6.3 with $j = k-1 \geq 1$,

$$f\big(c_{k-1} + tp^{k-1}\big) \equiv f(c_{k-1}) + tp^{k-1} f'(c_{k-1}) \pmod{p^k}.$$

By the induction hypothesis $p^{k-1} \mid f(c_{k-1})$, and $f'(c_{k-1}) \equiv f'(c) \pmod{p}$ since $c_{k-1} \equiv c \pmod{p}$ and $f' \in \mathbb{Z}[x]$. Hence $f\big(c_{k-1} + tp^{k-1}\big) \equiv 0 \pmod{p^k}$ if and only if

$$\frac{f(c_{k-1})}{p^{k-1}} + tf'(c) \equiv 0 \pmod{p}.$$

This is a linear congruence modulo p in t, and $f'(c) \not\equiv 0 \pmod{p}$, so by Remark 3.2 it has exactly one solution $t \equiv t_0 \pmod{p}$. Writing $t = t_0 + sp$ with $s \in \mathbb{Z}$,

$$x = c_{k-1} + (t_0 + sp)p^{k-1} = c_{k-1} + t_0 p^{k-1} + sp^k,$$

so every such solution satisfies $x \equiv c_{k-1} + t_0 p^{k-1} \pmod{p^k}$. Conversely, $c_k = c_{k-1} + t_0 p^{k-1}$ is a solution, and $c_k \equiv c_{k-1} \equiv c \pmod{p}$. Hence the solution is unique modulo p^k. ∎

Example 6.4. Solve $x^3 + 2x - 22 \equiv 0 \pmod{125}$.

Solution. Let $f(x) = x^3 + 2x - 22$, so $f'(x) = 3x^2 + 2$ and $125 = 5^3$.

Modulo 5. The values of $f(0), \ldots, f(4)$ modulo 5 are $3, 1, 0, 1, 0$, so $x \equiv 2$ or $x \equiv 4 \equiv -1 \pmod{5}$.

Case 1: $x \equiv 2 \pmod{5}$. Here $f'(2) = 14 \equiv -1 \not\equiv 0 \pmod{5}$, so by Theorem 6.2 there is exactly one solution modulo 125 in this class. Write $x = 5t + 2$ with $t \in \mathbb{Z}$. Modulo 25,

$$(5t+2)^3 + 2(5t+2) - 22 \equiv 60t + 8 + 10t + 4 - 22 = 70t - 10 \pmod{25},$$

and $70t - 10 \equiv 0 \pmod{25}$ if and only if $-5t \equiv 10 \pmod{25}$, that is, $t \equiv -2 \pmod{5}$. So $t = 5s - 2$ with $s \in \mathbb{Z}$, and $x = 5(5s-2) + 2 = 25s - 8$. Modulo 125, Lemma 6.3 gives

$$f(25s - 8) \equiv f(-8) + 25s f'(-8) = -550 + 25s \cdot 194 \equiv -50 + 25s \cdot 194 \pmod{125},$$

which vanishes modulo 125 if and only if $-2 + 194s \equiv 0 \pmod{5}$, that is, $s \equiv -2 \pmod{5}$. So $s = 5r - 2$ with $r \in \mathbb{Z}$, and $x = 25(5r-2) - 8 = 125r - 58$. Hence $x \equiv -58 \equiv 67 \pmod{125}$.

Case 2: $x \equiv -1 \pmod 5$. Here $f'(-1) = 5 \equiv 0 \pmod 5$, and Theorem 6.2 does not apply. By the proof of Theorem 6.2, if $c_{k-1} \equiv -1 \pmod 5$ solves the congruence modulo 5^{k-1}, then $c_{k-1} + t5^{k-1}$ solves it modulo 5^k if and only if $f(c_{k-1})/5^{k-1} + tf'(-1) \equiv 0 \pmod 5$, that is, if and only if $5^k \mid f(c_{k-1})$. So either every lift is a solution or none is.

Since $f(-1) = -25 \equiv 0 \pmod{25}$, every $x \equiv -1 \pmod 5$ solves $f(x) \equiv 0 \pmod{25}$, namely $x \equiv 4, 9, 14, 19, 24 \pmod{25}$. For each such c and $j \in \mathbb{Z}$, Lemma 6.3 and $5 \mid f'(c)$ give $f(c + 25j) \equiv f(c) \pmod{125}$. Now $f(4) = 50$, $f(9) = 725 \equiv 100$, $f(24) = 13850 \equiv 100$, while $f(14) = 2750 = 22 \cdot 125$ and $f(19) = 6875 = 55 \cdot 125$. So the solutions in this class are $x \equiv 14 + 25j$ and $x \equiv 19 + 25j \pmod{125}$ with $0 \le j \le 4$.

Altogether the congruence has 11 solutions modulo 125: $x \equiv 67$, and $x \equiv 14 + 25j$, $19 + 25j$ for $0 \le j \le 4$. ◊

7. Arithmetic functions

Definition 7.1 (Arithmetic function). An *arithmetic function* is a function $f : \mathbb{N} \to \mathbb{C}$. It is *multiplicative* if $f(ab) = f(a)f(b)$ for all $a, b \in \mathbb{N}$ with $(a, b) = 1$, and *additive* if $f(ab) = f(a) + f(b)$ for all $a, b \in \mathbb{N}$ with $(a, b) = 1$.

Example 7.2. Let $d(n)$, also written $\tau(n)$, be the number of positive divisors of n. If $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with distinct primes p_i, then $d(n) = \prod_{i=1}^{r} (1 + \alpha_i)$, because the divisors of n are exactly the products $p_1^{\beta_1} \cdots p_r^{\beta_r}$ with $0 \le \beta_i \le \alpha_i$. Hence d is multiplicative: for $(m, n) = 1$ the prime divisors of m and n are disjoint, and $d(mn) = d(m)d(n)$.

Example 7.3. Let $\sigma(n)$ be the sum of the positive divisors of n; for instance $\sigma(6) = 1 + 2 + 3 + 6 = 12$. For $n = \prod_{i=1}^{r} p_i^{\alpha_i}$ as above, summing over the divisors described in the previous example gives

$$\sigma(n) = \prod_{i=1}^{r} \sum_{j=0}^{\alpha_i} p_i^j.$$

As for d, this formula shows that σ is multiplicative.

Example 7.4. Euler's function φ is multiplicative (Remark 1.1). Let $\omega(n)$ be the number of distinct prime divisors of n; then ω is additive. Let $\Omega(n)$ be the number of prime divisors of n counted with multiplicity, so $\Omega(n) = \sum_{i=1}^{r} \alpha_i$ for n as above; then Ω is additive.

Proposition 7.5.

1) If f is multiplicative and not identically 0, then $f(1) = 1$.

2) If f is additive, then $f(1) = 0$.

Proof. 1) Choose a with $f(a) \neq 0$. Since $(1, a) = 1$, $f(a) = f(1 \cdot a) = f(1)f(a)$, so $f(1) = 1$.

2) Since $(1, 1) = 1$, $f(1) = f(1 \cdot 1) = f(1) + f(1)$, so $f(1) = 0$. ■

Definition 7.6 (Summation function). Let $f : \mathbb{N} \to \mathbb{C}$ be an arithmetic function. Its *summation function* is $f^+(n) = F(n) = \sum_{d|n} f(d)$, where d runs over the positive divisors of n.

Lemma 7.7. Let $a, b \in \mathbb{N}$ with $(a, b) = 1$ and let $d \mid ab$. Then there are unique $d_1, d_2 \in \mathbb{N}$ with $d_1 \mid a$, $d_2 \mid b$ and $d = d_1 d_2$. Hence $(d_1, d_2) \mapsto d_1 d_2$ is a bijection from the pairs of positive divisors of a and b onto the positive divisors of ab.

Proof. Put $d_1 = (d, a)$ and $d_2 = (d, b)$. They are coprime and both divide d, so $d_1 d_2 \mid d$. Conversely, let q^e be the exact power of a prime q dividing d. Since q divides at most one of a, b, say a, and $q^e \mid ab$, we get $q^e \mid a$ and so $q^e \mid d_1$. Hence $d \mid d_1 d_2$, and $d = d_1 d_2$. If also $d = e_1 e_2$ with $e_1 \mid a$ and $e_2 \mid b$, then $e_1 \mid (d, a) = d_1$ and $e_2 \mid d_2$, and $e_1 e_2 = d_1 d_2$ forces $e_1 = d_1$, $e_2 = d_2$. Every product $d_1 d_2$ with $d_1 \mid a$, $d_2 \mid b$ divides ab, which gives the bijection. ∎

Theorem 7.8. If $f : \mathbb{N} \to \mathbb{C}$ is multiplicative, then $F(n) = \sum_{d|n} f(d)$ is multiplicative.

Proof. Let $(m, n) = 1$. By Lemma 7.7, and since $(d_1, d_2) = 1$ for $d_1 \mid m$, $d_2 \mid n$,

$$F(mn) = \sum_{d|mn} f(d) = \sum_{d_1|m,\ d_2|n} f(d_1 d_2) = \sum_{d_1|m,\ d_2|n} f(d_1) f(d_2) = \sum_{d_1|m} f(d_1) \sum_{d_2|n} f(d_2) = F(m) F(n).$$

∎

Theorem 7.9. For every $n \in \mathbb{N}$, $\varphi^+(n) = \sum_{d|n} \varphi(d) = n$.

Proof. For $n = 1$ both sides equal 1. For a prime power $n = p^\alpha$ with $\alpha \geq 1$,

$$F(p^\alpha) = \sum_{d|p^\alpha} \varphi(d) = \sum_{i=0}^{\alpha} \varphi(p^i) = 1 + (p - 1) + (p^2 - p) + \cdots + (p^\alpha - p^{\alpha-1}) = p^\alpha.$$

For $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$, F is multiplicative by Theorem 7.8, so $F(n) = F(p_1^{\alpha_1}) \cdots F(p_r^{\alpha_r}) = p_1^{\alpha_1} \cdots p_r^{\alpha_r} = n$. ∎

Definition 7.10 (Möbius function). The *Möbius function* is

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1 \\ (-1)^r & \text{if } n = p_1 \cdots p_r \text{ with distinct primes } p_i \\ 0 & \text{otherwise.} \end{cases}$$

So $\mu(n) \neq 0$ if and only if n is square-free. For example, $\mu(10) = (-1)^2 = 1$.

Proposition 7.11. The function μ is multiplicative, and

$$\mu^+(n) = \sum_{d|n} \mu(d) = \begin{cases} 1 \text{ if } n = 1 \\ 0 \text{ otherwise.} \end{cases}$$

Proof. Let $(m,n)=1$. If m or n is not square-free, neither is mn, and both sides of $\mu(mn)=\mu(m)\mu(n)$ vanish. Otherwise m and n are products of r and s distinct primes, the primes of m and n are distinct, and $\mu(mn)=(-1)^{r+s}=\mu(m)\mu(n)$.

For $n=1$, $\mu^+(1)=\mu(1)=1$. For $n=p^\alpha$ with $\alpha\geq 1$, $\mu^+(p^\alpha)=\mu(1)+\mu(p)+\cdots+\mu(p^\alpha)=1+(-1)=0$. For $n=p_1^{\alpha_1}\cdots p_r^{\alpha_r}$ with $r\geq 1$, μ^+ is multiplicative by Theorem 7.8, so $\mu^+(n)=\mu^+(p_1^{\alpha_1})\cdots\mu^+(p_r^{\alpha_r})=0$. ∎

Theorem 7.12 (Möbius inversion formula). Let $f,F:\mathbb{N}\to\mathbb{C}$ be arithmetic functions with $F(n)=\sum_{d\mid n}f(d)$ for every $n\in\mathbb{N}$. Then $f(n)=\sum_{d\mid n}\mu(d)F(n/d)$ for every $n\in\mathbb{N}$.

Proof. The pairs (d,c) of positive integers with $d\mid n$ and $c\mid n/d$ are exactly those with $cd\mid n$, that is, with $c\mid n$ and $d\mid n/c$. Hence, using Proposition 7.11,

$$\sum_{d\mid n}\mu(d)F\left(\frac{n}{d}\right)=\sum_{d\mid n}\mu(d)\sum_{c\mid n/d}f(c)=\sum_{c\mid n}f(c)\sum_{d\mid n/c}\mu(d)=f(n),$$

since the inner sum is 1 for $c=n$ and 0 otherwise. ∎

Theorem 7.13. Let $f,F:\mathbb{N}\to\mathbb{C}$ with $F(n)=\sum_{d\mid n}f(d)$ for every n. If F is multiplicative, then f is multiplicative.

Proof. Let $(m,n)=1$. By Theorem 7.12 and Lemma 7.7, writing $d=d_1d_2$ with $d_1\mid m$, $d_2\mid n$,

$$\begin{aligned}f(mn)&=\sum_{d\mid mn}\mu(d)F\left(\frac{mn}{d}\right)=\sum_{d_1\mid m,\ d_2\mid n}\mu(d_1d_2)F\left(\frac{m}{d_1}\cdot\frac{n}{d_2}\right)\\&=\sum_{d_1\mid m}\mu(d_1)F\left(\frac{m}{d_1}\right)\sum_{d_2\mid n}\mu(d_2)F\left(\frac{n}{d_2}\right)=f(m)f(n).\end{aligned}$$

The third equality uses that μ and F are multiplicative, with $(d_1,d_2)=1$ and $(m/d_1,n/d_2)=1$. ∎

Example 7.14. Let $F(n)=n$. Compute f with $F=f^+$.

Solution. By Theorem 7.9, $F=\varphi^+$, so Theorem 7.12 gives

$$\varphi(n)=\sum_{d\mid n}\mu(d)F\left(\frac{n}{d}\right)=n\sum_{d\mid n}\frac{\mu(d)}{d}.$$

Both F and φ are multiplicative. For $n=p_1^{\alpha_1}\cdots p_r^{\alpha_r}$, only square-free d contribute, that is, products of distinct p_i, so

$$\varphi(n) = n - \frac{n}{p_1} - \cdots - \frac{n}{p_r} + \frac{n}{p_1 p_2} + \cdots + \frac{n}{p_{r-1} p_r} - \frac{n}{p_1 p_2 p_3} - \cdots + (-1)^r \frac{n}{p_1 \cdots p_r} = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right).$$

◊

8. p-adic numbers

Let p be a prime. Every $n \in \mathbb{Z}_{\geq 0}$ has a unique expansion $n = a_0 + a_1 p + \cdots + a_k p^k$ with $0 \leq a_i < p$.

Definition 8.1 (p-adic integer). A *p-adic integer* is a formal series $\alpha = a_0 + a_1 p + a_2 p^2 + \cdots$ with digits $a_i \in \{0, 1, \ldots, p-1\}$. The set of p-adic integers is denoted $\mathbb{Z}_p$. For $k \in \mathbb{N}$ the *truncation* of α is $\alpha_k = \sum_{i=0}^{k-1} a_i p^i$.

Remark 8.2. Let $(y_k)_{k \geq 1}$ be integers with $y_{k+1} \equiv y_k \pmod{p^k}$ for every k. Then there is exactly one $\alpha \in \mathbb{Z}_p$ with $\alpha_k \equiv y_k \pmod{p^k}$ for every k. Indeed, let x_k be the least non-negative residue of y_k modulo p^k. Then $x_{k+1} \equiv x_k \pmod{p^k}$ and $0 \leq x_k < p^k$ give $x_{k+1} = x_k + a_k p^k$ with $0 \leq a_k < p$, and $\alpha = \sum_k a_k p^k$ is the required series. Uniqueness holds because $0 \leq \alpha_k < p^k$ determines α_k from its residue, and the truncations determine the digits.

Definition 8.3 (Operations on $\mathbb{Z}_p$). For $\alpha, \beta \in \mathbb{Z}_p$, the sum $\alpha + \beta$ and the product $\alpha\beta$ are the p-adic integers with

$$(\alpha + \beta)_k \equiv \alpha_k + \beta_k \pmod{p^k}, \quad (\alpha\beta)_k \equiv \alpha_k \beta_k \pmod{p^k} \quad (k \in \mathbb{N}).$$

They exist and are unique by Remark 8.2, since the sequences $\alpha_k + \beta_k$ and $\alpha_k \beta_k$ satisfy its compatibility condition. Each ring axiom holds modulo every p^k, so $\mathbb{Z}_p$ is a commutative ring with zero $0 + 0p + \cdots$ and identity $1 + 0p + \cdots$.

Example 8.4. $-1 = (p-1) + (p-1)p + (p-1)p^2 + \cdots$ in $\mathbb{Z}_p$, because the truncations of the right side are $(p-1)\left(1 + p + \cdots + p^{k-1}\right) = p^k - 1 \equiv -1 \pmod{p^k}$.

Example 8.5. Let $\alpha, \beta \in \mathbb{Z}_3$ with $\alpha = 2 + 1 \cdot 3 + \cdots$ and $\beta = 1 + 2 \cdot 3 + \cdots$. Then $(\alpha + \beta)_1 \equiv 2 + 1 \equiv 0 \pmod{3}$. Further $\alpha_2 = 5$ and $\beta_2 = 7$, so $(\alpha + \beta)_2 \equiv 12 \equiv 3 \pmod{3^2}$. Hence $\alpha + \beta = 0 + 1 \cdot 3 + \cdots$.

Remark 8.6. $\mathbb{Z} \subset \mathbb{Z}_p$: the integer n is identified with the p-adic integer whose truncations satisfy $\alpha_k \equiv n \pmod{p^k}$. For $n \geq 0$ this is the base-p expansion of n followed by zero digits. The map is an injective ring homomorphism, since distinct integers differ modulo p^k for large k.

Example 8.7. The equation $2x + 3 = 0$ has no solution in $\mathbb{Z}$, but it has one in $\mathbb{Z}_3$. Formally,

$$-\frac{3}{2} = \frac{3}{-2} = \frac{3}{1-3} = 3 \cdot \frac{1}{1-3} = 3(1 + 3 + 3^2 + \cdots),$$

which suggests $\gamma = 0 + 1 \cdot 3 + 1 \cdot 3^2 + \cdots$. Indeed, $\gamma_k = 3 + 3^2 + \cdots + 3^{k-1} = \left(3^k - 3\right)/2$, so $2\gamma_k + 3 = 3^k \equiv 0 \pmod{3^k}$ for every k, and $2\gamma + 3 = 0$ in $\mathbb{Z}_3$.

Remark 8.8. Let $x = a/b$ with $a, b \in \mathbb{Z}$ and $p \nmid b$. Then there is exactly one $\xi \in \mathbb{Z}_p$ with $b\xi = a$, and we write $\xi = a/b$. Indeed, b is invertible modulo every p^k; if b_k^{-1} denotes an inverse modulo p^k, then $y_k = ab_k^{-1}$ satisfies $y_{k+1} \equiv y_k \pmod{p^k}$, and the ξ of Remark 8.2 satisfies $(b\xi)_k \equiv a \pmod{p^k}$ for all k. If $b\xi = b\xi'$, then $b(\xi - \xi')_k \equiv 0 \pmod{p^k}$, so $(\xi - \xi')_k = 0$ for all k and $\xi = \xi'$.

Definition 8.9 (p-adic number). A *p-adic number* is a formal series $x = \sum_{k=-N}^{\infty} a_k p^k$ with $N \in \mathbb{Z}_{\geq 0}$ and digits $a_k \in \{0, \ldots, p-1\}$; series that differ only by leading zero digits are identified. The set of p-adic numbers is denoted $\mathbb{Q}_p$, and $\mathbb{Z}_p \subset \mathbb{Q}_p$ consists of the series with $a_k = 0$ for $k < 0$.

Proposition 8.10. For $\alpha \in \mathbb{Z}_p$ and $N \in \mathbb{Z}_{\geq 0}$ let $p^{-N}\alpha$ denote the series whose digit at p^{k-N} is the digit of α at p^k. Every $x \in \mathbb{Q}_p$ has this form, and $p^{-N}\alpha = p^{-N-1}(p\alpha)$. Define

$$p^{-N}\alpha + p^{-N}\beta = p^{-N}(\alpha + \beta), \quad (p^{-N}\alpha)(p^{-N}\beta) = p^{-2N}(\alpha\beta).$$

Then $\mathbb{Q}_p$ is a field containing $\mathbb{Z}_p$ as a subring. An element $u \in \mathbb{Z}_p$ is a unit of $\mathbb{Z}_p$ if and only if its digit a_0 is nonzero, and every $x \in \mathbb{Q}_p$ with $x \neq 0$ can be written uniquely as $x = p^v u$ with $v \in \mathbb{Z}$ and u a unit of $\mathbb{Z}_p$.

Proof. Multiplication by p in $\mathbb{Z}_p$ shifts the digits up by one, since $(p\alpha)_{k+1} = p\alpha_k$. This gives $p^{-N}\alpha = p^{-N-1}(p\alpha)$, so the definitions do not depend on the choice of a common N, and the ring axioms follow from those of $\mathbb{Z}_p$.

If u has $a_0 \neq 0$, then every u_k is coprime to p. Let w_k be an inverse of u_k modulo p^k; then $w_{k+1} \equiv w_k \pmod{p^k}$, and the resulting $w \in \mathbb{Z}_p$ satisfies $(uw)_k \equiv 1$ for all k, so $uw = 1$. Conversely, $uw = 1$ gives $u_1 w_1 \equiv 1 \pmod{p}$, so $a_0 = u_1 \neq 0$.

For $x \neq 0$ let v be the least index of a nonzero digit. Shifting the digits gives $x = p^v u$ with $u \in \mathbb{Z}_p$ whose digit a_0 is nonzero, so u is a unit, and v and u are determined by x. Then $x^{-1} = p^{-v}u^{-1}$, so $\mathbb{Q}_p$ is a field. ∎

Remark 8.11. $\mathbb{Q} \subset \mathbb{Q}_p$: every nonzero rational number is $p^v a/b$ with $v \in \mathbb{Z}$ and $p \nmid ab$, and $a/b \in \mathbb{Z}_p$ by Remark 8.8. This gives an injective field homomorphism $\mathbb{Q} \to \mathbb{Q}_p$.

Example 8.12. Is $x^2 - 2 = 0$ solvable in $\mathbb{Z}_7$?

Solution. Let $\alpha = a_0 + a_1 \cdot 7 + a_2 \cdot 7^2 + \cdots$ be a solution. Then $a_0^2 \equiv 2 \pmod{7}$, so $a_0 = 3$ or $a_0 = 4$, that is, $\alpha_1 = 3$ or $\alpha_1 = 4$.

Let $\alpha_1 = 3$. We seek $\alpha_2 = a_0 + a_1 \cdot 7$ modulo 49:

$$\alpha_2^2 - 2 = (a_0 + a_1 \cdot 7)^2 - 2 = a_0^2 - 2 + 2 \cdot 7 a_0 a_1 + a_1^2 7^2 \equiv 0 \pmod{7^2}$$

if and only if $a_0^2 - 2 + 14 a_0 a_1 \equiv 0 \pmod{7^2}$, that is, $3^2 - 2 + 14 \cdot 3 a_1 \equiv 0 \pmod{49}$, which gives $1 + 6a_1 \equiv 0 \pmod 7$ and $a_1 = 1$. Continuing recursively, $\alpha = 3 + 1 \cdot 7 + 2 \cdot 7^2 + 6 \cdot 7^3 + \cdots$. The other solution is $\alpha = 4 + 5 \cdot 7 + 4 \cdot 7^2 + \cdots$.

Both solutions exist: $f(x) = x^2 - 2$ has $f'(3) = 6 \not\equiv 0$ and $f'(4) = 8 \not\equiv 0 \pmod 7$, so Theorem 6.2 gives compatible solutions c_k modulo 7^k, and by Remark 8.2 they define $\alpha \in \mathbb{Z}_7$ with $(\alpha^2 - 2)_k = 0$ for every k, that is, $\alpha^2 = 2$. ◊

Remark 8.13. Write $\mathbb{Q}_\infty = \mathbb{R}$, and let $p \le \infty$ range over the primes and ∞. If a polynomial equation with rational coefficients has a solution in $\mathbb{Q}$, then it has a solution in $\mathbb{Q}_p$ for every $p \le \infty$, since $\mathbb{Q} \subset \mathbb{Q}_p$.

Theorem 8.14 (Hasse–Minkowski) (cited)**.** Let $f(x_1, \ldots, x_n) \in \mathbb{Q}[x_1, \ldots, x_n]$ be a quadratic form. The equation $f(x_1, \ldots, x_n) = 0$ has a nontrivial solution in $\mathbb{Q}$ if and only if it has a nontrivial solution in $\mathbb{Q}_p$ for every $p \le \infty$.

9. The p-adic metric

Definition 9.1 (Absolute value). Let F be a field. A map $|\cdot| : F \to \mathbb{R}_{\ge 0}$ is an *absolute value* if for all $x, y \in F$:

1) $|x| = 0$ if and only if $x = 0$,
2) $|xy| = |x|\,|y|$,
3) $|x + y| \le |x| + |y|$.

It is *non-archimedean* if moreover $|x + y| \le \max\{|x|, |y|\}$ for all $x, y \in F$.

Remark 9.2. From $|1| = |1|^2$ and $|1| \neq 0$ we get $|1| = 1$, and from $|-1|^2 = |1|$ we get $|-1| = 1$. Hence an absolute value induces a metric $d(x, y) = |x - y|$ on F.

Definition 9.3 (Ultrametric space). A metric space whose metric is induced by a non-archimedean absolute value is called *ultrametric*. Its metric satisfies $d(x, z) \le \max\{d(x, y), d(y, z)\}$.

Definition 9.4 (p-adic valuation). The *p-adic valuation* is the function $v_p : \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ defined as follows. For $0 \neq x \in \mathbb{Z}$, $v_p(x)$ is the unique non-negative integer with $x = p^{v_p(x)} x'$ and $p \nmid x'$. For $x \in \mathbb{Q} \setminus \mathbb{Z}$, write $x = a/b$ with $a, b \in \mathbb{Z}$, $b \neq 0$, and put $v_p(x) = v_p(a) - v_p(b)$. Finally $v_p(0) = \infty$.

Proposition 9.5. The valuation v_p is well defined, and for all $x, y \in \mathbb{Q}$:

1) $v_p(xy) = v_p(x) + v_p(y)$,

2) $v_p(x+y) \geq \min\{v_p(x), v_p(y)\}$.

Proof. On nonzero integers, $v_p(ab) = v_p(a) + v_p(b)$ by unique factorization. If $a/b = a'/b'$, then $ab' = a'b$, so $v_p(a) + v_p(b') = v_p(a') + v_p(b)$, and the definition does not depend on the representation. Property 1 follows from the integer case. For 2 we may assume $x, y \neq 0$; write $x = p^v a/b$ and $y = p^w c/d$ with $p \nmid abcd$ and $v \leq w$. Then $x + y = p^v(ad + p^{w-v}cb)/(bd)$, and $p \nmid bd$, so $v_p(x+y) \geq v$. ∎

Definition 9.6 (p-adic absolute value). The *p-adic absolute value* $|\cdot|_p : \mathbb{Q} \to \mathbb{R}_{\geq 0}$ is $|x|_p = p^{-v_p(x)}$ for $x \neq 0$ and $|0|_p = 0$.

Theorem 9.7. $|\cdot|_p$ is a non-archimedean absolute value on $\mathbb{Q}$.

Proof. $|x|_p = 0$ only for $x = 0$. By Proposition 9.5, $|xy|_p = p^{-v_p(x)-v_p(y)} = |x|_p\,|y|_p$ and $|x+y|_p \leq p^{-\min\{v_p(x),v_p(y)\}} = \max\{|x|_p, |y|_p\}$, which also implies the triangle inequality. ∎

Remark 9.8. $\lim_{n\to\infty} p^n = 0$ in the p-adic metric. Indeed, let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $1/N < \varepsilon$. Since $p^N > N$, we have $1/p^N < \varepsilon$, so for $n \geq N$, $|p^n|_p = 1/p^n < \varepsilon$.

Definition 9.9 (Equivalent absolute values). Two absolute values on a field are *equivalent* if they induce the same topology. The *trivial* absolute value is $|x| = 1$ for $x \neq 0$, $|0| = 0$. The *standard* absolute value on $\mathbb{Q}$ is denoted $|\cdot|_\infty$.

Theorem 9.10 (Ostrowski). Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_\infty$ or to $|\cdot|_p$ for some prime p.

Proof. Let $|\cdot|$ be a nontrivial absolute value on $\mathbb{Q}$. It suffices to find $c > 0$ with $|x| = |x|_\infty^c$ for all x, or $|x| = |x|_p^c$ for all x: then the open balls of $|\cdot|$ of radius r are the balls of the other absolute value of radius $r^{1/c}$, so the topologies agree. Since $|-1| = 1$ and $|\cdot|$ is multiplicative, it suffices to determine $|n|$ for $n \in \mathbb{N}$. For $n \in \mathbb{N}$ the triangle inequality gives $|n| \leq n$.

Case 1: $|n| \leq 1$ for every $n \in \mathbb{N}$. If $|n| = 1$ for all $n \in \mathbb{N}$, then $|\cdot|$ is trivial by multiplicativity. So some $n \in \mathbb{N}$ has $|n| < 1$; let n_0 be the least one. Then $n_0 > 1$, and n_0 is a prime p: if $n_0 = ab$ with $1 < a, b < n_0$, then $|a| = |b| = 1$ and $|n_0| = 1$. Suppose a prime $q \neq p$ also has $|q| < 1$. Choose $M \in \mathbb{N}$ with $|p|^M < 1/2$ and $|q|^M < 1/2$. Since $(p^M, q^M) = 1$, there are $u, w \in \mathbb{Z}$ with $up^M + wq^M = 1$, and then $1 = |1| \leq |u|\,|p|^M + |w|\,|q|^M < 1/2 + 1/2 = 1$, because $|u|, |w| \leq 1$. This is a contradiction, so $|q| = 1$ for every prime $q \neq p$. Put $c = -\log|p| \,/ \log p > 0$, so $|p| = p^{-c}$. For $n = p^v n'$ with $p \nmid n'$ we get $|n| = p^{-cv} = |n|_p^c$.

Case 2: $|n_1| > 1$ for some $n_1 \in \mathbb{N}$. We first show that for all integers $m, n > 1$,

$$|m| \leq \max\{1, |n|\}^{\log m / \log n}.$$

Let $k \in \mathbb{N}$ and write $m^k = \sum_{i=0}^{s} d_i n^i$ in base n, with $0 \leq d_i < n$ and $s \leq k \log m / \log n$. Then $|d_i| \leq d_i < n$, and

$$|m|^k \le \sum_{i=0}^{s} |d_i|\ |n|^i \le (s+1)n \max\{1,|n|\}^s \le \left(1 + k\frac{\log m}{\log n}\right) n \max\{1,|n|\}^{k \log m/\log n}.$$

Taking k-th roots and letting $k \to \infty$ gives the inequality. Applying it with $m = n_1$ shows $1 < |n_1| \le \max\{1,|n|\}^{\log n_1/\log n}$, so $|n| > 1$ for every $n > 1$. Hence $|m|^{1/\log m} \le |n|^{1/\log n}$ for all $m, n > 1$, and by symmetry equality holds. So $|n|^{1/\log n} = e^c$ for a constant c, with $c > 0$ since $|n| > 1$. Then $|n| = n^c = |n|_\infty^c$ for $n > 1$, and also for $n = 1$. ∎

Remark 9.11 (Recall). A *completion* of a metric space X is a complete metric space $\hat{X}$ together with an isometric embedding $X \to \hat{X}$ whose image is dense. A completion exists and is unique up to an isometry that is the identity on X.

Theorem 9.12 (Construction of $\mathbb{Q}_p$ and $\mathbb{Z}_p$). For $0 \ne x \in \mathbb{Q}_p$ let $v(x)$ be the least index of a nonzero digit of x, and put $|x|_p = p^{-v(x)}$, $|0|_p = 0$. Then:

1) $|\cdot|_p$ is a non-archimedean absolute value on $\mathbb{Q}_p$ that extends the p-adic absolute value of $\mathbb{Q}$;
2) $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \le 1\}$;
3) $\mathbb{Q}$ is dense in $\mathbb{Q}_p$;
4) $\mathbb{Q}_p$ is complete.

Hence $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ with respect to $|\cdot|_p$. The field $\mathbb{Q}$ itself is not complete under the p-adic metric.

Proof. We use one fact throughout: for $\alpha, \beta \in \mathbb{Z}_p$ and $k \in \mathbb{N}$, $|\alpha - \beta|_p \le p^{-k}$ holds if and only if $\alpha_k = \beta_k$. Indeed, $(\alpha - \beta)_k \equiv \alpha_k - \beta_k \pmod{p^k}$ and $0 \le \alpha_k, \beta_k < p^k$, and $|\alpha - \beta|_p \le p^{-k}$ means that the first k digits of $\alpha - \beta$ vanish.

1) In the notation of Proposition 8.10, $x = p^v u$ with u a unit gives $v(x) = v$. For $x = p^v u$ and $y = p^w u'$, $xy = p^{v+w} uu'$ with uu' a unit, so $|xy|_p = |x|_p\ |y|_p$. For the ultrametric inequality, write $x = p^{-N}\alpha$ and $y = p^{-N}\beta$ with $\alpha, \beta \in \mathbb{Z}_p$. If the first k digits of α and β vanish, then $(\alpha + \beta)_k \equiv \alpha_k + \beta_k = 0$, so those of $\alpha + \beta$ vanish; hence $v(x + y) \ge \min\{v(x), v(y)\}$. For rational $x = p^v a/b$ with $p \nmid ab$, the element $a/b \in \mathbb{Z}_p$ has first digit $ab^{-1} \not\equiv 0 \pmod{p}$, so it is a unit and $v(x) = v = v_p(x)$.

2) $x \in \mathbb{Z}_p$ if and only if x has no nonzero digit of negative index, that is, $v(x) \ge 0$.

3) Let $x = \sum_{k \ge -N} a_k p^k$ and let $s_n = \sum_{k=-N}^{n-1} a_k p^k \in \mathbb{Q}$. The first nonzero digit of $x - s_n$ has index at least n, so $|x - s_n|_p \le p^{-n} \to 0$.

4) Let (x_n) be a Cauchy sequence in $\mathbb{Q}_p$. It is bounded, so there is N with $|x_n|_p \le p^N$ for all n; then $y_n = p^N x_n \in \mathbb{Z}_p$ by 2), and (y_n) is Cauchy. Choose $n_1 < n_2 < \cdots$ with $|y_n - y_m|_p \le p^{-k}$ for $n, m \ge n_k$. By the fact above, $(y_n)_k = \left(y_{n_k}\right)_k =: z_k$ for $n \ge n_k$. Then $z_{k+1} = \left(y_{n_{k+1}}\right)_{k+1} \equiv \left(y_{n_{k+1}}\right)_k = z_k \pmod{p^k}$, so by Remark 8.2 there is $\eta \in \mathbb{Z}_p$ with $\eta_k = z_k$. For $n \ge n_k$ we get $(y_n)_k = \eta_k$, that is, $|y_n - \eta|_p \le p^{-k}$. So $y_n \to \eta$ and $x_n \to p^{-N}\eta$.

By 1), 3) and 4), $\mathbb{Q}_p$ with $|\cdot|_p$ is a completion of $\mathbb{Q}$. Finally, $\mathbb{Q}$ is not complete: for $p = 7$, the truncations of the element α with $\alpha^2 = 2$ constructed in the example on $x^2 - 2$ form a Cauchy sequence in $\mathbb{Q}$ by 3). Its limit in $\mathbb{Q}_7$ is α, which is not rational, since 2 is not the square of a rational number. ∎

10. Worked exercises

Example 10.1. Show that 2 is a primitive root modulo 11 and solve $7x^8 \equiv 5 \pmod{11}$.

Solution. $O_{11}(2) \mid \varphi(11) = 10$, so $O_{11}(2) \in \{1, 2, 5, 10\}$. Since $2^1 = 2$, $2^2 = 4$ and $2^5 = 32 \equiv -1 \pmod{11}$, $O_{11}(2) = 10$, and 2 is a primitive root modulo 11. Taking indices, Theorem 2.3 gives $\operatorname{ind}_2(7x^8) \equiv \operatorname{ind}_2 5 \pmod{10}$, that is, $\operatorname{ind}_2 7 + 8 \operatorname{ind}_2 x \equiv \operatorname{ind}_2 5 \pmod{10}$. The tables of powers and indices are:

a	1	2	3	4	5	6	7	8	9	10
2^a	2	4	8	5	10	9	7	3	6	1
$\operatorname{ind}_2 a$	10	1	8	2	4	9	7	3	6	5

So $7 + 8 \operatorname{ind}_2 x \equiv 4 \pmod{10}$, that is, $8 \operatorname{ind}_2 x \equiv 7 \pmod{10}$. Since $(8, 10) = 2 \nmid 7$, there is no solution (Remark 3.2). ◊

Example 10.2. Solve $x^3 - x^2 - 1 \equiv 0 \pmod{27}$.

Solution. Here $27 = 3^3$. Let $f(x) = x^3 - x^2 - 1$. Modulo 3, $f(0) \equiv f(1) \equiv -1$ and $f(2) = 3 \equiv 0$, so $x \equiv -1 \pmod{3}$ is the only solution. Since $f'(-1) = 5 \not\equiv 0 \pmod{3}$, Theorem 6.2 gives exactly one solution modulo 27.

Write $x = 3b - 1$. Modulo 9,

$$(3b)^3 - 3(3b)^2 + 3(3b) - 1 - (3b)^2 + 2 \cdot 3b - 1 - 1 \equiv 6b - 3 \pmod{9},$$

and $6b - 3 \equiv 0 \pmod{9}$ if and only if $2b - 1 \equiv 0 \pmod{3}$, that is, $b \equiv 2 \pmod{3}$. So $b = 3c + 2$ and $x = 3(3c + 2) - 1 = 9c + 5$. Modulo 27,

$$(9c)^3 + 3 \cdot 5(9c)^2 + 3 \cdot 5^2(9c) + 5^3 - (9c)^2 - 2 \cdot 5(9c) - 5^2 - 1 \equiv 585c + 99 \equiv 18c + 18 \pmod{27},$$

and $18c + 18 \equiv 0 \pmod{27}$ if and only if $c \equiv -1 \pmod{3}$. Hence $x \equiv 9 \cdot (-1) + 5 = -4 \pmod{27}$. Equivalently, the solution is found digit by digit in the form $x = a + 3b + 3^2c$. ◊

Example 10.3. Let $p > 2$ be a prime. Show that $(p - 1)! \equiv -1 \pmod{p}$.

Solution. Let g be a primitive root modulo p. Both $1, 2, \ldots, p - 1$ and, by Theorem 1.17, $1, g, g^2, \ldots, g^{p-2}$ form reduced residue systems modulo p, so

$$(p - 1)! \equiv g^{1+2+\cdots+(p-2)} = g^{(p-2)(p-1)/2} = \left(g^{(p-1)/2}\right)^{p-2} \equiv (-1)^{p-2} = -1 \pmod{p}.$$

Here $g^{(p-1)/2} \equiv -1$ by Proposition 3.6, since $g^{(p-1)/2} \not\equiv 1$ because $O_p(g) = p - 1$; and $p - 2$ is odd. ◊

Example 10.4. Let $p > 2$ be a prime and g a primitive root modulo p. Show that g is a quadratic non-residue modulo p.

Solution. If g were a quadratic residue, then $g^{(p-1)/2} \equiv 1 \pmod{p}$ by Proposition 3.6, contradicting $O_p(g) = p-1$. ◊

Example 10.5. Decide whether $x^2 \equiv 66 \pmod{101}$, $x^2 \equiv 7! \pmod{83}$ and $x^2 \equiv 30 \pmod{77}$ are solvable.

Solution. Using $101 \equiv 1 \pmod{4}$, $101 \equiv -3 \pmod{8}$, $101 = 33 \cdot 3 + 2 = 9 \cdot 11 + 2$:

$$\left(\frac{66}{101}\right) = \left(\frac{2 \cdot 3 \cdot 11}{101}\right) = \left(\frac{2}{101}\right)\left(\frac{3}{101}\right)\left(\frac{11}{101}\right)$$
$$= -\left(\frac{101}{3}\right)\left(\frac{101}{11}\right) = -\left(\frac{2}{3}\right)\left(\frac{2}{11}\right) = -(-1)(-1) = -1,$$

so $x^2 \equiv 66 \pmod{101}$ has no solution.

Next, $7! = 1 \cdot 2 \cdot \cdots \cdot 7$ and $\left(\frac{4}{83}\right) = 1$, $\left(\frac{6}{83}\right) = \left(\frac{2}{83}\right)\left(\frac{3}{83}\right)$, so

$$\left(\frac{7!}{83}\right) = \left(\frac{1}{83}\right)\left(\frac{2}{83}\right)\cdots\left(\frac{7}{83}\right) = \left(\frac{5}{83}\right)\left(\frac{7}{83}\right)$$
$$= \left(\frac{83}{5}\right) \cdot \left(-\left(\frac{83}{7}\right)\right) = -\left(\frac{3}{5}\right)\left(\frac{6}{7}\right) = -\left(\frac{3}{5}\right)\left(\frac{-1}{7}\right) = -(-1)(-1) = -1,$$

using $5 \equiv 1 \pmod{4}$, $7 \equiv 83 \equiv -1 \pmod{4}$, $83 = 16 \cdot 5 + 3 = 11 \cdot 7 + 6$, $\left(\frac{3}{5}\right) = -1$ and $\left(\frac{-1}{7}\right) = -1$. So $x^2 \equiv 7! \pmod{83}$ has no solution.

Finally $77 = 7 \cdot 11$, and we check the two primes separately. Modulo 7,

$$\left(\frac{30}{7}\right) = \left(\frac{2 \cdot 3 \cdot 5}{7}\right) = \left(\frac{2}{7}\right)\left(\frac{3}{7}\right)\left(\frac{5}{7}\right)$$
$$= 1 \cdot \left(-\left(\frac{7}{3}\right)\right)\left(\frac{7}{5}\right) = -\left(\frac{1}{3}\right)\left(\frac{2}{5}\right) = -1 \cdot (-1) = 1,$$

so $x^2 \equiv 30 \pmod{7}$ is solvable. Modulo 11, $\left(\frac{30}{11}\right) = \left(\frac{8}{11}\right) = \left(\frac{2}{11}\right)^3 = (-1)^3 = -1$, so $x^2 \equiv 30 \pmod{11}$ is not solvable. A solution modulo 77 would be a solution modulo 11, so $x^2 \equiv 30 \pmod{77}$ has no solution. ◊

Example 10.6. Compute $\left(\frac{1234567}{225}\right)$ and $\left(\frac{589}{1999}\right)$.

Solution. Since 1234567 has digit sum 28 and last digit 7, $(1234567, 15) = 1$. By Proposition 5.3 5),

$$\left(\frac{1234567}{225}\right) = \left(\frac{1234567}{15^2}\right) = \left(\frac{1234567}{15}\right)^2 = 1.$$

Using $589 \equiv 1 \pmod{4}$, $1999 = 3 \cdot 589 + 232$, $232 = 2^3 \cdot 29$, $589 \equiv 5 \pmod{8}$, $29 \equiv 1 \pmod{4}$ and $589 = 20 \cdot 29 + 9$,

$$\left(\frac{589}{1999}\right) = \left(\frac{1999}{589}\right) = \left(\frac{232}{589}\right) = \left(\frac{2^3 \cdot 29}{589}\right) = \left(\frac{2}{589}\right)^3 \left(\frac{29}{589}\right) = (-1)^3 \left(\frac{29}{589}\right)$$
$$= -\left(\frac{589}{29}\right) = -\left(\frac{9}{29}\right) = -\left(\frac{3}{29}\right)^2 = -1.$$

◊

Definition 10.7. An arithmetic function f is *totally multiplicative* if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$. The *von Mangoldt function* is $\Lambda(n) = \log p$ if $n = p^k$ for a prime p and $k \in \mathbb{N}$, and $\Lambda(n) = 0$ otherwise. The *Liouville function* is $\lambda(n) = (-1)^{\Omega(n)}$. For a statement P, $\mathbb{1}\{P\}$ is 1 if P holds and 0 otherwise.

Example 10.8. Show that $\sum_{d|n} \Lambda(d) = \log n$ for $n \in \mathbb{N}$.

Solution. Let $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$. Only the prime-power divisors p_i^j with $1 \le j \le \alpha_i$ contribute, so

$$\sum_{d|n} \Lambda(d) = \sum_{p^\alpha | n} \Lambda(p^\alpha) = \sum_{i=1}^{r} \sum_{j=1}^{\alpha_i} \Lambda\left(p_i^j\right) = \sum_{i=1}^{r} \sum_{j=1}^{\alpha_i} \log p_i$$
$$= \sum_{i=1}^{r} \alpha_i \log p_i = \sum_{i=1}^{r} \log\left(p_i^{\alpha_i}\right) = \log\left(p_1^{\alpha_1} \cdots p_r^{\alpha_r}\right) = \log n.$$

For $n = 1$ both sides are 0. ◊

Example 10.9. Show that $\Lambda(n) = -\sum_{d|n} \mu(d) \log d$.

Solution. By the previous example and Theorem 7.12,

$$\Lambda(n) = \sum_{d|n} \mu(d) \log\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d)(\log n - \log d)$$
$$= \log n \sum_{d|n} \mu(d) - \sum_{d|n} \mu(d) \log d$$
$$= \log n \cdot \mathbb{1}\{n = 1\} - \sum_{d|n} \mu(d) \log d = -\sum_{d|n} \mu(d) \log d,$$

using Proposition 7.11 and $\log 1 = 0$. ◊

Example 10.10. Show that $\sum_{d|n} \lambda(d) = \mathbb{1}\{n \text{ is a square}\}$.

Solution. Since $\Omega(mn) = \Omega(m) + \Omega(n)$ for all $m, n \in \mathbb{N}$, $\lambda(mn) = (-1)^{\Omega(m)+\Omega(n)} = \lambda(m)\lambda(n)$, so λ is totally multiplicative, in particular multiplicative. By Theorem 7.8 its summation function F is multiplicative. For a prime power,

$$F(p^\alpha) = \sum_{d|p^\alpha} \lambda(d) = \lambda(1) + \lambda(p) + \cdots + \lambda(p^\alpha) = 1 - 1 + (-1)^2 + (-1)^3 + \cdots + (-1)^\alpha = \mathbb{1}\{2 \mid \alpha\}.$$

Hence $F(n) = F(p_1^{\alpha_1}) \cdots F(p_r^{\alpha_r}) = \mathbb{1}\{\text{every } \alpha_i \text{ is even}\} = \mathbb{1}\{n \text{ is a square}\}$. ◊

11. Geometry of $\mathbb{Q}_p$

Lemma 11.1. Let F be a field with a non-archimedean absolute value $|\cdot|$, and let $x, y \in F$ with $|x| \neq |y|$. Then $|x+y| = \max\{|x|, |y|\}$.

Proof. Assume $|x| > |y|$. Since $|\cdot|$ is non-archimedean, $|x+y| \leq \max\{|x|, |y|\} = |x|$. Also $x = (x+y) - y$, so $|x| \leq \max\{|x+y|, |-y|\} = \max\{|x+y|, |y|\}$. Since $|x| > |y|$, the maximum is $|x+y|$, and $|x| \leq |x+y|$. Hence $|x+y| = |x|$. ∎

Definition 11.2. In a metric space (X, d), a *triangle* is a triple of distinct points a, b, c, with side lengths $d(a,b)$, $d(a,c)$, $d(b,c)$. It is *isosceles* if two of its side lengths are equal. Three distinct points are *collinear* if, after relabelling, $d(a,c) = d(a,b) + d(b,c)$, and they form a *right triangle* if, after relabelling, $d(a,c)^2 = d(a,b)^2 + d(b,c)^2$. We write $d_p(x,y) = |x-y|_p$ on $\mathbb{Q}_p$.

Theorem 11.3. Every triangle in $\mathbb{Q}_p$ is isosceles. More precisely, its two longest sides have equal length.

Proof. Let $a, b, c \in \mathbb{Q}_p$ be distinct. If $d_p(a,b) = d_p(b,c)$, the triangle is isosceles. Otherwise $|a-b|_p \neq |b-c|_p$, and since $(a-b) + (b-c) = a-c$, Lemma 11.1 gives $|a-c|_p = \max\{|a-b|_p, |b-c|_p\}$. So $|a-c|_p$ equals $|a-b|_p$ or $|b-c|_p$, namely the larger of the two. In both cases the two longest sides are equal. ∎

Theorem 11.4. No three distinct points of $\mathbb{Q}_p$ are collinear.

Proof. Suppose $a, b, c \in \mathbb{Q}_p$ are distinct and $d_p(a,c) = d_p(a,b) + d_p(b,c)$. Since $d_p(a,b), d_p(b,c) > 0$, we get $d_p(a,c) > d_p(a,b), d_p(b,c)$. So the side ac is strictly the longest, which contradicts Theorem 11.3. ∎

Theorem 11.5. There are no right triangles in $\mathbb{Q}_p$.

Proof. Suppose $a, b, c \in \mathbb{Q}_p$ form a right triangle with $d_p(a,c)^2 = d_p(a,b)^2 + d_p(b,c)^2$. Again $d_p(a,c) > d_p(a,b), d_p(b,c)$. Since the triangle is isosceles, $d_p(a,b) = d_p(b,c)$, so ac is the base, it is the longest side, and $d_p(a,c) = \sqrt{2}\, d_p(a,b)$. By Theorem 11.3, however, the base of a triangle in $\mathbb{Q}_p$ that is not equilateral is its shortest side, a contradiction. ∎

12. Sequences and series in $\mathbb{Q}_p$

Theorem 12.1. A sequence (a_n) in $\mathbb{Q}_p$ is a Cauchy sequence if and only if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ with $|a_{n+1} - a_n|_p < \varepsilon$ for all $n \geq N$.

Proof. If (a_n) is Cauchy and $\varepsilon > 0$, there is $N \in \mathbb{N}$ with $|a_n - a_m|_p < \varepsilon$ for all $n, m \geq N$. For $n \geq N$ also $n+1 \geq N$, so $|a_{n+1} - a_n|_p < \varepsilon$.

Conversely, let $\varepsilon > 0$ and choose N with $|a_{n+1} - a_n|_p < \varepsilon$ for all $n \geq N$. Let $m, n \geq N$. If $n = m$, then $|a_n - a_m|_p = 0 < \varepsilon$. Otherwise assume $n > m$ and write $n = m + l$ with $l \in \mathbb{N}$. Then

$$|a_n - a_m|_p = |(a_{m+l} - a_{m+l-1}) + (a_{m+l-1} - a_{m+l-2}) + \cdots + (a_{m+1} - a_m)|_p.$$

Since the p-adic absolute value is non-archimedean,

$$|a_n - a_m|_p \le \max\{|a_{m+i} - a_{m+i-1}|_p : 1 \le i \le l\}.$$

The maximum is attained at some $1 \le i_0 \le l$, and $m + i_0 - 1 \ge N$, so $|a_n - a_m|_p \le |a_{m+i_0} - a_{m+i_0-1}|_p < \varepsilon$. ∎

Corollary 12.2. Let $(a_n)_{n\ge 0}$ be a sequence in $\mathbb{Q}_p$. Then $\sum_{n=0}^{\infty} a_n$ converges if and only if $\lim_{n\to\infty} a_n = 0$.

Proof. Let $S_n = \sum_{i=0}^{n} a_i$ be the partial sums. If the series converges, then (S_n) converges and hence is Cauchy. Let $\varepsilon > 0$; there is $N \in \mathbb{N}$ with $|S_{n+1} - S_n|_p < \varepsilon$ for all $n \ge N$. Since $|S_{n+1} - S_n|_p = |a_{n+1}|_p$, we get $|a_n|_p < \varepsilon$ for all $n > N$.

Conversely, let $\lim_{n\to\infty} a_n = 0$ and $\varepsilon > 0$. There is $N \in \mathbb{N}$ with $|a_n|_p < \varepsilon$ for all $n > N$, hence $|S_{n+1} - S_n|_p = |a_{n+1}|_p < \varepsilon$ for all $n \ge N$. By Theorem 12.1, (S_n) is Cauchy, and it converges because $\mathbb{Q}_p$ is complete (Theorem 9.12). ∎

Proposition 12.3. Every series $\sum_{n=n_0}^{\infty} a_n p^n$ with $n_0 \in \mathbb{Z}$ and $a_n \in \{0, 1, \ldots, p-1\}$ converges in $\mathbb{Q}_p$, and its sum is the p-adic number with digits a_n.

Proof. $|a_n p^n|_p \le p^{-n} \to 0$, so the series converges by Corollary 12.2. Its partial sums are the rational numbers s_n of part 3) of the proof of Theorem 9.12, which converge to the p-adic number with digits a_n. ∎

13. Topology on $\mathbb{Q}_p$

The p-adic metric induces a topology on $\mathbb{Q}_p$. For $a \in \mathbb{Q}_p$ and $r > 0$ write $B(a, r) = \{x \in \mathbb{Q}_p : |x - a|_p < r\}$ for the open ball and $\overline{B}(a, r) = \{x \in \mathbb{Q}_p : |x - a|_p \le r\}$ for the closed ball. As in every metric space, open balls are open and closed balls are closed. The values of $|\cdot|_p$ on $\mathbb{Q}_p$ are 0 and the powers p^k, $k \in \mathbb{Z}$.

Remark 13.1 (cited)**.** $\mathbb{Q}_p$ admits no ordering compatible with its field operations.

Theorem 13.2. In $\mathbb{Q}_p$ every open ball is closed, and every closed ball of positive radius is open.

Proof. Let $B(a, r)$ be an open ball and let n be the largest integer with $r \le p^{-n}$, so $p^{-(n+1)} < r \le p^{-n}$. No p-adic number has absolute value strictly between $p^{-(n+1)}$ and p^{-n}, hence

$$B(a, r) = \{x \in \mathbb{Q}_p : |x - a|_p < p^{-n}\} = \{x \in \mathbb{Q}_p : |x - a|_p \le p^{-(n+1)}\} = \overline{B}(a, p^{-(n+1)}),$$

which is closed.

Let $\overline{B}(a, r)$ be a closed ball with $r > 0$ and $x \in \overline{B}(a, r)$. For $y \in B(x, r)$, $|y - a|_p \le \max\{|y - x|_p, |x - a|_p\} \le r$, so $B(x, r) \subseteq \overline{B}(a, r)$. Hence $\overline{B}(a, r)$ is open. ∎

Corollary 13.3. $\mathbb{Z}_p$ is both closed and open in $\mathbb{Q}_p$.

Proof. By Theorem 9.12, $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\} = \overline{B}(0,1)$ is a closed ball, hence closed. By the theorem, it is open as well. ∎

Corollary 13.4. $\mathbb{Q}_p$ is totally disconnected, that is, its only connected subsets are the empty set and the singletons.

Proof. Let $A \subseteq \mathbb{Q}_p$ contain distinct points a, b and put $r = |a-b|_p$. The ball $U = B(a,r)$ is open and closed, contains a and not b. Hence $A = (A \cap U) \cup (A \setminus U)$ is a decomposition into two disjoint nonempty sets that are open in A, and A is not connected. ∎

Theorem 13.5. $\mathbb{Z}_p$ is compact.

Proof. $\mathbb{Z}_p$ is a metric space, so it suffices to show that every sequence (x_n) in $\mathbb{Z}_p$ has a convergent subsequence. Infinitely many x_n share the same first digit; among these, infinitely many share the same second digit; continuing, we obtain infinite index sets $I_1 \supseteq I_2 \supseteq \cdots$ such that all x_n with $n \in I_k$ have the same truncation $(x_n)_k$. Choose $n_1 < n_2 < \cdots$ with $n_k \in I_k$. For $j, l \geq k$ the truncations $\left(x_{n_j}\right)_k$ and $\left(x_{n_l}\right)_k$ agree, so $|x_{n_j} - x_{n_l}|_p \leq p^{-k}$, and $\left(x_{n_k}\right)$ is Cauchy. It converges in $\mathbb{Q}_p$ by Theorem 9.12, and its limit lies in $\mathbb{Z}_p$, which is closed. ∎

14. Finite fields

Definition 14.1 (Finite field). A *finite field* is a field F with $|F| < \infty$. The *characteristic* $\operatorname{char} F$ is the least $n \in \mathbb{N}$ with $n \cdot 1 = 0$ in F. A finite field with q elements is denoted $\mathbb{F}_q$, and $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ for a prime p.

Theorem 14.2. Let F be a finite field. Then $\operatorname{char} F$ is a prime p, and $|F| = p^n$ for some $n \in \mathbb{N}$. The multiplicative group $F^* = F^\times = F \setminus \{0\}$ is cyclic, $F^\times = \langle g \rangle$; a generator g is called a *primitive element* of F. Consequently $a^q = a$ for every $a \in \mathbb{F}_q$.

Proof. Since F is finite, the elements $n \cdot 1$ are not pairwise distinct, so $\operatorname{char} F$ exists. If $\operatorname{char} F = ab$ with $1 < a, b$, then $(a \cdot 1)(b \cdot 1) = 0$ with both factors nonzero, which is impossible in a field; so $\operatorname{char} F = p$ is prime. Then $\{n \cdot 1 : n \in \mathbb{Z}\}$ is a subfield isomorphic to $\mathbb{F}_p$, and F is a finite-dimensional vector space over it, of dimension n say, so $|F| = p^n$.

Let $|F| = q$. The proof of Theorem 1.8 in the case $m = p$ applies to $F^\times$, which has order $q-1$: every element has an order dividing $q-1$, a polynomial $x^d - 1$ over the field F has at most d roots, and $\sum_{d \mid q-1} \varphi(d) = q-1$ by Theorem 7.9. Hence $F^\times$ has $\varphi(q-1) \geq 1$ elements of order $q-1$, and it is cyclic. Finally $a^{q-1} = 1$ for $a \neq 0$, so $a^q = a$ for every $a \in F$. ∎

Lemma 14.3. Let p be a prime. Then $p \mid \binom{p}{j}$ for all $0 < j < p$.

Proof. We have $\binom{p}{j} = p!/((p-j)!j!)$, and

$$\frac{p}{j}\binom{p-1}{j-1} = \frac{p}{j}\cdot\frac{(p-1)!}{(j-1)!(p-j)!} = \frac{p!}{(p-j)!j!} = \binom{p}{j}.$$

Hence $j\binom{p}{j} = p\binom{p-1}{j-1}$, so $p \mid j\binom{p}{j}$. Since $(p,j)=1$, $p \mid \binom{p}{j}$. ∎

Theorem 14.4. Let $\mathbb{F}_q$ be a finite field with $\operatorname{char} \mathbb{F}_q = p$. Then $\varphi : \mathbb{F}_q \to \mathbb{F}_q$, $\varphi(a) = a^p$, is a ring homomorphism.

Proof. Let $a, b \in \mathbb{F}_q$; $\mathbb{F}_q$ is commutative. Then $\varphi(ab) = (ab)^p = a^p b^p = \varphi(a)\varphi(b)$, and by the binomial theorem and the lemma,

$$\varphi(a+b) = (a+b)^p = a^p + \sum_{j=1}^{p-1}\binom{p}{j}a^j b^{p-j} + b^p = a^p + b^p = \varphi(a) + \varphi(b).$$

From $\varphi(a-b) + \varphi(b) = \varphi(a)$ we get $\varphi(a-b) = a^p - b^p = \varphi(a) - \varphi(b)$. ∎

Remark 14.5 (Recall). Let $f \in \mathbb{F}_q[x]$ be irreducible of degree m. Then $\mathbb{F}_q[x]/(f)$ is a field containing a copy of $\mathbb{F}_q$, with q^m elements, in which the class of x is a root of f. If $K \supseteq \mathbb{F}_q$ is a field and $\alpha \in K$ is a root of f, then $\mathbb{F}_q(\alpha) = \{g(\alpha) : g \in \mathbb{F}_q[x], \deg g < m\}$ has exactly q^m elements. We write $\mathbb{F}_{q^m}$ for a field with q^m elements that contains $\mathbb{F}_q$.

Lemma 14.6. Let $f \in \mathbb{F}_q[x]$ be irreducible with $\deg f = m$. Then f has a root α in some $\mathbb{F}_{q^m}$, and for every such root the roots of f in $\mathbb{F}_{q^m}$ are exactly the m pairwise distinct elements $\alpha, \alpha^q, \alpha^{q^2}, \ldots, \alpha^{q^{m-1}}$.

Proof. A root exists in $\mathbb{F}_q[x]/(f)$ by Remark 14.5. Let $\beta \in \mathbb{F}_{q^m}$ be a root of f. We claim $f(\beta^q) = 0$. Let $f(x) = \sum_{i=0}^{m} a_i x^i$. Since $a_i \in \mathbb{F}_q$, $a_i^q = a_i$ by Theorem 14.2, and $x \mapsto x^q$ is a ring homomorphism, being an iterate of the map in Theorem 14.4 because q is a power of p. Hence

$$\begin{aligned} f(\beta^q) &= a_0 + a_1\beta^q + a_2(\beta^2)^q + \cdots + a_m(\beta^m)^q \\ &= a_0^q + (a_1\beta)^q + \cdots + (a_m\beta^m)^q \\ &= (a_0 + a_1\beta + \cdots + a_m\beta^m)^q = f(\beta)^q = 0. \end{aligned}$$

It follows that $\alpha, \alpha^q, (\alpha^q)^q, \ldots, \alpha^{q^{m-1}}$ are roots of f.

Suppose $\alpha^{q^i} = \alpha^{q^j}$ with $0 \le i < j \le m-1$. Raising to the power q^{m-j} gives $\alpha^{q^{m-j+i}} = \alpha^{q^m} = \alpha$, by Theorem 14.2 applied to $\mathbb{F}_{q^m}$. So α is a root of $x^{q^k} - x$ with $k = m - j + i$ and $1 \le k < m$. The set $L = \{\beta \in \mathbb{F}_{q^m} : \beta^{q^k} = \beta\}$ is a subfield, because $\beta \mapsto \beta^{q^k}$ is a ring homomorphism and $(\beta^{-1})^{q^k} = \left(\beta^{q^k}\right)^{-1}$. It contains $\mathbb{F}_q$ and α, hence $\mathbb{F}_q(\alpha)$, which has q^m elements by Remark 14.5. But L consists of roots of $x^{q^k} - x$, so $|L| \le q^k < q^m$, a contradiction, since $f(\alpha) = 0$, $\deg f = m$ and f is irreducible. Hence the m elements are pairwise distinct. A polynomial of degree m has at most m roots in a field, so these are all the roots of f in $\mathbb{F}_{q^m}$, and f splits completely over $\mathbb{F}_{q^m}$. ∎

Definition 14.7 (Conjugates). Let $\mathbb{F}_q \subseteq \mathbb{F}_{q^m}$ and $\alpha \in \mathbb{F}_{q^m}$. The elements $\alpha, \alpha^q, \alpha^{q^2}, \ldots, \alpha^{q^{m-1}}$ are called the *conjugates* of α with respect to $\mathbb{F}_q$.

Example 14.8. For fields $K \subseteq L$, the *degree* $[L : K]$ is the dimension of L as a vector space over K, and the *minimal polynomial* m_α of $\alpha \in L$ is the monic polynomial of least degree over K with root α. For $\mathbb{R} \subset \mathbb{C} = \mathbb{R}(i)$ we have $[\mathbb{C} : \mathbb{R}] = 2$, $m_i(x) = x^2 + 1$ and $\mathbb{C} \cong \mathbb{R}[x]/(x^2 + 1)$. The roots of $x^2 + 1$ are i and $-i$. Complex conjugation $\overline{a + bi} = a - bi$ satisfies $\bar{i} = -i$, so it interchanges the roots of the irreducible polynomial $x^2 + 1$.

Definition 14.9 (Automorphism over $\mathbb{F}_q$). An *automorphism of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$* is an isomorphism of $\mathbb{F}_{q^m}$ onto itself that leaves every element of $\mathbb{F}_q$ invariant.

Remark 14.10. Such a map is also called an *$\mathbb{F}_q$-automorphism*.

Example 14.11. For $\mathbb{R} \subset \mathbb{C}$, conjugation $z \mapsto \overline{z}$ fixes the real numbers, so it is an $\mathbb{R}$-automorphism of $\mathbb{C}$.

15. Worked exercises on p-adic numbers

Lemma 15.1. Let $x \in \mathbb{Q}_p$ with $|x|_p < 1$. Then $\sum_{n=0}^{\infty} x^n$ converges in $\mathbb{Q}_p$ and its sum is $1/(1 - x)$.

Proof. By Lemma 11.1, $|1 - x|_p = \max\{1, |x|_p\} = 1$, so $1 - x \neq 0$. The partial sums are $S_N = \sum_{n=0}^{N} x^n = (1 - x^{N+1})/(1 - x)$, hence

$$|S_N - \frac{1}{1-x}|_p = \frac{|x|_p^{N+1}}{|1-x|_p} = |x|_p^{N+1} \to 0.$$

■

Remark 15.2. For $x \in \mathbb{Q}_p$ we write $v_p(x) = v(x)$, with v as in Theorem 9.12. By part 1) of that theorem this extends the p-adic valuation of $\mathbb{Q}$, and $|x|_p = p^{-v_p(x)}$ for all $x \neq 0$.

Example 15.3. Find the p-adic expansions of $2/3$ in $\mathbb{Q}_2$, of $-1/6$ in $\mathbb{Q}_7$ and of $1/6$ in $\mathbb{Q}_3$.

Solution. **$2/3$ in $\mathbb{Q}_2$.** Since $|-2|_2 = 1/2 < 1$, Lemma 15.1 gives $\frac{1}{3} = \frac{1}{1-(-2)} = \sum_{n=0}^{\infty} (-2)^n$. This is a convergent series but not an expansion, since the digits must be 0 and 1. Similarly

$$-1 = \frac{1}{1-2} = \sum_{n=0}^{\infty} 2^n, \quad -\frac{1}{3} = \frac{1}{1-4} = \sum_{n=0}^{\infty} 4^n = 1 + 2^2 + 2^4 + \cdots,$$

and the latter is an expansion. Hence

$$\frac{2}{3} = -2 \cdot \left(-\frac{1}{3}\right) = -2 \sum_{n=0}^{\infty} 4^n = -(2 + 2^3 + 2^5 + \cdots).$$

By Example 15.5 with $p = 2$, lowest digit $a_1 = 1$ and $a_n = 1$ exactly for odd n, the negation has digits $2 - a_1 = 1$ at 2^1 and $1 - a_n$ at 2^n for $n \geq 2$:

$$\frac{2}{3} = 2 + 2^2 + 2^4 + 2^6 + \cdots.$$

Alternatively, $2/3 \in \mathbb{Z}_2$ by Remark 8.8, and its truncations x_k satisfy $3x_k \equiv 2 \pmod{2^k}$, so they are found by solving congruences. If $3x \equiv 2 \pmod 2$, then $x \equiv 0 \pmod 2$ and $x = 2y$. Then $3(2y) \equiv 2 \pmod 4$ gives $2y \equiv 2 \pmod 4$, so $y \equiv 1 \pmod 2$, $y = 2z + 1$ and $x = 4z + 2$. Then $3(4z + 2) \equiv 2 \pmod 8$ gives $4z + 6 \equiv 2 \pmod 8$, that is, $2z + 3 \equiv 1 \pmod 4$, so $z \equiv 1 \pmod 2$, $z = 2w + 1$ and

$$x = 4(2w + 1) + 2 = 8w + 6.$$

So $x_3 = 6 = 0 + 1 \cdot 2 + 1 \cdot 2^2$, in agreement with the first three digits above.

$-1/6$ in $\mathbb{Q}_7$. By Lemma 15.1, $-\frac{1}{6} = \frac{1}{1-7} = \sum_{n=0}^{\infty} 7^n = 1 + 7 + 7^2 + \cdots$, which is an expansion. Alternatively, the truncations satisfy $6x_k \equiv -1 \pmod{7^k}$. Modulo 7, $6x \equiv -1$ gives $x \equiv 1 \pmod 7$, so $x = 7y + 1$. Then $6(7y + 1) \equiv -1 \pmod{49}$ gives $42y + 6 \equiv -1 \pmod{49}$, that is, $6y + 1 \equiv 0 \pmod 7$, so $y \equiv 1 \pmod 7$ and $x_2 = 1 + 1 \cdot 7$.

$1/6$ in $\mathbb{Q}_3$. We have $\frac{1}{6} = \frac{1}{2} \cdot \frac{1}{3}$. The formula $\frac{1}{2} = \frac{1}{1-(-1)}$ does not help: $|-1|_3 = 1$, so the terms of $\sum (-1)^n$ do not tend to 0 and the series diverges by Corollary 12.2. Instead,

$$\frac{1}{2} = \frac{2}{4} = \frac{2}{1-(-3)} = 2\sum_{n=0}^{\infty} (-3)^n.$$

The digits must be $0, 1, 2$, and the terms with odd n have coefficient -2. Using $-2 \cdot 3^{2k+1} = (1 - 3) \cdot 3^{2k+1} = 3^{2k+1} - 3^{2k+2}$,

$$2(-3)^{2k+1} + 2(-3)^{2k+2} = 3^{2k+1} - 3^{2k+2} + 2 \cdot 3^{2k+2} = 3^{2k+1} + 3^{2k+2}.$$

Grouping the terms in consecutive pairs after the first replaces the sequence of partial sums by a subsequence, which has the same limit. Hence

$$\frac{1}{2} = 2 + \sum_{k=0}^{\infty} (3^{2k+1} + 3^{2k+2}) = 2 + 3 + 3^2 + 3^3 + \cdots,$$

and multiplying by 3^{-1} shifts the digits:

$$\frac{1}{6} = 2 \cdot 3^{-1} + 1 + 3 + 3^2 + \cdots.$$

◊

Example 15.4. Let $a = a_0 + a_1 p + a_2 p^2 + \cdots \in \mathbb{Z}_p$. Show that a is a unit of $\mathbb{Z}_p$ if and only if $a_0 \neq 0$, and that this holds if and only if $|a|_p = 1$.

Solution. The first equivalence is part of Proposition 8.10. For $a \in \mathbb{Z}_p$ we have $v_p(a) \geq 0$, and $v_p(a) = 0$ holds if and only if $a_0 \neq 0$. Hence $a_0 \neq 0$ if and only if $|a|_p = p^0 = 1$. ◊

Example 15.5. Let $0 \neq a \in \mathbb{Q}_p$ with expansion $a = \sum_{n=-m}^{\infty} a_n p^n$, where $m \in \mathbb{Z}$ and $a_{-m} \neq 0$. Find the expansion of $-a$.

Solution. Write $-a = \sum_{n=-m}^{\infty} b_n p^n$; its lowest index is at least $-m$ because $v_p(-a) = v_p(a)$. Then $a + (-a) = 0$. Multiplying by p^m,

$$\alpha = p^m a = a_{-m} + a_{-m+1}p + \cdots + a_0 p^m + a_1 p^{m+1} + \cdots, \quad \beta = p^m(-a) = b_{-m} + b_{-m+1}p + \cdots,$$

with $\alpha, \beta \in \mathbb{Z}_p$ and $\alpha + \beta = 0$. So the truncations satisfy $\alpha_k + \beta_k \equiv 0 \pmod{p^k}$ for every $k \in \mathbb{N}$.

For $k = 1$: $a_{-m} + b_{-m} \equiv 0 \pmod{p}$, and $1 \le a_{-m} + b_{-m} \le 2p - 1$, so $a_{-m} + b_{-m} = p$ and $b_{-m} = p - a_{-m}$. Hence $\alpha_1 + \beta_1 = p$.

Continuing the process, suppose $\alpha_k + \beta_k = p^k$. Then

$$\alpha_{k+1} + \beta_{k+1} = p^k + (a_{k-m} + b_{k-m})p^k \equiv 0 \pmod{p^{k+1}},$$

so $1 + a_{k-m} + b_{k-m} \equiv 0 \pmod{p}$. Since $1 \le 1 + a_{k-m} + b_{k-m} \le 2p - 1$, this gives $1 + a_{k-m} + b_{k-m} = p$, that is, $b_{k-m} = p - 1 - a_{k-m}$, and $\alpha_{k+1} + \beta_{k+1} = p^{k+1}$. By induction,

$$-a = (p - a_{-m})p^{-m} + (p - 1 - a_{-m+1})p^{-m+1} + \cdots + (p - 1 - a_0) + (p - 1 - a_1)p + \cdots.$$

◊

Example 15.6 (Legendre's formula). Let $n \in \mathbb{N}$ and let p be a prime. Show that

$$v_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor,$$

where the sum is finite because its terms vanish for $p^i > n$. Consequently $n! = \prod_{p \le n} p^{\alpha_p}$ with $\alpha_p = \sum_{i \ge 1} \lfloor n/p^i \rfloor$.

Solution. Since $n! = 1 \cdot 2 \cdot \cdots \cdot n$, Proposition 9.5 gives $v_p(n!) = \sum_{j=1}^{n} v_p(j)$, and $v_p(j)$ is the number of $i \ge 1$ with $p^i \mid j$. Exchanging the order of summation,

$$v_p(n!) = \sum_{j=1}^{n} \sum_{i \ge 1,\ p^i \mid j} 1 = \sum_{i \ge 1} \#\{1 \le j \le n : p^i \mid j\}.$$

The factors divisible by p are $p, 2p, \ldots, tp$, where $tp \le n < (t+1)p$, so there are $t = \lfloor n/p \rfloor$ of them. In the same way, exactly $\lfloor n/p^2 \rfloor$ factors are divisible by p^2, and $\lfloor n/p^i \rfloor$ by p^i. The prime divisors of $n!$ are the primes $p \le n$, which gives the factorization. ◊

Example 15.7. Show that $x^2 = 6$ has a solution in $\mathbb{Q}_5$.

Solution. $x^2 \equiv 6 \pmod{5}$ has the solution $x \equiv 1 \pmod{5}$. We prove by induction that there are integers x_n, $n \ge 1$, with $x_n^2 \equiv 6 \pmod{5^n}$, $x_1 = 1$ and $x_{n+1} \equiv x_n \pmod{5^n}$. Given x_n with $x_n \equiv 1 \pmod{5}$, write $x_n^2 = 5^n s + 6$ and $x_{n+1} = x_n + t5^n$ with $t \in \mathbb{Z}$. Then

$$x_{n+1}^2 = x_n^2 + 2x_n t 5^n + t^2 5^{2n} \equiv 6 \pmod{5^{n+1}}$$

holds if and only if

$$2x_n t + \frac{x_n^2 - 6}{5^n} \equiv 0 \pmod{5}.$$

Since $(2x_n, 5) = 1$, this linear congruence has a solution t (Remark 3.2), which is the lifting step of Theorem 6.2. Then $x_{n+1} \equiv x_n \equiv 1 \pmod 5$.

Since $|x_{n+1} - x_n|_5 \le 5^{-n}$, the sequence (x_n) is Cauchy by Theorem 12.1 and converges to some $x \in \mathbb{Q}_5$ by Theorem 9.12; $x \in \mathbb{Z}_5$ since $\mathbb{Z}_5$ is closed. Finally $|x + x_n|_5 \le 1$, so

$$|x^2 - 6|_5 \le \max\{|x - x_n|_5 \ |x + x_n|_5, |x_n^2 - 6|_5\} \le \max\{|x - x_n|_5, 5^{-n}\} \to 0,$$

and $x^2 = 6$. ◊

Example 15.8. Consider $\mathbb{Q}$ with the 5-adic metric and the triangle with vertices $A = 2/15$, $B = 1/5$, $C = 7/15$. Compute its side lengths.

Solution. Recall that $v_p(a/b) = v_p(a) - v_p(b)$ and $|x|_p = p^{-v_p(x)}$ for $x \neq 0$. Here $v_5(-1) = 0$ and $v_5(15) = 1$. Hence

$$|A - B|_5 = |\frac{2}{15} - \frac{3}{15}|_5 = |-\frac{1}{15}|_5 = \frac{1}{5^{-1}} = 5,$$
$$|B - C|_5 = |-\frac{4}{15}|_5 = 5, \quad |C - A|_5 = |\frac{5}{15}|_5 = |\frac{1}{3}|_5 = 1.$$

In agreement with Theorem 11.3, the two longest sides are equal. ◊

Example 15.9. Characterize the units of $\mathbb{Z}_p$ in terms of v_p. Prove that $0 \neq x \in \mathbb{Q}_p$ is a square in $\mathbb{Q}_p$ if and only if $x = p^{2n}y^2$ with $n \in \mathbb{Z}$ and y a unit of $\mathbb{Z}_p$.

Solution. By Theorem 9.12 2), $x \in \mathbb{Q}_p$ lies in $\mathbb{Z}_p$ if and only if $v_p(x) \ge 0$. By Example 15.4, an element of $\mathbb{Z}_p$ is a unit if and only if $v_p = 0$. Hence $\mathbb{Z}_p^\times = \{x \in \mathbb{Q}_p : v_p(x) = 0\}$.

If $x = p^{2n}y^2$, then $p^{2n} = (p^n)^2$ and $x = (p^n y)^2$ with $p^n y \in \mathbb{Q}_p$. Conversely, let $x = z^2$ with $z \in \mathbb{Q}_p$; then $z \neq 0$. By Proposition 8.10, $z = p^n y$ with $n = v_p(z) \in \mathbb{Z}$ and $y \in \mathbb{Z}_p^\times$. Hence $x = z^2 = (p^n y)^2 = p^{2n}y^2$, and $y^2 = y \cdot y \in \mathbb{Z}_p^\times$. ◊

Example 15.10. Prove that $(x^2 - 2)(x^2 - 17)(x^2 - 34) = 0$ has a root in $\mathbb{Q}_p$ for every $p \le \infty$, but no root in $\mathbb{Q}$.

Solution. None of $2, 17, 34$ is the square of a rational number, since $v_q(z^2) = 2v_q(z)$ is even for every prime q and every $0 \neq z \in \mathbb{Q}$, whereas $v_2(2) = v_{17}(17) = v_2(34) = 1$. So there is no rational root.

$p = \infty$. $\mathbb{Q}_\infty = \mathbb{R}$ and $\sqrt{2} \in \mathbb{R}$ is a root.

p odd, $p \neq 17$. Then $p \nmid 2 \cdot 17 \cdot 34$. If $\left(\frac{2}{p}\right) = 1$ or $\left(\frac{17}{p}\right) = 1$, one of $2, 17$ is a quadratic residue modulo p. Otherwise $\left(\frac{2}{p}\right) = \left(\frac{17}{p}\right) = -1$, and by Proposition 4.3 2)

$$\left(\frac{34}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{17}{p}\right) = (-1)(-1) = 1,$$

so there is x with $x^2 \equiv 34 \pmod p$. In each case some $a \in \{2, 17, 34\}$ and $c \in \mathbb{Z}$ satisfy $c^2 \equiv a \pmod p$, and $p \nmid c$. For $f(x) = x^2 - a$ we have $f'(c) = 2c \not\equiv 0 \pmod p$, so as in the solution of Example 8.12, Theorem 6.2 yields $\alpha \in \mathbb{Z}_p$ with $\alpha^2 = a$.

$p = 17$. $6^2 = 36 \equiv 2 \pmod{17}$ and $f'(6) = 12 \not\equiv 0 \pmod{17}$ for $f(x) = x^2 - 2$, so 2 is a square in $\mathbb{Q}_{17}$ in the same way.

$p = 2$. We show that 17 is a square in $\mathbb{Q}_2$; Theorem 6.2 does not apply, since $f'(x) = 2x \equiv 0 \pmod 2$. We construct odd integers x_k, $k \geq 3$, with $x_k^2 \equiv 17 \pmod{2^k}$ and $x_{k+1} \equiv x_k \pmod{2^{k-1}}$. Take $x_3 = 1$. Given x_k, write $x_k^2 - 17 = 2^k s$ and put $x_{k+1} = x_k + t2^{k-1}$ with $t \in \{0, 1\}$. Then

$$x_{k+1}^2 = x_k^2 + tx_k 2^k + t^2 2^{2k-2} \equiv 17 + 2^k(s + tx_k) \pmod{2^{k+1}},$$

since $2k - 2 \geq k + 1$ for $k \geq 3$. As x_k is odd, $t \equiv s \pmod 2$ makes $s + tx_k$ even, and $x_{k+1}^2 \equiv 17 \pmod{2^{k+1}}$. Now $|x_{k+1} - x_k|_2 \leq 2^{-(k-1)}$, so (x_k) converges to some $x \in \mathbb{Z}_2$, and the argument at the end of the solution of Example 15.7 gives $x^2 = 17$. ◊

Example 15.11. Let $f \in \mathbb{Z}[x]$. Is it true that f is irreducible in $\mathbb{Q}[x]$ if and only if f is irreducible in $\mathbb{Q}_p[x]$ for every prime p?

Solution. One implication holds. If f is reducible over $\mathbb{Q}$, that is, $f = gh$ with $g, h \in \mathbb{Q}[x]$ of positive degree, then this is a factorization in $\mathbb{Q}_p[x]$, so f is reducible over $\mathbb{Q}_p$. Hence irreducibility over every $\mathbb{Q}_p$ implies irreducibility over $\mathbb{Q}$.

The other implication fails. $x^2 + 1$ is irreducible over $\mathbb{Q}$, since -1 is not a rational square. It is reducible over $\mathbb{Q}_5$: $2^2 \equiv -1 \pmod 5$ and $f'(2) = 4 \not\equiv 0 \pmod 5$ for $f(x) = x^2 + 1$, so Theorem 6.2 gives $\iota \in \mathbb{Z}_5$ with $\iota^2 = -1$, and $x^2 + 1 = (x - \iota)(x + \iota)$.

A stronger example: $x^4 + 1$ is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{Q}_p$ for every odd prime p.

Irreducibility over $\mathbb{Q}$. It has no rational root, since $x^4 \geq 0 > -1$ for real x. In $\mathbb{R}[x]$,

$$x^4 + 1 = (x^2 + 1)^2 - 2x^2 = \left(x^2 - \sqrt{2}x + 1\right)\left(x^2 + \sqrt{2}x + 1\right),$$

and both factors are irreducible over $\mathbb{R}$ (discriminant $2 - 4 < 0$). By unique factorization in $\mathbb{R}[x]$, a monic quadratic factor of $x^4 + 1$ over $\mathbb{Q}$ would equal one of them, but neither has rational coefficients.

Reducibility over $\mathbb{Q}_p$, p odd. Recall from Proposition 4.3 that

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod 4 \\ -1 & \text{if } p \equiv 3 \pmod 4 \end{cases}, \quad \left(\frac{2}{p}\right) = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod 8 \\ -1 & \text{if } p \equiv \pm 3 \pmod 8 \end{cases}.$$

Since $\left(\frac{-1}{p}\right)\left(\frac{2}{p}\right) = \left(\frac{-2}{p}\right)$, at least one of $-1, 2, -2$ is a quadratic residue modulo p; in particular $\left(\frac{-2}{p}\right) = 1$ if and only if $p \equiv 1, 3 \pmod 8$. As in the previous example, Theorem 6.2 lifts a square root modulo p of -1, 2 or -2 to $\mathbb{Q}_p$, since p is odd.

- If $\left(\frac{2}{p}\right) = 1$, there is $a \in \mathbb{Z}_p$ with $a^2 = 2$, and $x^4 + 1 = (x^2 + 1)^2 - 2x^2 = (x^2 + 1 - ax)(x^2 + 1 + ax)$.
- If $\left(\frac{-2}{p}\right) = 1$, there is $b \in \mathbb{Z}_p$ with $b^2 = -2$, and

$$x^4+1=(x^2-1)^2+2x^2=(x^2-1)^2-(-2)x^2=(x^2-1-bx)(x^2-1+bx).$$

- If $\left(\frac{-1}{p}\right)=1$, that is, $p\equiv 1 \pmod 4$, there is $c\in\mathbb{Z}_p$ with $c^2=-1$, and $x^4+1=x^4-(-1)=(x^2-c)(x^2+c)$.

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Remark 15.12 (cited). The prime 2 is a genuine exception: x^4+1 is irreducible over $\mathbb{Q}_2$, although $x^4+1\equiv(x+1)^4 \pmod 2$. Likewise x^2+1 is irreducible over $\mathbb{Q}_2$, although $x^2+1\equiv(x+1)^2 \pmod 2$.

16. Trace and norm

Throughout, q is a prime power, $m\in\mathbb{N}$, and $\mathbb{F}_q\subseteq\mathbb{F}_{q^m}$ as in Remark 14.5.

Definition 16.1 (Trace). Let $\alpha\in\mathbb{F}_{q^m}$. The *relative trace* of α over $\mathbb{F}_q$ is

$$\operatorname{tr}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)=\alpha+\alpha^q+\cdots+\alpha^{q^{m-1}}.$$

If $\mathbb{F}_q=\mathbb{F}_p$ with p prime, then $\operatorname{tr}_{\mathbb{F}_{p^m}\mid\mathbb{F}_p}$ is called the *trace*.

Definition 16.2 (Norm). Let $L=\mathbb{F}_{q^m}$, $K=\mathbb{F}_q$ and $\alpha\in L$. The *relative norm* of α over K is

$$\mathrm{N}_{L\mid K}(\alpha)=\alpha\cdot\alpha^q\cdot\cdots\cdot\alpha^{q^{m-1}}.$$

Lemma 16.3. Let $\alpha\in\mathbb{F}_{q^m}$. Then $\operatorname{tr}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)\in\mathbb{F}_q$ and $\mathrm{N}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)\in\mathbb{F}_q$. More precisely, if $m_\alpha(x)=\sum_{i=0}^{r}a_ix^i$ is the minimal polynomial of α over $\mathbb{F}_q$, then $r\mid m$ and

$$\operatorname{tr}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)=-\frac{m}{r}\cdot a_{r-1},\quad \mathrm{N}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)=(-1)^m a_0^{m/r}.$$

Proof. α is a root of $x^{q^m}-x$ by Theorem 14.2, so it has a minimal polynomial $m_\alpha\in\mathbb{F}_q[x]$, which is monic and irreducible, of degree $r=[\mathbb{F}_q(\alpha):\mathbb{F}_q]$; so $a_r=1$. By Remark 14.5, $\mathbb{F}_q(\alpha)$ is a field with q^r elements containing $\mathbb{F}_q$. Since $\mathbb{F}_{q^m}$ is a vector space over $\mathbb{F}_q(\alpha)$ of some dimension s, we get $q^m=(q^r)^s$, so $r\mid m$ and $m=rs$.

By Lemma 14.6 applied to m_α and the field $\mathbb{F}_q(\alpha)$, $m_\alpha(x)=\prod_{i=0}^{r-1}\left(x-\alpha^{q^i}\right)$. By Theorem 14.2 applied to $\mathbb{F}_q(\alpha)$, $\alpha^{q^r}=\alpha$, hence $\alpha^{q^{jr}}=\alpha$ and $\alpha^{q^{i+jr}}=\left(\alpha^{q^{jr}}\right)^{q^i}=\alpha^{q^i}$ for all $i,j\geq 0$. Splitting $0\leq i\leq m-1$ into m/r blocks of length r,

$$\prod_{i=0}^{m-1}\left(x-\alpha^{q^i}\right)=\prod_{i=0}^{r-1}\left(x-\alpha^{q^i}\right)\cdot\prod_{i=0}^{r-1}\left(x-\alpha^{q^{i+r}}\right)\cdot\cdots\cdot\prod_{i=0}^{r-1}\left(x-\alpha^{q^{i+r(m/r-1)}}\right)=m_\alpha(x)^{m/r}.$$

So the coefficients of the left side lie in $\mathbb{F}_q$. Its coefficient of x^{m-1} is $-\left(\alpha+\alpha^q+\cdots+\alpha^{q^{m-1}}\right)=-\operatorname{tr}(\alpha)$, and its constant term is $(-1)^m\alpha\cdot\alpha^q\cdot\cdots\cdot\alpha^{q^{m-1}}=(-1)^m\,\mathrm{N}(\alpha)$. On the right, the coefficient of x^{m-1} in $\left(x^r+a_{r-1}x^{r-1}+\cdots+a_0\right)^{m/r}$ is $(m/r)a_{r-1}$, and the constant term is $a_0^{m/r}$. Comparing coefficients gives both formulas. ■

17. Field extensions

17.1. The Frobenius automorphism

Definition 17.1 (Frobenius automorphism). The map $\Phi : \mathbb{F}_{q^m} \to \mathbb{F}_{q^m}$, $\Phi(\alpha) = \alpha^q$, is the *Frobenius automorphism* of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$.

Theorem 17.2. The $\mathbb{F}_q$-automorphisms of $\mathbb{F}_{q^m}$ are exactly $\Phi, \Phi^2, ..., \Phi^m = \mathrm{id}$, and these are pairwise distinct.

Proof. Φ is an iterate of $a \mapsto a^p$, where $p = \mathrm{char}\ \mathbb{F}_q$, since q is a power of p; so it is a ring homomorphism by Theorem 14.4. It fixes $\mathbb{F}_q$, since $a^q = a$ for $a \in \mathbb{F}_q$ by Theorem 14.2. It is injective: $\Phi(\alpha) = \Phi(\beta)$ if and only if $\Phi(\alpha - \beta) = 0$, if and only if $(\alpha - \beta)^q = 0$, if and only if $\alpha = \beta$. An injective map of the finite set $\mathbb{F}_{q^m}$ into itself is surjective. So Φ is an $\mathbb{F}_q$-automorphism, and so are its powers. By Theorem 14.2 applied to $\mathbb{F}_{q^m}$, $\Phi^m(\alpha) = \alpha^{q^m} = \alpha$, so $\Phi^m = \mathrm{id}$.

Let β be a generator of $\mathbb{F}_{q^m}^\times$ (Theorem 14.2). Then $\mathbb{F}_q(\beta) = \mathbb{F}_{q^m}$, so by Remark 14.5 the minimal polynomial $f(x) = x^m + a_{m-1}x^{m-1} + \cdots + a_0 \in \mathbb{F}_q[x]$ of β has degree m. By Lemma 14.6 its roots in $\mathbb{F}_{q^m}$ are the pairwise distinct elements $\beta, \beta^q, ..., \beta^{q^{m-1}}$. Hence $\Phi^k(\beta) = \beta^{q^k}$, $1 \le k \le m$, are pairwise distinct, and so are $\Phi, ..., \Phi^m$.

Now let ψ be an $\mathbb{F}_q$-automorphism of $\mathbb{F}_{q^m}$. Then

$$0 = \psi(0) = \psi(f(\beta)) = \psi(\beta^m + a_{m-1}\beta^{m-1} + \cdots + a_0).$$

Since ψ is an $\mathbb{F}_q$-automorphism, it fixes the coefficients, hence

$$\psi(f(\beta)) = \psi(\beta)^m + a_{m-1}\psi(\beta)^{m-1} + \cdots + a_1\psi(\beta) + a_0 = f(\psi(\beta)).$$

So $\psi(\beta)$ is a root of f, that is, $\psi(\beta) = \beta^{q^k} = \Phi^k(\beta)$ for some $1 \le k \le m$. Two $\mathbb{F}_q$-automorphisms that agree on β agree on every $g(\beta)$ with $g \in \mathbb{F}_q[x]$, that is, on $\mathbb{F}_q(\beta) = \mathbb{F}_{q^m}$. Hence $\psi = \Phi^k$. ∎

17.2. Separable extensions

Remark 17.3 (Recall). Let F be a field. We write $E \mid F$ for a field extension $F \subseteq E$.

1) $[E : F]$ is the dimension of E over F, and $E \mid F$ is *finite* if it is finite. For $F \subseteq K \subseteq E$, $[E : F] = [E : K][K : F]$ (tower law).
2) $\alpha \in E$ is *algebraic* over F if $g(\alpha) = 0$ for some $0 \neq g \in F[x]$. Then its minimal polynomial $m_{\alpha,F}$ is irreducible, divides every $g \in F[x]$ with $g(\alpha) = 0$, and $[F(\alpha) : F] = \deg m_{\alpha,F}$. Every element of a finite extension is algebraic; $E \mid F$ is *algebraic* if every element of E is.
3) If $f \in F[x]$ is irreducible, then $F[x]/(f)$ is an extension of degree $\deg f$ containing a root of f. If α is a root of f in an extension, then $g(x) + (f) \mapsto g(\alpha)$ is an isomorphism $F[x]/(f) \to F(\alpha)$.
4) Every nonzero $f \in F[x]$ has a *splitting field*, an extension in which f is a product of linear factors and which is generated by the roots of f; it is unique up to F-isomorphism.
5) Every field F has an *algebraic closure* $\overline{F}$, an algebraic extension in which every nonconstant polynomial has a root. $\mathbb{C}$ is algebraically closed.

6) The monic greatest common divisor of $f, g \in F[x]$ is the same in $K[x]$ for every extension $K \mid F$, since the Euclidean algorithm runs in $F[x]$.

7) If S is an infinite subset of a field L, a nonzero $P \in L[c_1, \ldots, c_k]$ does not vanish on all of S^k.

A ring homomorphism between fields is injective; it is called an *embedding* or *monomorphism*, and an *F-monomorphism* if it fixes F pointwise.

Definition 17.4 (Separable polynomial). A nonzero polynomial $f \in F[x]$ is *separable* if every root of f in its splitting field is simple.

Example 17.5. $x^2 + 1 \in \mathbb{R}[x]$ has splitting field $\mathbb{C}$ and simple roots $\pm i$, so it is separable. $(x-1)^2(x-2) \in \mathbb{Q}[x]$ is not separable, since 1 is a double root.

Proposition 17.6. A nonzero $f \in F[x]$ is not separable if and only if $\deg \gcd(f, f') \geq 1$.

Proof. Let E be the splitting field of f; by Remark 17.3 6) the gcd may be computed in $E[x]$.

($\Rightarrow$) Let $\alpha \in E$ be a multiple root. Then $f(x) = (x-\alpha)^r h(x)$ with $r \geq 2$ and $h \in E[x]$, so

$$f'(x) = r(x-\alpha)^{r-1} h(x) + (x-\alpha)^r h'(x).$$

Hence $x - \alpha$ is a common divisor of f and f', and $\deg \gcd(f, f') \geq 1$.

($\Leftarrow$) Let $g = \gcd(f, f')$ with $\deg g \geq 1$. Since $g \mid f$, g splits in E; let $\alpha \in E$ be a root of g. Then $x - \alpha$ divides f and f'. Write $f(x) = (x-\alpha)^r h(x)$ with $r \geq 1$ and $h(\alpha) \neq 0$, and suppose $r = 1$. Then $f'(x) = h(x) + (x-\alpha)h'(x)$, and $f'(\alpha) = 0$ forces $h(\alpha) = 0$, a contradiction. So α is a multiple root of f. ∎

Lemma 17.7. An irreducible $f \in F[x]$ is separable if and only if $f' \neq 0$. In particular, if $\operatorname{char} F = 0$, every irreducible polynomial is separable.

Proof. Since f is irreducible, $\gcd(f, f')$ is 1 or f up to a unit. If $f' \neq 0$, then $\deg f' < \deg f$ and $f \nmid f'$, so the gcd is 1 and f is separable by Proposition 17.6. If $f' = 0$, the gcd is f, of degree at least 1. In characteristic 0, $f' \neq 0$ whenever $\deg f \geq 1$. ∎

Theorem 17.8. Over a finite field every irreducible polynomial is separable.

Proof. Let F be finite, $f \in F[x]$ irreducible, and suppose that f is not separable. By Lemma 17.7, $f' = 0$. Let $\operatorname{char} F = p$ (Theorem 14.2). Since $(x^p)' = p x^{p-1} = 0$, the condition $f' = \sum_i i a_i x^{i-1} = 0$ holds exactly when $a_i = 0$ for $p \nmid i$, so

$$f(x) = a_0 + a_1 x^p + \cdots + a_n (x^p)^n.$$

By Theorem 17.2 with $q = p$, the map $b \mapsto b^p$ is surjective on F, so $a_i = b_i^p$ with $b_i \in F$. The binomial argument of Theorem 14.4 holds in every commutative ring of characteristic p, in particular in $F[x]$, hence

$$f(x) = b_0^p + b_1^p x^p + \cdots + b_n^p x^{np} = (b_0 + b_1 x + \cdots + b_n x^n)^p.$$

As $\deg f \geq 1$ and $p \geq 2$, this contradicts the irreducibility of f. ■

Definition 17.9 (Separable extension). Let $E \mid F$ be an extension. An element $\alpha \in E$ is *separable* over F if it is algebraic over F and its minimal polynomial is separable. The extension is *separable* if every element of E is separable over F.

Theorem 17.10. Let $F \subseteq K \subseteq E$ be fields with E separable over F. Then E is separable over K, and K is separable over F.

Proof. 1) $K \mid F$ is separable: every $\beta \in K$ lies in E, and is therefore separable over F.

2) $E \mid K$ is separable: let $\alpha \in E$. Then α is separable over F, in particular algebraic over F, hence over K. Since $m_{\alpha,F} \in F[x] \subseteq K[x]$ vanishes at α, Remark 17.3 2) gives $m_{\alpha,K} \mid m_{\alpha,F}$. So every root of $m_{\alpha,K}$ in a splitting field of $m_{\alpha,F}$ is a root of $m_{\alpha,F}$ of at least the same multiplicity. As $m_{\alpha,F}$ has only simple roots, so does $m_{\alpha,K}$, and α is separable over K. ■

Example 17.11 (A non-separable extension). Fix a prime p, let t be an indeterminate and $F = \mathbb{F}_p(t)$, the field of rational functions over $\mathbb{F}_p$. Let E be obtained by adjoining to F a root s of $x^p - t$. Then $x^p - t$ is irreducible over F, and $E \mid F$ is not separable: in $E[x]$,

$$x^p - t = x^p - s^p = (x - s)^p.$$

Irreducibility of $x^p - t$. First, $s \notin F$: if $s = g/h$ with $g, h \in \mathbb{F}_p[t]$, $h \neq 0$, then $g^p = th^p$, and comparing degrees in t gives $p \deg g = 1 + p \deg h$, which is impossible. A monic factor of $x^p - t$ in $F[x]$ divides $(x - s)^p$ in $E[x]$, so it equals $(x - s)^k$ with $0 \leq k \leq p$. Its coefficient of x^{k-1} is $-ks$, which lies in F. For $1 \leq k \leq p - 1$, k is invertible in $\mathbb{F}_p$, so $s \in F$, a contradiction. Hence the only monic factors are 1 and $x^p - t$. ■

Theorem 17.12. Let $E \mid F$ be an extension, $\sigma : E \to E$ an F-monomorphism, and $f \in F[x]$ a polynomial that splits over E. If $\alpha \in E$ is a root of f, then so is $\sigma(\alpha)$. Hence σ permutes the roots of f in E.

Proof. Let $f(x) = b_0 + b_1 x + \cdots + b_n x^n \in F[x]$ with $f(\alpha) = b_0 + b_1\alpha + \cdots + b_n\alpha^n = 0$. Then

$$\begin{aligned} 0 = \sigma(0) &= \sigma(b_0 + b_1\alpha + \cdots + b_n\alpha^n) = \sigma(b_0) + \sigma(b_1\alpha) + \cdots + \sigma(b_n\alpha^n) \\ &= b_0 + b_1\sigma(\alpha) + \cdots + b_n\sigma(\alpha)^n = f(\sigma(\alpha)), \end{aligned}$$

since σ is an F-monomorphism and fixes the coefficients. So σ maps the finite set of roots of f in E into itself, and it is injective, hence it permutes this set. ■

17.3. Embeddings and primitive elements

Theorem 17.13. Let $E \mid F$ be a finite separable extension with $[E:F]=n$, let L be an algebraically closed field, and let $\sigma : F \to L$ be an embedding. Then σ has exactly n extensions to embeddings $E \to L$. In particular this holds for $L = \overline{F}$.

Proof. We proceed by induction on n. For $n=1$, $E=F$ and σ is its own unique extension.

Let $n>1$, choose $\alpha \in E \setminus F$, and let $r = \deg m_{\alpha,F} > 1$. In the tower $F \subseteq F(\alpha) \subseteq E$,

$$n = [E:F] = [E:F(\alpha)][F(\alpha):F] = \frac{n}{r}\cdot r.$$

Let $g = \sigma\big(m_{\alpha,F}\big) \in \sigma(F)[x]$ be the polynomial obtained by applying σ to the coefficients. Since $\sigma : F \to \sigma(F)$ is an isomorphism, it induces a ring isomorphism $F[x] \to \sigma(F)[x]$ compatible with derivatives and gcds, so g is irreducible and $\deg \gcd(g,g') = \deg \gcd\big(m_{\alpha,F}, m'_{\alpha,F}\big)$, which is 0 by Proposition 17.6 because α is separable over F. By Proposition 17.6, g is separable, so it has exactly r distinct roots in the algebraically closed field L.

For each root $\beta \in L$ of g define

$$\psi : F(\alpha) \to \sigma(F)(\beta), \quad \psi\big(b_0 + b_1\alpha + \cdots + b_k\alpha^k\big) = \sigma(b_0) + \sigma(b_1)\beta + \cdots + \sigma(b_k)\beta^k.$$

This is well defined and an isomorphism, being the composite $F(\alpha) \cong F[x]/\big(m_{\alpha,F}\big) \cong \sigma(F)[x]/(g) \cong \sigma(F)(\beta)$ of the isomorphisms of Remark 17.3 3). Its restriction to F is σ. Conversely, every extension ψ of σ to $F(\alpha)$ satisfies $g(\psi(\alpha)) = \psi\big(m_{\alpha,F}(\alpha)\big) = 0$, by the computation in the proof of Theorem 17.12, and is determined by $\psi(\alpha)$. So σ has exactly r extensions to $F(\alpha)$, one for each root β of g.

$E \mid F(\alpha)$ is separable by Theorem 17.10, and $[E:F(\alpha)] = n/r < n$. By the induction hypothesis each of the r embeddings ψ has exactly n/r extensions to E. Distinct ψ have no common extension, and every extension of σ to E restricts to some ψ. Hence σ has exactly $\frac{n}{r}\cdot r = n$ extensions. ∎

Theorem 17.14 (Primitive element theorem). If $E \mid F$ is a finite separable extension, then $E = F(\gamma)$ for some $\gamma \in E$. Such a γ is called a *primitive element* of $E \mid F$.

Proof. If F is finite, so is E, and a generator γ of $E^\times$ (Theorem 14.2) satisfies $E = F(\gamma)$.

Let F be infinite and $n = [E:F]$. Write $E = F(\alpha_1, \ldots, \alpha_k)$, for instance with a basis. By Theorem 17.13 applied to the inclusion $F \to \overline{F}$, there are exactly n distinct F-embeddings $\sigma_1, \ldots, \sigma_n : E \to \overline{F}$. An F-embedding of E is determined by its values on $\alpha_1, \ldots, \alpha_k$, so for $i \neq j$ some l has $\sigma_i(\alpha_l) \neq \sigma_j(\alpha_l)$. Hence

$$P(c_1, \ldots, c_k) = \prod_{i<j} \sum_{l=1}^{k} c_l\big(\sigma_i(\alpha_l) - \sigma_j(\alpha_l)\big) \in \overline{F}[c_1, \ldots, c_k]$$

is a product of nonzero linear forms, so $P \neq 0$. By Remark 17.3 7) there is $(c_1, \ldots, c_k) \in F^k$ with $P(c_1, \ldots, c_k) \neq 0$. Put $\gamma = \sum_l c_l\alpha_l \in E$. Then the elements $\sigma_i(\gamma) = \sum_l c_l\sigma_i(\alpha_l)$ are pairwise

distinct, and each is a root of $m_{\gamma,F}$ by the computation in the proof of Theorem 17.12. So $[F(\gamma) : F] = \deg m_{\gamma,F} \geq n = [E : F]$, and since $F(\gamma) \subseteq E$, $F(\gamma) = E$. ∎

Remark 17.15. For a finite field E, a generator of $E^\times$, called a primitive element of E in Theorem 14.2, is a primitive element of $E \mid F$ for every subfield F. The converse fails: $x^2 + 1$ is irreducible over $\mathbb{F}_3$, so $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$, but i has order 4 in $\mathbb{F}_9^\times$, which has order 8.

17.4. Normal extensions

Definition 17.16 (Normal extension). An algebraic extension $E \mid F$ is *normal* if every irreducible polynomial in $F[x]$ that has at least one root in E splits into linear factors in $E[x]$.

Example 17.17. $\mathbb{Q}\left(\sqrt[3]{2}\right) \mid \mathbb{Q}$ is not normal. The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$: it has degree 3 and no rational root, since $z^3 = 2$ would give $3v_2(z) = 1$. Its roots are $\sqrt[3]{2}$, $\varepsilon\sqrt[3]{2}$, $\varepsilon^2\sqrt[3]{2}$, where $\varepsilon = e^{2\pi i/3}$ is a primitive third root of unity. The last two are not real, while $\mathbb{Q}\left(\sqrt[3]{2}\right) \subseteq \mathbb{R}$, so they do not lie in $\mathbb{Q}\left(\sqrt[3]{2}\right)$.

$\mathbb{Q}\left(\sqrt{2}\right) \mid \mathbb{Q}$ is normal; the roots of $x^2 - 2$ are $\pm\sqrt{2}$. More generally, every extension $E \mid F$ with $[E : F] = 2$ is normal. If $f \in F[x]$ is irreducible with a root $\alpha \in E$, then f is a scalar multiple of $m_{\alpha,F}$ and $\deg f = [F(\alpha) : F]$ divides 2. If $\deg f = 1$, f is linear. If $\deg f = 2$, then $f = c(x - \alpha)(x - \alpha')$ in $E[x]$, where $x - \alpha'$ is the quotient of f/c by $x - \alpha$ in $E[x]$, so $\alpha' \in E$.

Theorem 17.18. Let $F \subseteq E \subseteq \mathbb{C}$ be fields with $[E : F]$ finite. Then $E \mid F$ is normal if and only if every F-monomorphism $\tau : E \to \mathbb{C}$ is an F-automorphism of E, that is, $\tau(E) = E$.

Proof. ($\Rightarrow$) Let $E \mid F$ be normal and $\tau : E \to \mathbb{C}$ an F-monomorphism. For $\gamma \in E$, the irreducible polynomial $m_{\gamma,F}$ has the root $\gamma \in E$, so it splits into linear factors in $E[x]$. By the computation in the proof of Theorem 17.12, $\tau(\gamma)$ is a root of $m_{\gamma,F}$, hence $\tau(\gamma) \in E$. So $\tau(E) \subseteq E$. Since τ is an injective F-linear map of the finite-dimensional F-space E into itself, it is surjective, and τ is an F-automorphism of E.

($\Leftarrow$) Let $f \in F[x]$ be irreducible with a root $\alpha \in E$, and let $\beta \in \mathbb{C}$ be any root of f, that is, a conjugate of α over F. As in the proof of Theorem 17.13, $g(\alpha) \mapsto g(\beta)$ defines an F-embedding $\psi : F(\alpha) \to \mathbb{C}$. Since $\operatorname{char} F = 0$, $E \mid F(\alpha)$ is separable by Lemma 17.7, and $\mathbb{C}$ is algebraically closed, so by Theorem 17.13 ψ extends to an F-monomorphism $\tau : E \to \mathbb{C}$, with $\tau(\alpha) = \beta$. By hypothesis $\tau(E) = E$, so $\beta \in E$. Thus every root of f in $\mathbb{C}$ lies in E; since f splits into linear factors over $\mathbb{C}$, it splits into linear factors in $E[x]$. Hence $E \mid F$ is normal. ∎

Lemma 17.19. Let $E \mid F$ be a finite extension, L an algebraically closed field and $\sigma : F \to L$ an embedding. Then σ extends to an embedding $E \to L$.

Proof. We follow the proof of Theorem 17.13, without separability. We proceed by induction on $[E : F]$; for $[E : F] = 1$ there is nothing to prove. Otherwise choose $\alpha \in E \setminus F$. The polynomial $g = \sigma\left(m_{\alpha,F}\right)$ has a root β in L, since L is algebraically closed, and $g(\alpha) \mapsto g(\beta)$ extends σ to an

embedding $\psi : F(\alpha) \to L$, as in that proof. Since $[E : F(\alpha)] < [E : F]$, the induction hypothesis extends ψ to E. ∎

Proposition 17.20. Let $E \mid F$ be a finite extension and Ω an algebraically closed field containing E. Then $E \mid F$ is normal if and only if $\tau(E) = E$ for every F-monomorphism $\tau : E \to \Omega$.

Proof. This is the proof of Theorem 17.18 with $\mathbb{C}$ replaced by Ω and Theorem 17.13 replaced by Lemma 17.19, which needs no separability. In ($\Leftarrow$), the irreducible f splits into linear factors over the algebraically closed field Ω, and every root of f in Ω lies in E. ∎

Theorem 17.21. A finite extension $E \mid F$ is normal if and only if E is a splitting field over F of some $f \in F[x]$.

Proof. ($\Rightarrow$) Let $[E : F] = n$, let $\alpha_1, ..., \alpha_n \in E$ be a basis of E over F, and let $f_i \in F[x]$ be the minimal polynomial of α_i. Since $E \mid F$ is normal and f_i is irreducible with the root $\alpha_i \in E$, f_i splits over E. Hence $f = f_1 \cdots f_n$ splits over E. It remains to show that E is generated by the roots of f. If $F \subseteq K \subseteq E$ and f splits over K, then $\alpha_i \in K$ for every i, and since the α_i span E, $K = E$.

($\Leftarrow$) Let E be a splitting field of $f \in F[x]$, let $\alpha_1, ..., \alpha_n$ be the roots of f in E, so that $E = F(\alpha_1, ..., \alpha_n)$, and let Ω be an algebraic closure of E (Remark 17.3). Let $\tau : E \to \Omega$ be an F-monomorphism. By the computation in the proof of Theorem 17.12, each $\tau(\alpha_i)$ is a root of f in Ω. Since f is a product of linear factors over E, its roots in Ω are $\alpha_1, ..., \alpha_n$, so $\tau(\alpha_i) = \alpha_j$ for some j. Hence $\tau(E) = F(\tau(\alpha_1), ..., \tau(\alpha_n)) \subseteq E$. As τ is F-linear and injective, $\dim_F \tau(E) = n = \dim_F E$, so $\tau(E) = E$. By Proposition 17.20, $E \mid F$ is normal. ∎

Corollary 17.22. Let $F \subseteq K \subseteq E$ be fields with $[E : F]$ finite. If $E \mid F$ is normal, then so is $E \mid K$.

Proof. By Theorem 17.21, E is a splitting field over F of some $f \in F[x]$, so $E = F(\alpha_1, ..., \alpha_n)$, where $\alpha_1, ..., \alpha_n$ are the roots of f. Since $f \in F[x] \subseteq K[x]$ and $F \subseteq K \subseteq E$, also $E = K(\alpha_1, ..., \alpha_n)$. Hence E is a splitting field of f over K, and $E \mid K$ is normal by Theorem 17.21. ∎

18. Galois theory

Definition 18.1 (Galois extension). An algebraic extension $E \mid F$ is a *Galois extension* if it is normal and separable.

Definition 18.2 (Galois group). Let $E \mid F$ be a field extension. The *Galois group* $\mathrm{Gal}(E \mid F)$ is the set of F-automorphisms of E. It is a group under composition, since compositions and inverses of F-automorphisms are F-automorphisms.

Proposition 18.3. Let $E \mid F$ be a finite Galois extension with $[E : F] = n$. Then $|\mathrm{Gal}(E \mid F)| = n$.

Proof. Let Ω be an algebraic closure of E and $\iota : E \to \Omega$ the inclusion. By Theorem 17.13, applied to the inclusion $F \to \Omega$, there are exactly n F-monomorphisms $E \to \Omega$. Since $E \mid F$ is normal, each of them has image E by Proposition 17.20, so it equals $\iota \circ \sigma$ for a unique $\sigma \in \mathrm{Gal}(E \mid F)$. Conversely $\iota \circ \sigma$ is an F-monomorphism for every $\sigma \in \mathrm{Gal}(E \mid F)$. Hence $\sigma \mapsto \iota \circ \sigma$ is a bijection from $\mathrm{Gal}(E \mid F)$ onto the F-monomorphisms $E \to \Omega$. ∎

Example 18.4. $\mathrm{Gal}\left(\mathbb{Q}\left(\sqrt[3]{2}\right) \mid \mathbb{Q}\right) = \{\mathrm{id}\}$. Let σ be a $\mathbb{Q}$-automorphism of $E = \mathbb{Q}\left(\sqrt[3]{2}\right)$. By Theorem 17.12, $\sigma\left(\sqrt[3]{2}\right)$ is a root of $x^3 - 2$ in $E \subseteq \mathbb{R}$. Since $t \mapsto t^3$ is strictly increasing on $\mathbb{R}$, the only real root is $\sqrt[3]{2}$, so $\sigma\left(\sqrt[3]{2}\right) = \sqrt[3]{2}$, and σ is the identity on $\mathbb{Q}\left(\sqrt[3]{2}\right)$. Here $|\mathrm{Gal}(E \mid \mathbb{Q})| = 1 < 3 = [E : \mathbb{Q}]$; Proposition 18.3 does not apply, since $E \mid \mathbb{Q}$ is not normal (Example 17.17).

Definition 18.5 (Fixed field). Let $E \mid F$ be an extension, $G = \mathrm{Gal}(E \mid F)$ and $H \leq G$ a subgroup. The *fixed field* of H is

$$\mathcal{F}(H) = \{x \in E : \sigma(x) = x \text{ for all } \sigma \in H\}.$$

It is a subfield of E, since each $\sigma \in H$ is a ring homomorphism, and it contains F, since each $\sigma \in H$ fixes F.

Definition 18.6. Let $E \mid F$ be an extension, $G = \mathrm{Gal}(E \mid F)$, and let K be an *intermediate field*, that is, $F \subseteq K \subseteq E$. Put

$$\mathcal{G}(K) = \mathrm{Gal}(E \mid K) = \{\sigma \in G : \sigma(x) = x \text{ for all } x \in K\},$$

the subgroup of G fixing K pointwise.

Example 18.7. $\mathbb{C} \mid \mathbb{R}$ is a Galois extension with $\mathrm{Gal}(\mathbb{C} \mid \mathbb{R}) = \{\mathrm{id}, \text{complex conjugation}\} \cong \mathbb{Z}/2\mathbb{Z}$. Indeed, $[\mathbb{C} : \mathbb{R}] = 2$, so $\mathbb{C} \mid \mathbb{R}$ is normal (Example 17.17), and separable by Lemma 17.7, since $\mathrm{char}\ \mathbb{R} = 0$. By Proposition 18.3, $|\mathrm{Gal}(\mathbb{C} \mid \mathbb{R})| = 2$, and complex conjugation is an $\mathbb{R}$-automorphism different from the identity.

Theorem 18.8. Let $E \mid F$ be a finite Galois extension with $G = \mathrm{Gal}(E \mid F)$.

1) The fixed field of G is F.

2) If H is a proper subgroup of G, then $\mathcal{F}(H) \supsetneq F$.

Proof. 1) Let $F_0 = \mathcal{F}(G)$, so $F \subseteq F_0 \subseteq E$. Every $\sigma \in G$ fixes F_0 pointwise, so $G \subseteq \mathrm{Gal}(E \mid F_0)$. Conversely every F_0-automorphism of E fixes $F \subseteq F_0$, so $\mathrm{Gal}(E \mid F_0) \subseteq G$, and $G = \mathrm{Gal}(E \mid F_0)$. The extension $E \mid F_0$ is Galois: it is normal by Corollary 17.22 and separable by Theorem 17.10. By Proposition 18.3, $[E : F_0] = |\mathrm{Gal}(E \mid F_0)| = |G| = [E : F]$. By the tower law $[E : F] = [E : F_0][F_0 : F]$, so $[F_0 : F] = 1$ and $F_0 = F$.

2) Suppose $F = \mathcal{F}(H)$. Since $E \mid F$ is finite and separable, Theorem 17.14 gives $\alpha \in E$ with $E = F(\alpha)$. Consider

$$f(x) = \prod_{\sigma \in H} (x - \sigma(\alpha)) \in E[x].$$

We show $f \in F[x]$. For $\tau \in H$ the map $\sigma \mapsto \tau\sigma$ is a bijection of H, so

$$\prod_{\sigma\in H}(x - \tau(\sigma(\alpha))) = \prod_{\sigma\in H}(x - \sigma(\alpha)).$$

Applying τ to the coefficients of f therefore leaves f unchanged, so every coefficient lies in $\mathcal{F}(H) = F$, and $f \in F[x]$. Moreover $f(\alpha) = 0$, since $\mathrm{id} \in H$. Then $\deg f = |H| < |G| = [E : F]$ by Proposition 18.3. On the other hand $m_{\alpha,F}$ divides f, so $\deg f \geq [F(\alpha) : F] = [E : F]$, a contradiction. Hence $F \neq \mathcal{F}(H)$, and since $F \subseteq \mathcal{F}(H)$, $\mathcal{F}(H) \supsetneq F$. ■

Remark 18.9. For a finite Galois extension $E \mid F$ with $G = \mathrm{Gal}(E \mid F)$ there are two maps between the intermediate fields and the subgroups of G:

$$\mathcal{G} : \{\text{intermediate fields}\} \to \{\text{subgroups of } G\}, \quad K \mapsto \mathcal{G}(K) = \mathrm{Gal}(E \mid K),$$

$$\mathcal{F} : \{\text{subgroups of } G\} \to \{\text{intermediate fields}\}, \quad H \mapsto \mathcal{F}(H).$$

By Theorem 18.11 they are mutually inverse bijections.

Example 18.10. Let $E = \mathbb{Q}\left(i, \sqrt{5}\right)$ and $F = \mathbb{Q}$. Determine $G = \mathrm{Gal}(E \mid \mathbb{Q})$, its subgroups and the corresponding intermediate fields.

Solution. E is the splitting field of $(x^2 + 1)(x^2 - 5)$ over $\mathbb{Q}$, so $E \mid \mathbb{Q}$ is normal by Theorem 17.21, and separable since $\mathrm{char}\ \mathbb{Q} = 0$. We have $[\mathbb{Q}(i) : \mathbb{Q}] = 2$, and $\sqrt{5} \notin \mathbb{Q}(i)$: an element $a + bi$ with $a, b \in \mathbb{Q}$ is real only if $b = 0$, and $\sqrt{5} \notin \mathbb{Q}$. Hence $[E : \mathbb{Q}(i)] = 2$ and $[E : \mathbb{Q}] = 4$, so $|G| = 4$ by Proposition 18.3. Each $\sigma \in G$ is determined by $\sigma(i) \in \{i, -i\}$ and $\sigma\left(\sqrt{5}\right) \in \left\{\sqrt{5}, -\sqrt{5}\right\}$, by Theorem 17.12. These are four possibilities, so all of them occur:

$$\sigma_1 : \left(i, \sqrt{5}\right) \mapsto \left(i, \sqrt{5}\right), \quad \sigma_2 : \left(i, \sqrt{5}\right) \mapsto \left(i, -\sqrt{5}\right),$$
$$\sigma_3 : \left(i, \sqrt{5}\right) \mapsto \left(-i, \sqrt{5}\right), \quad \sigma_4 : \left(i, \sqrt{5}\right) \mapsto \left(-i, -\sqrt{5}\right).$$

Each σ_j satisfies $\sigma_j^2 = \sigma_1 = \mathrm{id}$. The proper subgroups are $\{\sigma_1\}$, $\{\sigma_1, \sigma_2\}$, $\{\sigma_1, \sigma_3\}$ and $\{\sigma_1, \sigma_4\}$. Since σ_2 fixes i, σ_3 fixes $\sqrt{5}$, and $\sigma_4\left(i\sqrt{5}\right) = (-i)\left(-\sqrt{5}\right) = i\sqrt{5}$, each of the three subgroups of order 2 fixes a field of degree 2 over $\mathbb{Q}$:

$$\mathcal{F}(\{\sigma_1, \sigma_2\}) = \mathbb{Q}(i), \quad \mathcal{F}(\{\sigma_1, \sigma_3\}) = \mathbb{Q}\left(\sqrt{5}\right), \quad \mathcal{F}(\{\sigma_1, \sigma_4\}) = \mathbb{Q}\left(i\sqrt{5}\right).$$

Equality holds in each case. For instance $K = \mathcal{F}(\{\sigma_1, \sigma_2\})$ satisfies $\mathbb{Q}(i) \subseteq K \subsetneq E$, since $\sigma_2\left(\sqrt{5}\right) \neq \sqrt{5}$. As $[E : \mathbb{Q}(i)] = 2$, the tower law forces $K = \mathbb{Q}(i)$. The same argument applies to $\mathbb{Q}\left(\sqrt{5}\right)$ and $\mathbb{Q}\left(i\sqrt{5}\right)$, both of degree 2 over $\mathbb{Q}$ because $\sqrt{5}, i\sqrt{5} \notin \mathbb{Q}$ are roots of $x^2 - 5$ and $x^2 + 5$. Finally $\mathcal{F}(\{\sigma_1\}) = E$ and $\mathcal{F}(G) = \mathbb{Q}$ by Theorem 18.8.

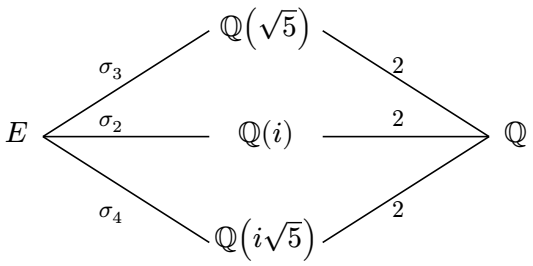


Figure 18.1. *Intermediate fields of $\mathbb{Q}(i, \sqrt{5}) \mid \mathbb{Q}$. An edge from E is labelled with the nontrivial element of $\mathrm{Gal}(E \mid K)$, an edge to $\mathbb{Q}$ with $[K : \mathbb{Q}]$.*

◊

Theorem 18.11 (Fundamental theorem of Galois theory, part I). Let $E \mid F$ be a finite Galois extension with $G = \mathrm{Gal}(E \mid F)$.

1) The map $\mathcal{F}$ is a bijection from the subgroups of G onto the intermediate fields, with inverse $\mathcal{G}$.
2) Let $H \leq G$, $K = \mathcal{F}(H)$ and $\sigma \in G$. Then $\sigma(K) = \{\sigma(x) : x \in K\}$ is the intermediate field fixed by $\sigma H \sigma^{-1}$, that is, $\sigma(K) = \mathcal{F}(\sigma H \sigma^{-1})$.

Proof. 1) We show $\mathcal{G}(\mathcal{F}(H)) = H$ for every subgroup H and $\mathcal{F}(\mathcal{G}(K)) = K$ for every intermediate field K.

Let $H \leq G$. Every $\sigma \in H$ fixes $\mathcal{F}(H)$, so $\sigma \in \mathrm{Gal}(E \mid \mathcal{F}(H)) = \mathcal{G}(\mathcal{F}(H))$, and $H \subseteq \mathcal{G}(\mathcal{F}(H))$. The extension $E \mid \mathcal{F}(H)$ is Galois, by Corollary 17.22 and Theorem 17.10, with Galois group $\mathcal{G}(\mathcal{F}(H))$. If H were a proper subgroup of it, Theorem 18.8 2), applied to $E \mid \mathcal{F}(H)$, would give that the fixed field of H strictly contains $\mathcal{F}(H)$. But the fixed field of H is $\mathcal{F}(H)$ by definition. Hence $H = \mathcal{G}(\mathcal{F}(H))$.

Let K be an intermediate field. Then $\mathcal{F}(\mathcal{G}(K))$ is the fixed field of $\mathrm{Gal}(E \mid K)$. Since $E \mid K$ is Galois, by Corollary 17.22 and Theorem 17.10, this fixed field is K by Theorem 18.8 1).

2) Since σ is an automorphism of E,

$$\begin{aligned}\mathcal{F}(\sigma H \sigma^{-1}) &= \{x \in E : (\sigma\tau\sigma^{-1})(x) = x \text{ for all } \tau \in H\} \\ &= \{x \in E : \tau(\sigma^{-1}(x)) = \sigma^{-1}(x) \text{ for all } \tau \in H\} \\ &= \{x \in E : \sigma^{-1}(x) \in \mathcal{F}(H)\} = \sigma(\mathcal{F}(H)) = \sigma(K).\end{aligned}$$

■

Theorem 18.12 (Fundamental theorem of Galois theory, part II). Let $E \mid F$ be a finite Galois extension with $G = \mathrm{Gal}(E \mid F)$, let $H \leq G$, and let $K = \mathcal{F}(H)$ be the intermediate field fixed by H. Then:

1) $E \mid K$ is a Galois extension.
2) $K \mid F$ is normal if and only if H is a normal subgroup of G.
3) If H is normal in G, then $\mathrm{Gal}(K \mid F) \cong G/H = \mathrm{Gal}(E \mid F)/\mathrm{Gal}(E \mid K)$.
4) $[K : F] = |G : H|$, whether or not $K \mid F$ is normal.

Proof. 1) $E \mid K$ is normal by Corollary 17.22 and separable by Theorem 17.10.

2) Let Ω be an algebraic closure of E. We first show that the F-monomorphisms $K \to \Omega$ are exactly the restrictions $\sigma|_K$ with $\sigma \in G$. Each $\sigma|_K$ is one. Conversely, let $\tau : K \to \Omega$ be an F-monomorphism. By Lemma 17.19, applied to $E \mid K$, τ extends to an F-monomorphism $\tau' : E \to \Omega$. Since $E \mid F$ is normal, $\tau'(E) = E$ by Proposition 17.20, so $\tau' \in G$ and $\tau = \tau'|_K$.

By Proposition 17.20 applied to $K \mid F$, it follows that $K \mid F$ is normal if and only if $\sigma(K) = K$ for every $\sigma \in G$. Since $K = \mathcal{F}(H)$ and $\sigma(\mathcal{F}(H)) = \mathcal{F}(\sigma H \sigma^{-1})$ by Theorem 18.11 2), this holds if and only if $\mathcal{F}(\sigma H \sigma^{-1}) = \mathcal{F}(H)$ for every $\sigma \in G$. As $\mathcal{F}$ is injective by Theorem 18.11 1), this is equivalent to $\sigma H \sigma^{-1} = H$ for every $\sigma \in G$, that is, to H being normal in G.

3) Let H be normal in G. By 2), $\sigma(K) = K$ for every $\sigma \in G$, so $\sigma|_K \in \mathrm{Gal}(K \mid F)$, and

$$G \to \mathrm{Gal}(K \mid F), \quad \sigma \mapsto \sigma|_K,$$

is a group homomorphism. Its kernel is $\{\sigma \in G : \sigma|_K = \mathrm{id}\} = \mathrm{Gal}(E \mid K) = \mathcal{G}(\mathcal{F}(H)) = H$, by Theorem 18.11 1). It is surjective: every $\rho \in \mathrm{Gal}(K \mid F)$, composed with the inclusion $K \to \Omega$, is an F-monomorphism $K \to \Omega$, hence the restriction of some $\sigma \in G$ by the first paragraph of 2). By the first isomorphism theorem, $G/H \cong \mathrm{Gal}(K \mid F)$.

4) By Theorem 18.11 1), $H = \mathrm{Gal}(E \mid K)$. Since $E \mid F$ and $E \mid K$ are Galois, Proposition 18.3 gives $|G| = [E : F]$ and $|H| = [E : K]$. By the tower law $[E : F] = [E : K][K : F]$, so $|G| = |H|\,[K : F]$, and by Lagrange's theorem $|G : H| = |G| \,/\, |H| = [K : F]$. ∎

19. Worked exercises on fields

Throughout the first two exercises, q is a prime power, $m \in \mathbb{N}$, $L = \mathbb{F}_{q^m}$ and $K = \mathbb{F}_q$; we write $\mathrm{N} = \mathrm{N}_{L \mid K}$ and $\mathrm{tr} = \mathrm{tr}_{L \mid K}$. By Theorem 14.2, $c^q = c$ for every $c \in K$.

Example 19.1. Show that for $\alpha, \beta \in L$ and $a, c \in K$:

1) $\mathrm{N}(\alpha\beta) = \mathrm{N}(\alpha)\,\mathrm{N}(\beta)$;

2) $\mathrm{N}(c\alpha) = c^m\,\mathrm{N}(\alpha)$;

3) $\mathrm{N}(a) = a^m$.

Solution. 1) Rearranging the factors,

$$\mathrm{N}(\alpha)\,\mathrm{N}(\beta) = \alpha\alpha^q \cdots \alpha^{q^{m-1}} \cdot \beta\beta^q \cdots \beta^{q^{m-1}} = (\alpha\beta)(\alpha\beta)^q \cdots (\alpha\beta)^{q^{m-1}} = \mathrm{N}(\alpha\beta).$$

2) Since $c^{q^i} = c$ for every i,

$$\mathrm{N}(c\alpha) = c\alpha \cdot (c\alpha)^q \cdots (c\alpha)^{q^{m-1}} = \left(cc^q \cdots c^{q^{m-1}}\right)\left(\alpha\alpha^q \cdots \alpha^{q^{m-1}}\right) = c^m\,\mathrm{N}(\alpha).$$

3) $\mathrm{N}(a) = aa^q \cdots a^{q^{m-1}} = a^m$, since $a^{q^i} = a$. ◊

Example 19.2. Show that for $\alpha, \beta \in L$ and $a, c \in K$:

1) $\mathrm{tr}(\alpha + \beta) = \mathrm{tr}(\alpha) + \mathrm{tr}(\beta)$;

2) $\mathrm{tr}(c\alpha) = c\,\mathrm{tr}(\alpha)$;

3) $\operatorname{tr}(a) = ma$.

Solution. 1) $\alpha \mapsto \alpha^{q^i}$ is a ring homomorphism (Theorem 17.2), so

$$\begin{aligned}\operatorname{tr}(\alpha+\beta) &= (\alpha+\beta) + (\alpha+\beta)^q + \cdots + (\alpha+\beta)^{q^{m-1}} \\ &= \alpha + \beta + \alpha^q + \beta^q + \cdots + \alpha^{q^{m-1}} + \beta^{q^{m-1}} = \operatorname{tr}(\alpha) + \operatorname{tr}(\beta).\end{aligned}$$

2) Using $c^{q^i} = c$,

$$\operatorname{tr}(c\alpha) = c\alpha + (c\alpha)^q + \cdots + (c\alpha)^{q^{m-1}} = c\alpha + c\alpha^q + \cdots + c\alpha^{q^{m-1}} = c\operatorname{tr}(\alpha).$$

3) $\operatorname{tr}(a) = a + a^q + \cdots + a^{q^{m-1}} = ma$, since $a^{q^i} = a$. ◊

Definition 19.3 (Perfect field). A field F is *perfect* if every irreducible polynomial in $F[x]$ is separable.

By Lemma 17.7 and Theorem 17.8, fields of characteristic 0 and finite fields are perfect.

Example 19.4. Show that an algebraic extension of a perfect field is perfect.

Solution. Let K be perfect and $L \mid K$ algebraic. Let $f \in L[x]$ be irreducible; we show that f is separable. Let E be a splitting field of f over L and $\alpha_1 \in E$ a root of f. Since f is irreducible, it is a scalar multiple of $m_{\alpha_1, L}$.

The element α_1 is algebraic over K. Indeed, the coefficients of f are algebraic over K, so they generate a finite extension K' of K (tower law). As α_1 is algebraic over K', $[K'(\alpha_1) : K]$ is finite, and α_1 is algebraic over K.

Let $g = m_{\alpha_1, K}$. Then $g \in K[x] \subseteq L[x]$ and $g(\alpha_1) = 0$, so $f \mid g$ in $L[x]$. Since K is perfect, g is separable: in a splitting field E' of g over E, $g = (x-\alpha_1)\cdots(x-\alpha_r)$ with pairwise distinct α_i. As $f \mid g$, f splits in $E'[x]$ into pairwise distinct linear factors. Hence f is separable, and L is perfect. ◊

Example 19.5. Let p be a prime and $a \in \mathbb{F}_p$. Show that $f(x) = x^p - x + a$ is separable over $\mathbb{F}_p$.

Solution. $f'(x) = px^{p-1} - 1 = -1$ in $\mathbb{F}_p[x]$. Hence $\gcd(f, f') = 1$, and f is separable by Proposition 17.6. ◊

Example 19.6. Is $x^4 - 7$ irreducible over $\mathbb{F}_5$? Determine the irreducible factors of $x^4 + 1$ over $\mathbb{F}_3$.

Solution. Over $\mathbb{F}_5$, $x^4 - 7 = x^4 - 2$. By Theorem 14.2, $x^4 \in \{0, 1\}$ for $x \in \mathbb{F}_5$, so $x^4 - 2$ has no root and no linear factor. If it were reducible, it would therefore be a product of two monic quadratics:

$$x^4 - 2 = (x^2 + ax + b)(x^2 + cx + d) = x^4 + (a+c)x^3 + (b + d + ac)x^2 + (ad + bc)x + bd.$$

Comparing coefficients, $a + c = 0$, $b + d + ac = 0$, $ad + bc = 0$, $bd = -2$. So $c = -a$, $b + d = a^2$, $a(d - b) = 0$ and $bd = -2$.

- If $a = 0$, then $b + d = 0$ and $-b^2 = -2$, so $b^2 = 2$. But $\left(\frac{2}{5}\right) = -1$, since $5 \equiv -3 \pmod 8$.
- If $a \neq 0$, then $b = d$ and $b^2 = -2$. But $\left(\frac{-2}{5}\right) = \left(\frac{-1}{5}\right)\left(\frac{2}{5}\right) = 1 \cdot (-1) = -1$.

So $x^4 - 7$ is irreducible over $\mathbb{F}_5$.

Over $\mathbb{F}_3$, $x^4 + 1 \in \{1, 2\}$ for every $x \in \mathbb{F}_3$, again since $x^4 \in \{0, 1\}$, so there is no linear factor. With the same ansatz, $c = -a$, $b + d = a^2$, $a(d - b) = 0$ and $bd = 1$. In $\mathbb{F}_3$ the units ± 1 are their own inverses, so $bd = 1$ forces $b = d = 1$ or $b = d = -1$.
- If $b = d = 1$, then $a^2 = 2$, which has no solution in $\mathbb{F}_3$.
- If $b = d = -1$, then $a^2 = -2 = 1$, so $a = 1$, $c = -1$ is a solution.

Hence

$$x^4 + 1 = (x^2 + x - 1)(x^2 - x - 1) \quad \text{in } \mathbb{F}_3[x],$$

which can be checked directly: the product is $(x^2 - 1)^2 - x^2 = x^4 - 3x^2 + 1 = x^4 + 1$. Both factors are irreducible, since neither has a root in $\mathbb{F}_3$: $x^2 + x - 1$ takes the values $-1, 1, 2$ and $x^2 - x - 1$ the values $-1, -1, 1$ at $x = 0, 1, 2$. ◊

Definition 19.7 (Purely inseparable polynomial). Let F be a field with algebraic closure $\overline{F}$. A nonconstant $f \in F[x]$ is *purely inseparable* if it has exactly one root in $\overline{F}$.

Example 19.8. Let F be a finite field and $f \in F[x]$ purely inseparable, with root $\alpha \in \overline{F}$. Prove that $f = c \cdot m_{\alpha,F}^k$ for some $k \in \mathbb{N}$ and $c \in F^\times$.

Solution. We proceed by induction on $n = \deg f \geq 1$. Since $m_{\alpha,F}$ divides f in $F[x]$, write $f = m_{\alpha,F} g$ with $g \in F[x]$ and $\deg g < n$. If $\deg g = 0$, then $g = c \in F^\times$ and $f = c \cdot m_{\alpha,F}$. Otherwise g has a root in $\overline{F}$, and every root of g is a root of f, so g is purely inseparable with the root α. By the induction hypothesis $g = a_0 \cdot m_{\alpha,F}^{k_0}$ with $k_0 \in \mathbb{N}$ and $a_0 \in F^\times$, hence

$$f = m_{\alpha,F} \cdot g = a_0 \cdot m_{\alpha,F}^{k_0+1}.$$

The case $n = 1$ is the case $\deg g = 0$. The argument uses only that F is a field. For finite F more is true: $m_{\alpha,F}$ is separable by Theorem 17.8, and its only root is α, so $m_{\alpha,F} = x - \alpha$ and $\alpha \in F$. ◊

Example 19.9. Show that no irreducible polynomial of degree at least 2 over $\mathbb{F}_2$ has an even number of nonzero terms.

Solution. Let $f = x^k + \varepsilon_{k-1}x^{k-1} + \cdots + \varepsilon_1 x + \varepsilon_0$ with $k \geq 2$ and $\varepsilon_i \in \{0, 1\}$. Then $f(1)$ is the number of nonzero terms modulo 2. If this number is even, then $f(1) = 0$, so $x - 1$ divides f, and since $\deg f \geq 2$, f is reducible. The degree condition is needed: $x + 1$ is irreducible with two nonzero terms. ◊

Example 19.10. Let q be a prime power and $d \in \mathbb{N}$. Prove that $x^{q^d} - x$ is the product of all monic irreducible $f \in \mathbb{F}_q[x]$ with $\deg f \mid d$, each with multiplicity 1. Design an algorithm that tests whether a polynomial over $\mathbb{F}_q$ is irreducible.

Solution. Let $P = x^{q^d} - x$. Since $P' = q^d x^{q^d-1} - 1 = -1$, $\gcd(P, P') = 1$, so by Proposition 17.6 P has no multiple roots and no irreducible factor of multiplicity greater than 1. As P is monic, it is the product of its distinct monic irreducible factors. It remains to show that a monic irreducible $f \in \mathbb{F}_q[x]$ divides P if and only if $\deg f \mid d$. Let $r = \deg f$.

Suppose $r \mid d$. In the field $\mathbb{F}_q[x]/(f)$, which has q^r elements (Remark 14.5), the class α of x is a root of f and satisfies $\alpha^{q^r} = \alpha$ by Theorem 14.2. Iterating d/r times gives $\alpha^{q^d} = \alpha$, so $P(\alpha) = 0$, and $f = m_{\alpha,\mathbb{F}_q}$ divides P.

Conversely, suppose $f \mid P$. Let E be a splitting field of P over $\mathbb{F}_q$ and $R \subseteq E$ the set of roots of P; it has exactly q^d elements, since the roots are simple. R is a subfield, as in the proof of Lemma 14.6, and it contains $\mathbb{F}_q$. A root $\alpha \in E$ of f lies in R, so $\mathbb{F}_q(\alpha) \subseteq R$. Then R is a vector space over $\mathbb{F}_q(\alpha)$, which has q^r elements, so $q^d = (q^r)^s$ for some $s \in \mathbb{N}$, and $r \mid d$.

Algorithm. Let $f \in \mathbb{F}_q[x]$ be monic of degree $n \geq 1$.

1) If $x^{q^n} \not\equiv x \pmod f$, answer NO.
2) If $\gcd\left(f, x^{q^{n/r}} - x\right) \neq 1$ for some prime $r \mid n$, answer NO.
3) Otherwise answer YES.

The powers $x^{q^k} \bmod f$ are computed by repeated q-th powering modulo f.

Correctness. Let f be irreducible. By the first part $f \mid x^{q^n} - x$, so step 1 passes. For a prime $r \mid n$, $\gcd\left(f, x^{q^{n/r}} - x\right)$ is 1 or f, and it is not f, since $n \nmid n/r$. So the answer is YES. Conversely, suppose the answer is YES. By step 1, $f \mid x^{q^n} - x$, so by the first part f is a product of distinct monic irreducible factors whose degrees divide n. If f were reducible, it would have a factor g of degree $e < n$ with $e \mid n$. Choose a prime r dividing n/e; then $e \mid n/r$, so $g \mid x^{q^{n/r}} - x$ and step 2 answers NO. Hence f is irreducible.

Step 2 cannot be replaced by the congruence test $x^{q^{n/r}} \equiv x \pmod f$. Over $\mathbb{F}_2$, $f = x(x^2 + x + 1)(x^3 + x + 1)$ has degree 6 and passes step 1, since all factor degrees divide 6. Neither $x^{2^3} \equiv x$ nor $x^{2^2} \equiv x \pmod f$ holds, because $2 \nmid 3$ and $3 \nmid 2$; yet f is reducible. ◊

Example 19.11. Give fields $F \subseteq K \subseteq L$ such that $L \mid K$ and $K \mid F$ are normal, but $L \mid F$ is not normal.

Solution. Take $\mathbb{Q} \subseteq \mathbb{Q}\left(\sqrt{2}\right) \subseteq \mathbb{Q}\left(\sqrt[4]{2}\right)$, where $\sqrt[4]{2}$ is the positive real fourth root of 2 and $\sqrt{2} = \sqrt[4]{2}^2$.

First, $x^4 - 2$ is irreducible over $\mathbb{Q}$. Its complex roots are $\pm a$ and $\pm ia$ with $a = \sqrt[4]{2}$, and none is rational, since $v_2(z^4) = 4v_2(z) \neq 1$. A monic quadratic factor over $\mathbb{Q}$ would be $(x - r)(x - r')$ for two of these roots, with constant term $rr' \in \left\{-\sqrt{2}, \sqrt{2}, \pm i\sqrt{2}\right\}$, none of which is rational. Hence $\left[\mathbb{Q}\left(\sqrt[4]{2}\right) : \mathbb{Q}\right] = 4$, and by the tower law $\left[\mathbb{Q}\left(\sqrt[4]{2}\right) : \mathbb{Q}\left(\sqrt{2}\right)\right] = 2$.

$\mathbb{Q}\left(\sqrt{2}\right) \mid \mathbb{Q}$ and $\mathbb{Q}\left(\sqrt[4]{2}\right) \mid \mathbb{Q}\left(\sqrt{2}\right)$ have degree 2, so both are normal (Example 17.17); the relevant polynomials are $x^2 - 2$ and $x^2 - \sqrt{2}$. But $\mathbb{Q}\left(\sqrt[4]{2}\right) \mid \mathbb{Q}$ is not normal: the irreducible $x^4 - 2$ has the root $\sqrt[4]{2} \in \mathbb{Q}\left(\sqrt[4]{2}\right)$, while its roots $\pm i\sqrt[4]{2}$ are not real and so do not lie in $\mathbb{Q}\left(\sqrt[4]{2}\right) \subseteq \mathbb{R}$. ◊

Example 19.12. Prove that $\mathbb{Q}\left(\sqrt[3]{3}\right) \mid \mathbb{Q}$ is not normal.

Solution. $x^3 - 3$ is irreducible over $\mathbb{Q}$, since it has degree 3 and no rational root ($z^3 = 3$ would give $3v_3(z) = 1$). Its roots are $\sqrt[3]{3}$, $\varepsilon\sqrt[3]{3}$, $\varepsilon^2\sqrt[3]{3}$, where ε is a primitive third root of unity. The root $\varepsilon\sqrt[3]{3}$ is not real, so $\varepsilon\sqrt[3]{3} \notin \mathbb{Q}\left(\sqrt[3]{3}\right) \subseteq \mathbb{R}$. ◊

20. Cyclotomic fields

Definition 20.1 (Cyclotomic extension). Let F be a field and $n \in \mathbb{N}$ with $\operatorname{char} F \nmid n$. A *cyclotomic extension* of F is a splitting field E of $x^n - 1$ over F.

Proposition 20.2. Let E be a cyclotomic extension of F for $x^n - 1$. The roots of $x^n - 1$ in E, the *n-th roots of unity*, form a cyclic subgroup μ_n of order n of $E^\times$. A generator of μ_n is a *primitive n-th root of unity*, denoted ζ_n, and $E = F(\zeta_n)$.

Proof. The derivative of $x^n - 1$ is nx^{n-1}, which is nonzero because $\operatorname{char} F \nmid n$ and vanishes only at 0. Hence $\gcd(x^n - 1, nx^{n-1}) = 1$, and $x^n - 1$ has n distinct roots in E by Proposition 17.6. They form a subgroup of $E^\times$, since $(\alpha\beta^{-1})^n = \alpha^n\beta^{-n} = 1$. A finite subgroup of the multiplicative group of a field is cyclic: the second paragraph of the proof of Theorem 14.2 uses only that a polynomial $x^d - 1$ has at most d roots in a field. So $\mu_n = \{1, \zeta_n, \ldots, \zeta_n^{n-1}\}$, and E, being generated over F by μ_n, equals $F(\zeta_n)$. ■

Example 20.3. For a prime p, $\mathbb{Q}(\zeta_p)$ is a cyclotomic extension of $\mathbb{Q}$.

Proposition 20.4. Let $E = F(\zeta_n)$ be a cyclotomic extension. For $\sigma \in \operatorname{Gal}(E \mid F)$ there is $r_\sigma \in \mathbb{Z}$ with $\gcd(r_\sigma, n) = 1$ and $\sigma(\zeta_n) = \zeta_n^{r_\sigma}$, and $\sigma \mapsto r_\sigma \bmod n$ is an injective group homomorphism $\operatorname{Gal}(E \mid F) \to U_n$, where $U_n = (\mathbb{Z}/n\mathbb{Z})^\times$. In particular $\operatorname{Gal}(E \mid F)$ is abelian. If $\operatorname{char} F = 0$ or F is finite, then $E \mid F$ is Galois and $[E : F] = |\operatorname{Gal}(E \mid F)|$ divides $\varphi(n)$.

Proof. By Theorem 17.12, σ maps μ_n into itself, and it preserves orders of elements, since it is an injective homomorphism of $E^\times$. So $\sigma(\zeta_n)$ is a primitive n-th root of unity, $\sigma(\zeta_n) = \zeta_n^{r_\sigma}$ with $\gcd(r_\sigma, n) = 1$, and r_σ is determined modulo n. For $\sigma, \tau \in \operatorname{Gal}(E \mid F)$,

$$(\tau\sigma)(\zeta_n) = \tau(\zeta_n^{r_\sigma}) = \tau(\zeta_n)^{r_\sigma} = \zeta_n^{r_\tau r_\sigma},$$

so the map is a homomorphism. It is injective, since σ is determined by $\sigma(\zeta_n)$ on $E = F(\zeta_n)$.

If $\operatorname{char} F = 0$ or F is finite, every irreducible polynomial over F is separable by Lemma 17.7 and Theorem 17.8, so $E \mid F$ is separable. It is normal by Theorem 17.21, being a splitting field. Hence it is Galois, and the last statement follows from Proposition 18.3 and Lagrange's theorem, as $|U_n| = \varphi(n)$. ■

From now on $F = \mathbb{Q}$ and $\zeta_n = e^{2\pi i/n} \in \mathbb{C}$. The primitive n-th roots of unity in $\mathbb{C}$ are ζ_n^r with $1 \le r \le n$ and $\gcd(r, n) = 1$.

Definition 20.5 (Cyclotomic polynomial). The *n-th cyclotomic polynomial* is

$$\Phi_n(x) = \prod_{1 \le r \le n,\ \gcd(r,n)=1} (x - \zeta_n^r), \quad \deg \Phi_n = \varphi(n).$$

Proposition 20.6. For every $n \in \mathbb{N}$, $x^n - 1 = \prod_{d \mid n} \Phi_d(x)$.

Proof. Both sides are monic with simple roots, so it suffices to compare roots. If $\zeta^n = 1$ and ζ has order d in $\mathbb{C}^\times$, then $d \mid n$ and ζ is a primitive d-th root of unity, that is, a root of Φ_d. Conversely, if $d \mid n$, every root ζ of Φ_d satisfies $\zeta^n = \left(\zeta^d\right)^{n/d} = 1$. Since every root of unity has exactly one order, the roots of the Φ_d, $d \mid n$, are pairwise distinct. ∎

Example 20.7. If $n = p$ is prime, then $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1$, by Proposition 20.6 with $\Phi_1(x) = x - 1$.

Proposition 20.8. $\Phi_n \in \mathbb{Z}[x]$ for every $n \in \mathbb{N}$.

Proof. Induction on n. For $n = 1$, $\Phi_1(x) = x - 1$. Let $n > 1$ and assume $\Phi_k \in \mathbb{Z}[x]$ for every $k \le n - 1$. By Proposition 20.6,

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x) = \Phi_n(x) \prod_{d \mid n,\ d \ne n} \Phi_d(x), \quad \text{so} \quad \Phi_n(x) = \frac{x^n - 1}{\prod_{d \mid n,\ d \ne n} \Phi_d(x)}.$$

The denominator is monic and lies in $\mathbb{Z}[x]$ by the induction hypothesis. Division with remainder by a monic polynomial in $\mathbb{Z}[x]$ stays in $\mathbb{Z}[x]$, and the remainder is 0, so $\Phi_n \in \mathbb{Z}[x]$. ∎

Remark 20.9 (Recall). *Gauss's lemma.* If $h \in \mathbb{Z}[x]$ is monic and $h = fg$ with monic $f, g \in \mathbb{Q}[x]$, then $f, g \in \mathbb{Z}[x]$.

Theorem 20.10. Φ_n is irreducible over $\mathbb{Q}$. Consequently $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$ and $\mathrm{Gal}(\mathbb{Q}(\zeta_n) \mid \mathbb{Q}) \cong U_n$.

Proof. Let f be the minimal polynomial over $\mathbb{Q}$ of a primitive n-th root of unity θ. Then f is monic and $f \mid x^n - 1$, so $x^n - 1 = fg$ with $f, g \in \mathbb{Z}[x]$ monic, by Remark 20.9. We show that θ^p is a root of f for every prime $p \nmid n$.

Suppose $f(\theta^p) \ne 0$. Since $(\theta^p)^n - 1 = 0 = f(\theta^p) g(\theta^p)$, we get $g(\theta^p) = 0$. So θ is a root of $g(x^p)$, and $f(x) \mid g(x^p)$ in $\mathbb{Q}[x]$, hence $g(x^p) = f(x)h(x)$ with $h \in \mathbb{Z}[x]$ by Remark 20.9. Reduce modulo p and work in $\mathbb{F}_p[x]$; for $q = \sum a_i x^i \in \mathbb{Z}[x]$ write $\overline{q} = \sum (a_i \bmod p) x^i$. By Theorem 14.4 and $a^p = a$ for $a \in \mathbb{F}_p$,

$$\overline{g}(x^p) = \overline{g}(x)^p = \overline{f}(x)\overline{h}(x) \in \mathbb{F}_p[x].$$

Hence any irreducible factor of $\overline{f}$ divides $\overline{g}^p$, hence $\overline{g}$, so $\overline{f}$ and $\overline{g}$ have a common factor. Reducing $x^n - 1 = fg$ gives $x^n - 1 = \overline{f}\,\overline{g}$ in $\mathbb{F}_p[x]$, so $x^n - 1$ has a multiple root over $\mathbb{F}_p$. But its derivative nx^{n-1} is nonzero in $\mathbb{F}_p[x]$, since $p \nmid n$, and $\gcd(x^n - 1, nx^{n-1}) = 1$, contradicting Proposition 17.6.

Since θ^p is again a primitive n-th root of unity and f, being monic and irreducible with the root θ^p, is also its minimal polynomial, the argument applies to θ^p in place of θ. Every primitive n-th root of unity is ζ_n^r with $\gcd(r,n)=1$, and writing $r = p_1 \cdots p_k$ as a product of primes not dividing n, it is obtained from ζ_n by successive prime powers. Hence all primitive n-th roots of unity are roots of $f = m_{\zeta_n,\mathbb{Q}}$, so Φ_n, whose roots are simple, divides f in $\mathbb{Q}[x]$. Since f is irreducible and Φ_n is monic and nonconstant, $f = \Phi_n$. The last statement follows from Proposition 20.4, since $|\operatorname{Gal}(\mathbb{Q}(\zeta_n) \mid \mathbb{Q})| = \varphi(n) = |U_n|$. ∎

21. Ruler and compass constructions

The two admissible operations in the plane $\mathbb{R}^2$ are:

1) *ruler*: draw the straight line through two given points;

2) *compass*: draw the circle with a given point as centre passing through a given point.

Definition 21.1 (Constructible point). Let $P_0 \subseteq \mathbb{R}^2$. A point $r \in \mathbb{R}^2$ is *constructible* from P_0 if there is a sequence $r_1, \dots, r_n = r$ of points such that for each $i = 1, \dots, n$, the point r_i is an intersection point of two distinct lines or circles drawn by operations 1 or 2 from points of $P_{i-1} = P_0 \cup \{r_1, \dots, r_{i-1}\}$. A real number x is *constructible* if $(x, 0)$ is constructible from $\{(0,0), (1,0)\}$.

With each stage we associate a subfield of $\mathbb{R}$. Let $\mathbb{K}_0$ be the subfield of $\mathbb{R}$ generated by the coordinates of the points of P_0. If $r_i = (x_i, y_i)$, put inductively $\mathbb{K}_i = \mathbb{K}_{i-1}(x_i, y_i)$. Then $\mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \cdots \subseteq \mathbb{K}_n \subseteq \mathbb{R}$.

Lemma 21.2. For each i, the coordinates x_i and y_i are zeros of polynomials of degree at most 2 over $\mathbb{K}_{i-1}$.

Proof. All points of P_{i-1} have coordinates in $\mathbb{K}_{i-1}$. The line through $A = (p, q)$ and $B = (r, s)$ is

$$(s - q)(x - p) = (r - p)(y - q),$$

a linear equation with coefficients in $\mathbb{K}_{i-1}$. The circle with centre $C = (t, u)$ through $D = (v, w)$ is

$$(x - t)^2 + (y - u)^2 = \rho^2, \quad \rho^2 = (v - t)^2 + (w - u)^2 \in \mathbb{K}_{i-1},$$

that is, $x^2 + y^2 + ax + by + c = 0$ with $a, b, c \in \mathbb{K}_{i-1}$.

Two lines. The intersection point solves a 2×2 linear system over $\mathbb{K}_{i-1}$ with nonzero determinant, so $x_i, y_i \in \mathbb{K}_{i-1}$ by Cramer's rule.

A line and a circle. If the line is not vertical, it is $y = mx + k$ with $m, k \in \mathbb{K}_{i-1}$. Substituting into the circle gives a polynomial of degree 2 over $\mathbb{K}_{i-1}$ vanishing at x_i, and $y_i = mx_i + k$ satisfies the polynomial obtained by substituting $x = (y - k)/m$ if $m \neq 0$, or $y_i = k$ if $m = 0$. A vertical line $x = k$ gives $x_i = k$ and a quadratic equation for y_i.

Two circles. Subtracting the equations of two distinct circles $x^2 + y^2 + ax + by + c = 0$ and $x^2 + y^2 + a'x + b'y + c' = 0$ gives the linear equation $(a - a')x + (b - b')y + (c - c') = 0$. If $a = a'$

and $b = b'$, the circles are concentric and distinct and do not meet. Otherwise this is a line over $\mathbb{K}_{i-1}$ through every common point, and we are in the previous case. ∎

Theorem 21.3. If $r = (x, y)$ is constructible from $P_0 \subseteq \mathbb{R}^2$ and $\mathbb{K}_0$ is the subfield of $\mathbb{R}$ generated by the coordinates of the points of P_0, then $[\mathbb{K}_0(x) : \mathbb{K}_0]$ and $[\mathbb{K}_0(y) : \mathbb{K}_0]$ are powers of 2.

Proof. By Lemma 21.2, $[\mathbb{K}_{i-1}(x_i) : \mathbb{K}_{i-1}] \in \{1, 2\}$, and y_i is a zero of a polynomial of degree at most 2 over $\mathbb{K}_{i-1} \subseteq \mathbb{K}_{i-1}(x_i)$, so $[\mathbb{K}_{i-1}(x_i, y_i) : \mathbb{K}_{i-1}(x_i)] \in \{1, 2\}$. By the tower law

$$[\mathbb{K}_{i-1}(x_i, y_i) : \mathbb{K}_{i-1}] = [\mathbb{K}_{i-1}(x_i, y_i) : \mathbb{K}_{i-1}(x_i)][\mathbb{K}_{i-1}(x_i) : \mathbb{K}_{i-1}] \in \{1, 2, 4\}.$$

Since $\mathbb{K}_i = \mathbb{K}_{i-1}(x_i, y_i)$, the tower law gives that $[\mathbb{K}_n : \mathbb{K}_0]$ is a power of 2. As $x = x_n \in \mathbb{K}_n$, $[\mathbb{K}_n : \mathbb{K}_0] = [\mathbb{K}_n : \mathbb{K}_0(x)][\mathbb{K}_0(x) : \mathbb{K}_0]$, so $[\mathbb{K}_0(x) : \mathbb{K}_0]$ divides a power of 2 and is itself a power of 2. The same holds for y. ∎

Below, $P_0 = \{(0, 0), (1, 0)\}$, so $\mathbb{K}_0 = \mathbb{Q}$. We use the elementary fact that if a segment of length ℓ has been constructed, so has $(\ell, 0)$: transfer the segment to the origin and intersect the circle of radius ℓ about $(0, 0)$ with the x-axis.

Theorem 21.4 (Duplication of the cube). The cube cannot be duplicated with ruler and compass: there is no construction from P_0 of the side of a cube whose volume is twice the volume of the unit cube.

Proof. The unit cube has volume 1. A cube of volume 2 has side $\alpha = \sqrt[3]{2}$, so a duplication would make $(\alpha, 0)$ constructible from P_0. By Theorem 21.3, $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ would be a power of 2. But α is a root of $x^3 - 2$, which is irreducible over $\mathbb{Q}$, since it has degree 3 and no rational root ($3v_2(z) = 1$ is impossible). So $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$, which is not a power of 2. ∎

Theorem 21.5 (Trisection of the angle). The angle $60°$ cannot be trisected with ruler and compass.

Proof. The angle $60°$ is constructible from P_0: the circles of radius 1 about $(0, 0)$ and $(1, 0)$ meet at $\left(1/2, \sqrt{3}/2\right)$. A trisection would make the point $(\cos 20°, \sin 20°)$, and hence $\alpha = \cos 20°$, constructible from P_0. Recall $\cos(3\beta) = 4\cos^3\beta - 3\cos\beta$. With $\beta = 20°$,

$$\frac{1}{2} = \cos 60° = 4\cos^3 20° - 3\cos 20° \iff 8\cos^3 20° - 6\cos 20° - 1 = 0,$$

so α is a root of $f(x) = 8x^3 - 6x - 1$. Putting $y = 2x$, $f(x) = y^3 - 3y - 1$. A rational root of the monic integer polynomial $y^3 - 3y - 1$ would be an integer dividing 1, but $1 - 3 - 1 = -3$ and $-1 + 3 - 1 = 1$. So $y^3 - 3y - 1$ has no rational root, and being cubic, it is irreducible over $\mathbb{Q}$; hence so is f. Therefore $[\mathbb{Q}(\cos 20°) : \mathbb{Q}] = 3$, not a power of 2, contradicting Theorem 21.3. ∎

Theorem 21.6 (Squaring the circle). The circle cannot be squared with ruler and compass: there is no construction of a square whose area equals that of a given circle.

Proof. Take the unit circle with centre $(0, 0)$, so P_0 is as above. A square of area π has side $\sqrt{\pi}$, so squaring the circle would make $(\sqrt{\pi}, 0)$ constructible. By Theorem 21.3, $[\mathbb{Q}(\sqrt{\pi}) : \mathbb{Q}]$ would be

a power of 2, in particular finite, so $\sqrt{\pi}$ and hence $\pi = (\sqrt{\pi})^2$ would be algebraic over $\mathbb{Q}$. This contradicts the following theorem. ∎

Theorem 21.7 (Lindemann) (cited)**.** π is transcendental over $\mathbb{Q}$.

Lemma 21.8. Let $t \in \mathbb{N}$. If $2^t + 1$ is prime, then t is a power of 2.

Proof. If t is odd, then $2 \equiv -1 \pmod 3$ gives $2^t \equiv (-1)^t = -1 \pmod 3$, so $3 \mid 2^t + 1$; as $2^t + 1$ is prime, $2^t + 1 = 3$ and $t = 1 = 2^0$. In general write $t = 2^k r$ with $k \geq 0$ and r odd. Then

$$2^t + 1 = \left(2^{2^k}\right)^r + 1 \equiv (-1)^r + 1 = 0 \pmod{2^{2^k} + 1},$$

so $2^{2^k} + 1$ divides the prime $2^t + 1$. Since $2^{2^k} + 1 > 1$, it equals $2^t + 1$, so $r = 1$ and $t = 2^k$. ∎

Definition 21.9 (Fermat numbers). The *Fermat numbers* are $F_n = 2^{2^n} + 1$, $n \geq 0$. A prime of this form is a *Fermat prime*. The first five, $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$, $F_4 = 65537$, are prime.

Definition 21.10. Let $n \geq 3$. The regular n-gon is *constructible* by ruler and compass if the point $(\cos(2\pi/n), \sin(2\pi/n))$ is constructible from $P_0 = \{(0,0), (1,0)\}$.

Theorem 21.11 (Regular n-gon). Let $n \geq 3$. The regular n-gon is constructible by ruler and compass if and only if $n = 2^k p_1 \cdots p_r$ with $k, r \geq 0$ and $p_1, \ldots, p_r$ pairwise distinct Fermat primes.

Proof. We prove necessity; sufficiency is cited below. Suppose the regular n-gon is constructible. Then $c = \cos(2\pi/n)$ is a coordinate of a constructible point, so $[\mathbb{Q}(c) : \mathbb{Q}]$ is a power of 2 by Theorem 21.3. Let $s = \sin(2\pi/n)$ and $\varepsilon_n = c + is = \zeta_n$. Since

$$(is)^2 = -s^2 = c^2 - 1 \in \mathbb{Q}(c),$$

the degree $[\mathbb{Q}(c)(is) : \mathbb{Q}(c)]$ is 1 or 2, so $[\mathbb{Q}(c, is) : \mathbb{Q}]$ is a power of 2. As $\varepsilon_n \in \mathbb{Q}(c, is)$, the tower law and Theorem 20.10 show that $\varphi(n) = [\mathbb{Q}(\varepsilon_n) : \mathbb{Q}]$ divides a power of 2, so $\varphi(n) = 2^m$ for some m.

Write $n = 2^k p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with distinct odd primes p_i and $\alpha_i \geq 1$. Then

$$\varphi(n) = \varphi\big(2^k\big)\big(p_1^{\alpha_1} - p_1^{\alpha_1 - 1}\big) \cdots \big(p_r^{\alpha_r} - p_r^{\alpha_r - 1}\big) = \varphi\big(2^k\big) p_1^{\alpha_1 - 1} \cdots p_r^{\alpha_r - 1} (p_1 - 1) \cdots (p_r - 1) = 2^m.$$

As the p_i are odd, $\alpha_i = 1$ and $p_i - 1 = 2^{\beta_i}$ for every i. By Lemma 21.8 each $p_i = 2^{\beta_i} + 1$ is a Fermat prime. ∎

Theorem 21.12 (Gauss) (cited)**.** If $n = 2^k p_1 \cdots p_r$ with pairwise distinct Fermat primes p_i, then the regular n-gon is constructible.

22. Worked exercises on Galois groups

Remark 22.1 (Recall). *Schönemann–Eisenstein criterion.* Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ and p a prime with $p \nmid a_n$, $p \mid a_i$ for $i < n$ and $p^2 \nmid a_0$. Then f is irreducible over $\mathbb{Q}$.

Example 22.2. Determine $G = \mathrm{Gal}\big(\mathbb{Q}\big(\sqrt[4]{2}, i\big) \mid \mathbb{Q}\big)$.

Solution. Let $a = \sqrt[4]{2} > 0$. Over $\mathbb{C}$,

$$x^4 - 2 = \big(x^2 + \sqrt{2}\big)\big(x^2 - \sqrt{2}\big) = (x-a)(x+a)(x-ia)(x+ia),$$

so $E = \mathbb{Q}(a, i)$ is the splitting field of $x^4 - 2$, and $E \mid \mathbb{Q}$ is normal by Theorem 17.21 and Galois since $\mathrm{char}\ \mathbb{Q} = 0$. By Remark 22.1 with $p = 2$, $x^4 - 2$ is irreducible over $\mathbb{Q}$, so $[\mathbb{Q}(a) : \mathbb{Q}] = 4$. Since $i \notin \mathbb{Q}(a) \subseteq \mathbb{R}$, $[E : \mathbb{Q}(a)] = 2$, and $|G| = [E : \mathbb{Q}] = 8$ by Proposition 18.3. By the tower law $[E : \mathbb{Q}(i)] = 4$, so $x^4 - 2$ is the minimal polynomial of a over $\mathbb{Q}(i)$.

Let τ be complex conjugation restricted to E; it maps E onto E by Proposition 17.20, so $\tau \in G$, with $\tau(a) = a$, $\tau(i) = -i$. By Lemma 17.19, the $\mathbb{Q}(i)$-embedding $\mathbb{Q}(i)(a) \to \mathbb{C}$ with $a \mapsto ia$ exists, and it maps E onto E by Proposition 17.20; call it σ. So $\sigma(a) = ia$, $\sigma(i) = i$, and

$$\sigma^2(a) = \sigma(ia) = -a, \quad \sigma^3(a) = \sigma(-a) = -ia, \quad \sigma^4(a) = a.$$

Hence σ has order 4, τ has order 2, and the eight elements $1, \sigma, \sigma^2, \sigma^3, \tau, \sigma\tau, \sigma^2\tau, \sigma^3\tau$ are pairwise distinct, since they differ on a or on i. So

$$G = \{1, \sigma, \sigma^2, \sigma^3, \tau, \sigma\tau, \sigma^2\tau, \sigma^3\tau\}.$$

Checking on the generators, $\tau^{-1}\sigma\tau(a) = \tau\sigma(a) = \tau(ia) = -ia = \sigma^3(a)$ and $\tau^{-1}\sigma\tau(i) = i = \sigma^3(i)$, so $\tau^{-1}\sigma\tau = \sigma^3 = \sigma^{-1}$. Therefore $G \cong D_4 = \langle \sigma, \tau \mid \sigma^4 = \tau^2 = 1, \tau\sigma\tau = \sigma^{-1} \rangle$, the dihedral group of order 8. ◊

Example 22.3. Determine $G = \mathrm{Gal}\Big(\mathbb{Q}\Big(\sqrt{2+\sqrt{2}}\Big) \mid \mathbb{Q}\Big)$.

Solution. Let $\alpha = \sqrt{2+\sqrt{2}}$. Then

$$\alpha^2 = 2 + \sqrt{2} \iff \alpha^2 - 2 = \sqrt{2} \implies (\alpha^2 - 2)^2 = 2 \iff \alpha^4 - 4\alpha^2 + 2 = 0.$$

By Remark 22.1 with $p = 2$, $x^4 - 4x^2 + 2$ is irreducible, so $m_{\alpha,\mathbb{Q}} = x^4 - 4x^2 + 2$ and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Its roots are $\pm\sqrt{2 \pm \sqrt{2}}$. Let $\beta = \sqrt{2 - \sqrt{2}}$; since $\alpha\beta = \sqrt{4-2} = \sqrt{2}$, we get $\beta = \sqrt{2}/\alpha = (\alpha^2 - 2)/\alpha \in \mathbb{Q}(\alpha)$. So all roots lie in $\mathbb{Q}(\alpha)$, which is their splitting field; $\mathbb{Q}(\alpha) \mid \mathbb{Q}$ is normal and separable, hence Galois, and $|G| = 4$.

Let $\sigma \in G$ with $\sigma(\alpha) = \beta$; it exists by Theorem 17.13 and Proposition 17.20, since β is a root of $m_{\alpha,\mathbb{Q}}$. Then

$$2 + \sigma\big(\sqrt{2}\big) = \sigma\big(2 + \sqrt{2}\big) = \sigma(\alpha^2) = \sigma(\alpha)^2 = \beta^2 = 2 - \sqrt{2},$$

so $\sigma\big(\sqrt{2}\big) = -\sqrt{2}$, and

$$\sigma^2(\alpha) = \sigma(\beta) = \sigma\left(\frac{\sqrt{2}}{\alpha}\right) = \frac{\sigma\left(\sqrt{2}\right)}{\sigma(\alpha)} = \frac{-\sqrt{2}}{\beta} = -\alpha.$$

Hence $\sigma^2 \neq \mathrm{id}$, σ has order 4, and $G = \langle\sigma\rangle \cong \mathbb{Z}_4$. ◊

Example 22.4. Determine the Galois groups of $x^3 + 2$ and $x^4 - 7x^2 + 10$ over $\mathbb{Q}$.

Solution. $x^3 + 2$. Since $x^3 + 2 = x^3 + \left(\sqrt[3]{2}\right)^3 = \left(x + \sqrt[3]{2}\right)\left(x^2 - \sqrt[3]{2}x + \sqrt[3]{4}\right)$, the roots are

$$x_1 = -\sqrt[3]{2}, \quad x_2 = \frac{\sqrt[3]{2}}{2} + \frac{i\sqrt{3}\sqrt[3]{2}}{2}, \quad x_3 = \frac{\sqrt[3]{2}}{2} - \frac{i\sqrt{3}\sqrt[3]{2}}{2}.$$

The splitting field is $E = \mathbb{Q}\left(\sqrt[3]{2}, i\sqrt{3}\right)$. By Remark 22.1 with $p = 2$, $x^3 + 2$ is irreducible, so $\left[\mathbb{Q}\left(\sqrt[3]{2}\right) : \mathbb{Q}\right] = 3$, and $i\sqrt{3} \notin \mathbb{Q}\left(\sqrt[3]{2}\right) \subseteq \mathbb{R}$ gives $[E : \mathbb{Q}] = 6 = |G|$ by Proposition 18.3. Each $\sigma \in G$ permutes $\{x_1, x_2, x_3\}$ by Theorem 17.12 and is determined by this permutation, since E is generated by the roots. This gives an injective homomorphism $G \to S_3$, and as $|S_3| = 3! = 6 = |G|$, $G \cong S_3$.

$x^4 - 7x^2 + 10 = (x^2 - 5)(x^2 - 2)$. The roots are $\pm\sqrt{5}, \pm\sqrt{2}$, and the splitting field is $E = \mathbb{Q}\left(\sqrt{2}, \sqrt{5}\right)$. Here $\sqrt{5} \notin \mathbb{Q}\left(\sqrt{2}\right)$: if $\sqrt{5} = u + v\sqrt{2}$ with $u, v \in \mathbb{Q}$, then $5 = u^2 + 2v^2 + 2uv\sqrt{2}$ forces $uv = 0$, and $v = 0$ or $u = 0$ would make $\sqrt{5}$ or $\sqrt{5/2}$ rational. Hence $[E : \mathbb{Q}] = 4 = |G|$. As in the example on $\mathbb{Q}\left(i, \sqrt{5}\right)$ in Section 18, the elements of G are given by sign changes:

$$\sigma : \sqrt{5} \mapsto -\sqrt{5},\ \sqrt{2} \mapsto \sqrt{2}; \quad \tau : \sqrt{2} \mapsto -\sqrt{2},\ \sqrt{5} \mapsto \sqrt{5}; \quad \sigma\tau : \sqrt{2} \mapsto -\sqrt{2},\ \sqrt{5} \mapsto -\sqrt{5}.$$

Each satisfies $\sigma^2 = \tau^2 = (\sigma\tau)^2 = \mathrm{id}$, so $G = \{\mathrm{id}, \sigma, \tau, \sigma\tau\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, the Klein four-group. ◊

Definition 22.5 (Galois group of a polynomial). Let F be a field and $f \in F[x]$ nonconstant. The *Galois group of f over F* is $\mathrm{Gal}(E \mid F)$, where E is a splitting field of f over F.

Example 22.6. Let F be a field, $f \in F[x]$ and $c \in F$. Show that the Galois groups of $f(x)$ and $f(x + c)$ over F are isomorphic.

Solution. Let $\alpha_1, \ldots, \alpha_n$ be the roots of f in an algebraic closure $\overline{F}$. Then $f(\beta + c) = 0$ if and only if $\beta + c = \alpha_i$ for some i, so the roots of $f(x + c)$ are $\alpha_1 - c, \ldots, \alpha_n - c$. Their splitting fields in $\overline{F}$ are

$$L = F(\alpha_1, \ldots, \alpha_n), \quad L' = F(\alpha_1 - c, \ldots, \alpha_n - c),$$

and $L = L'$, since $c \in F$. So both Galois groups equal $\mathrm{Gal}(L \mid F)$. ◊

Example 22.7. Let F be a finite field with $\mathrm{char}\ F = p$ and $|F| = p^n$. Show that $F \mid \mathbb{F}_p$ is a Galois extension whose Galois group is generated by the Frobenius automorphism $\sigma(x) = x^p$.

Solution. Let $f(x) = x^{p^n} - x$. Then $f'(x) = p^n x^{p^n - 1} - 1 = -1$, so f is separable by Proposition 17.6. By Theorem 14.2 every element of F is a root of f, and f has at most p^n roots, so F is exactly the set of roots of f and is a splitting field of f over $\mathbb{F}_p$. Hence $F \mid \mathbb{F}_p$ is normal (Theorem 17.21) and separable (Theorem 17.8), so it is a Galois extension.

Let $G = \mathrm{Gal}(F \mid \mathbb{F}_p)$. By Theorem 17.2 with $q = p$, $\sigma \in G$, so $\mathbb{F}_p \subseteq \mathcal{F}(\langle\sigma\rangle)$. The fixed field of $\langle\sigma\rangle$ consists of the roots of $x^p - x$ in F, so it has at most p elements; hence $\mathcal{F}(\langle\sigma\rangle) = \mathbb{F}_p$. By Theorem 18.11, $\langle\sigma\rangle = \mathcal{G}(\mathbb{F}_p) = G$. This reproves part of Theorem 17.2 for $q = p$. ◊

Remark 22.8 (Recall). For angles, measured in degrees: an angle α is *constructible* if $(\cos\alpha, \sin\alpha)$ is constructible from $P_0 = \{(0,0),(1,0)\}$. Sums and differences of constructible angles are constructible, and so is half of a constructible angle.

Example 22.9. For which $n \in \mathbb{N}$ is the angle $n°$ constructible?

Solution. We show that $n°$ is constructible if and only if $3 \mid n$. By Theorem 21.11 the regular 5-gon is constructible, 5 being a Fermat prime, so $72°$ is constructible. The angle $60°$ is constructible, hence so are $12° = 72° - 60°$, $3°$ (bisecting twice) and every multiple $3k°$.

Conversely, if $(3k+1)°$ were constructible, so would be $1° = (3k+1)° - 3k°$. If $(3k+2)°$ were constructible, so would be $1° = 3(k+1)° - (3k+2)°$. Then $20°$, a sum of twenty copies of $1°$, would be constructible, and the angle $60°$ could be trisected, contradicting the trisection theorem of Section 21. ◊

Example 22.10. [p4: statement of the exercise not recorded; the solution concerns an isosceles triangle with legs a, base $2b$, height f_a and inscribed circle of radius r]

Solution.

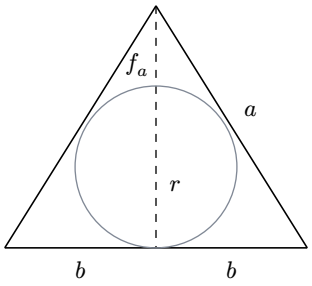


Figure 22.1. *Isosceles triangle with legs a, base $2b$, height f_a and inradius r.*

The height is $f_a = \sqrt{a^2 - b^2}$ and the semiperimeter is $S = (2a + 2b)/2 = a + b$. Computing the area in two ways,

$$\frac{2b \cdot f_a}{2} = b f_a, \quad rS = r(a+b),$$

where r is the radius of the inscribed circle. Hence $b\sqrt{a^2 - b^2} = r(a+b)$, so $b^2(a^2 - b^2) = r^2(a+b)^2$, and dividing by $a + b > 0$,

$$b^2(a-b) = r^2(a+b) \iff b^3 - ab^2 + r^2 a + r^2 b = 0.$$

With $r = 1$, b is a root of $f(x) = x^3 - ax^2 + x + a$.

Whether f is irreducible over $\mathbb{Q}$ depends on a: for $a = 10/3$, $f(2) = 8 - 40/3 + 2 + 10/3 = 0$. Take $a = 5$. Then $f(x) = x^3 - 5x^2 + x + 5$ has no rational root, since $f(1) = 2$, $f(-1) = -2$, $f(5) =$

10, $f(-5) = -150$, so it is irreducible over $\mathbb{Q}$. It has roots in $(1,2)$ and $(4,5)$, since $f(1) = 2 > 0 > -5 = f(2)$ and $f(4) = -7 < 0 < 10 = f(5)$; each gives an isosceles triangle with leg 5 and inradius 1. For both, $[\mathbb{Q}(b) : \mathbb{Q}] = 3$, which is not a power of 2, so b is not constructible from the data $a = 5$, $r = 1$ by Theorem 21.3. ◊

23. Solvable groups

Remark 23.1 (Recall). For a group G, the *commutator subgroup* $G' = \langle g^{-1}h^{-1}gh : g, h \in G \rangle$ is a normal subgroup of G. For $N \trianglelefteq G$, the quotient G/N is abelian if and only if $G' \leq N$. If $\varphi : G \to H$ is a surjective homomorphism, then $\varphi(G') = H'$.

Definition 23.2 (Derived series, solvable group). The *derived series* (commutator series) of a group G is $G^{(0)} = G$, $G^{(n+1)} = \left(G^{(n)}\right)'$ for $n \geq 0$. The group G is *solvable* if $G^{(k)} = 1$ for some $k \in \mathbb{N}$.

Definition 23.3 (Normal series). A *normal series* of G is a chain $1 = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_n = G$ of subgroups, each normal in the next. Its *factors* are the groups H_{i+1}/H_i.

Proposition 23.4. A group G is solvable if and only if it has a normal series with abelian factors.

Proof. ($\Rightarrow$) If $G^{(k)} = 1$, then $1 = G^{(k)} \trianglelefteq G^{(k-1)} \trianglelefteq \cdots \trianglelefteq G^{(0)} = G$, and $G^{(i)}/G^{(i+1)} = G^{(i)}/\left(G^{(i)}\right)'$ is abelian.

($\Leftarrow$) Let $1 = H_0 \trianglelefteq \cdots \trianglelefteq H_n = G$ have abelian factors. We show $G^{(i)} \leq H_{n-i}$ by induction on i. For $i = 0$ this is $G = H_n$. If $G^{(i)} \leq H_{n-i}$, then $G^{(i+1)} = \left(G^{(i)}\right)' \leq (H_{n-i})' \leq H_{n-i-1}$, the last inclusion because H_{n-i}/H_{n-i-1} is abelian. For $i = n$ we get $G^{(n)} \leq H_0 = 1$. ■

Proposition 23.5. If G and H are solvable groups, then so is $G \times H$.

Proof. Commutators in $G \times H$ are formed componentwise, so $(G \times H)' = G' \times H'$, and by induction $(G \times H)^{(i)} = G^{(i)} \times H^{(i)}$. If $G^{(k)} = 1$ and $H^{(l)} = 1$, then $(G \times H)^{(\max(k,l))} = 1$. ■

Proposition 23.6. A finite group G is solvable if and only if it has a normal series whose factors are cyclic.

Proof. Cyclic groups are abelian, so one direction is Proposition 23.4. Conversely, take a normal series with abelian factors. It suffices to refine each step $H_i \trianglelefteq H_{i+1}$ with $A = H_{i+1}/H_i$ finite abelian. By induction on $|A|$, A has a chain $1 = A_0 \leq A_1 \leq \cdots \leq A_s = A$ with cyclic factors: for $A \neq 1$ choose $1 \neq g \in A$, put $A_1 = \langle g \rangle$, and lift a chain of A/A_1, which is smaller. Every subgroup of the abelian group A is normal in it, and the preimages of the A_j in H_{i+1} form a chain from H_i to H_{i+1}, each normal in the next, with factors A_{j+1}/A_j. ■

Definition 23.7 (Alternating group). A_n is the group of even permutations of $\{1, \dots, n\}$; $|A_n| = n!/2$ for $n \geq 2$. A group is *simple* if its only normal subgroups are 1 and itself.

Theorem 23.8 (cited)**.** A_n is simple for $n \geq 5$.

Theorem 23.9. S_n is solvable if and only if $n \leq 4$.

Proof. S_1 and S_2 are abelian. $1 \trianglelefteq A_3 \trianglelefteq S_3$ has factors of orders 3 and 2. For S_4, let $V = \{\mathrm{id}, (12)(34), (13)(24), (14)(23)\}$, which is normal in S_4, being the union of id and a conjugacy class. Then $1 \trianglelefteq V \trianglelefteq A_4 \trianglelefteq S_4$ has factors $V \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, A_4/V of order 3, and S_4/A_4 of order 2, all abelian. By Proposition 23.4 these groups are solvable.

Let $n \geq 5$. Since $(123)(345) \neq (345)(123)$, A_n is not abelian, so $A_{n'} \neq 1$. As $A_{n'}$ is normal in the simple group A_n, $A_{n'} = A_n$, and the derived series of A_n is constant. Since $A_n \leq S_n$, induction gives $A_n = A_n^{(i)} \leq S_n^{(i)}$ for every i, so the derived series of S_n never reaches 1, and S_n is not solvable. ∎

24. Solvability by radicals

Throughout, all fields considered over a field $\mathbb{K}$ lie in a fixed algebraic closure Ω of $\mathbb{K}$.

Definition 24.1 (Radical extension). An extension $\mathbb{K} \subseteq \mathbb{K}(b)$ with $b^n = a \in \mathbb{K}$ for some $n \in \mathbb{N}$ is a *radical extension*; we write $\mathbb{K}(b) = \mathbb{K}(\sqrt[n]{a})$.

Remark 24.2. The notation $\sqrt[n]{a} = b$ only says $b^n = a$, $b \in \Omega$, and is ambiguous: $x^n - a$ has several roots. When $\mathbb{K}$ contains the n-th roots of unity, $\mathbb{K}(\sqrt[n]{a})$ contains all roots of $x^n - a$, namely $\sqrt[n]{a} \cdot \zeta$ with $\zeta^n = 1$, so the field does not depend on the choice.

Definition 24.3 (Radical tower, root extension). A tower $\mathbb{K} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \cdots \subseteq \mathbb{K}_\ell$ in which every $\mathbb{K}_i \mid \mathbb{K}_{i-1}$ is a radical extension is a *radical tower*. Then $\mathbb{K}_\ell \mid \mathbb{K}$ is a *root extension*, and the elements of $\mathbb{K}_\ell$ are said to be *expressible by radicals* over $\mathbb{K}$.

Definition 24.4 (Solvable by radicals). A polynomial $f \in \mathbb{K}[x]$ is *solvable by radicals* over $\mathbb{K}$ if all its roots in Ω lie in a root extension of $\mathbb{K}$.

Definition 24.5 (Cyclic extension). A *cyclic extension* is a Galois extension $\mathbb{F} \mid \mathbb{K}$ whose Galois group $\mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$ is cyclic.

Lemma 24.6 (Dedekind). Let $\mathbb{F}$ be a field and $\sigma_1, \dots, \sigma_n$ pairwise distinct automorphisms of $\mathbb{F}$. If $c_1, \dots, c_n \in \mathbb{F}$ satisfy $c_1\sigma_1(x) + \cdots + c_n\sigma_n(x) = 0$ for all $x \in \mathbb{F}$, then $c_1 = \cdots = c_n = 0$.

Proof. Suppose not, and take a nontrivial relation with the least number of nonzero coefficients; renumbering, $c_1\sigma_1 + \cdots + c_k\sigma_k = 0$ with all $c_i \neq 0$. Then $k \geq 2$, since $\sigma_1(1) = 1 \neq 0$. Choose $y \in$

$\mathbb{F}$ with $\sigma_1(y) \neq \sigma_k(y)$. For all x, $\sum_i c_i\sigma_i(yx) = \sum_i c_i\sigma_i(y)\sigma_i(x) = 0$, and also $\sigma_k(y)\sum_i c_i\sigma_i(x) = 0$. Subtracting,

$$\sum_{i=1}^{k-1} c_i(\sigma_i(y) - \sigma_k(y))\sigma_i(x) = 0 \quad \text{for all } x \in \mathbb{F},$$

a relation with fewer nonzero terms whose first coefficient $c_1(\sigma_1(y) - \sigma_k(y))$ is nonzero. This contradicts minimality. ■

Theorem 24.7 (Kummer). Let $n \in \mathbb{N}$, let $\operatorname{char} \mathbb{K} = 0$, and assume that $\mathbb{K}$ contains the n-th roots of unity.

1) For $a \in \mathbb{K}$ let $\mathbb{F} = \mathbb{K}(\sqrt[n]{a})$. Then $\mathbb{F} \mid \mathbb{K}$ is cyclic of degree d for some $d \mid n$.

2) If $\mathbb{F} \mid \mathbb{K}$ is cyclic of degree n, then $\mathbb{F} = \mathbb{K}(\sqrt[n]{a})$ for some $a \in \mathbb{K}$.

Proof. By Proposition 20.2, $\mathbb{K}$ contains a primitive n-th root of unity ξ, and the n-th roots of unity are ξ^i, $0 \leq i \leq n-1$, pairwise distinct.

1) If $a = 0$, then $\mathbb{F} = \mathbb{K}$ and the claim holds with $d = 1$. Let $a \neq 0$ and $b = \sqrt[n]{a}$, so $b \neq 0$. The roots of $x^n - a$ are the n distinct elements $b\xi^i$, $0 \leq i \leq n-1$, all in $\mathbb{F}$. So $\mathbb{F}$ is the splitting field of $x^n - a$ over $\mathbb{K}$, hence normal (Theorem 17.21). Since $\operatorname{char} \mathbb{K} = 0$, $\mathbb{F} \mid \mathbb{K}$ is separable by Lemma 17.7, so it is Galois, and $|\operatorname{Gal}(\mathbb{F} \mid \mathbb{K})| = [\mathbb{F} : \mathbb{K}]$ by Proposition 18.3. Each $\sigma \in \operatorname{Gal}(\mathbb{F} \mid \mathbb{K})$ is determined by $\sigma(b) = b\xi^{i_\sigma}$, with $i_\sigma \in \mathbb{Z}_n$, by Theorem 17.12. The map $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K}) \to \mathbb{Z}_n$, $\sigma \mapsto i_\sigma$, is injective, and it is a homomorphism: since $\xi \in \mathbb{K}$ is fixed,

$$(\tau\sigma)(b) = \tau(b\xi^{i_\sigma}) = \tau(b)\xi^{i_\sigma} = b\xi^{i_\tau + i_\sigma}.$$

The map need not be surjective. Hence $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K})$ is isomorphic to a subgroup of $\mathbb{Z}_n$, that is, cyclic of some order $d \mid n$, and $[\mathbb{F} : \mathbb{K}] = d$.

2) Let $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K}) = \langle\sigma\rangle$ with $|\sigma| = n$. For $\alpha \in \mathbb{F}$ define the *Lagrange resolvent*

$$(\alpha, \xi) = \alpha + \xi\sigma(\alpha) + \xi^2\sigma^2(\alpha) + \cdots + \xi^{n-1}\sigma^{n-1}(\alpha).$$

The automorphisms $1, \sigma, \ldots, \sigma^{n-1}$ are pairwise distinct, so by Lemma 24.6 the map $1 + \xi\sigma + \cdots + \xi^{n-1}\sigma^{n-1}$ is not identically 0; choose $\alpha \in \mathbb{F}$ with $\rho = (\alpha, \xi) \neq 0$. Since $\xi \in \mathbb{K}$ and $\sigma^n = 1$,

$$\sigma(\rho) = \sigma(\alpha) + \xi\sigma^2(\alpha) + \cdots + \xi^{n-1}\sigma^n(\alpha) = \xi^{-1}(\xi\sigma(\alpha) + \cdots + \xi^{n-1}\sigma^{n-1}(\alpha) + \xi^n\alpha) = \xi^{-1}\rho.$$

Hence $\sigma(\rho^n) = \xi^{-n}\rho^n = \rho^n$, so ρ^n lies in the fixed field of $\langle\sigma\rangle = \operatorname{Gal}(\mathbb{F} \mid \mathbb{K})$, which is $\mathbb{K}$ by Theorem 18.8. For $0 < i < n$, $\sigma^i(\rho) = \xi^{-i}\rho \neq \rho$, since ξ is primitive and $\rho \neq 0$; so $\sigma^i \notin \operatorname{Gal}(\mathbb{F} \mid \mathbb{K}(\rho))$. Thus $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K}(\rho)) = \{\mathrm{id}\}$, and by Theorem 18.11, $\mathbb{K}(\rho) = \mathcal{F}(\{\mathrm{id}\}) = \mathbb{F}$. With $a = \rho^n \in \mathbb{K}$, $\mathbb{F} = \mathbb{K}(\rho) = \mathbb{K}(\sqrt[n]{a})$. ■

Remark 24.8. Let $\operatorname{char} \mathbb{K} = 0$ and ζ_n a primitive n-th root of unity. Then $\mathbb{K} \subseteq \mathbb{K}(\zeta_n)$ is a Galois extension with abelian Galois group, by Proposition 20.4.

Definition 24.9 (Compositum, normal closure). For fields $\mathbb{F}, \mathbb{E} \subseteq \Omega$, the *compositum* $\mathbb{F}\mathbb{E}$ is the smallest subfield of Ω containing $\mathbb{F} \cup \mathbb{E}$. For a finite extension $\mathbb{F} = \mathbb{K}(\gamma_1, ..., \gamma_s)$, the *normal closure* of $\mathbb{F} \mid \mathbb{K}$ is the splitting field in Ω over $\mathbb{K}$ of $\prod_j m_{\gamma_j, \mathbb{K}}$.

Lemma 24.10. Let $\operatorname{char} \mathbb{K} = 0$.

1) If $\mathbb{E} \mid \mathbb{K}$ is a root extension and $\mathbb{F} \supseteq \mathbb{K}$ is any field, then $\mathbb{F}\mathbb{E} \mid \mathbb{F}$ is a root extension.

2) If $\mathbb{F} \mid \mathbb{K}$ and $\mathbb{E} \mid \mathbb{K}$ are root extensions, then so is $\mathbb{F}\mathbb{E} \mid \mathbb{K}$.

3) If $\mathbb{F} \mid \mathbb{K}$ is a root extension, then so is its normal closure $\mathbb{L} \mid \mathbb{K}$.

Proof. 1) Let $\mathbb{K} = \mathbb{E}_0 \subseteq \cdots \subseteq \mathbb{E}_\ell = \mathbb{E}$ with $\mathbb{E}_i = \mathbb{E}_{i-1}(b_i)$ and $b_i^{n_i} \in \mathbb{E}_{i-1}$. Then $\mathbb{F}\mathbb{E} = \mathbb{F}(b_1, ..., b_\ell)$, and $b_i^{n_i} \in \mathbb{E}_{i-1} \subseteq \mathbb{F}(b_1, ..., b_{i-1})$. So $\mathbb{F} \subseteq \mathbb{F}(b_1) \subseteq \cdots \subseteq \mathbb{F}(b_1, ..., b_\ell)$ is a radical tower. For instance, if $\mathbb{E} = \mathbb{K}(\sqrt[n]{a})$, then $\mathbb{F}\mathbb{E} = \mathbb{F}(\sqrt[n]{a})$.

2) By 1), $\mathbb{F}\mathbb{E} \mid \mathbb{F}$ is a root extension. Appending its radical tower to one of $\mathbb{F} \mid \mathbb{K}$ gives a radical tower from $\mathbb{K}$ to $\mathbb{F}\mathbb{E}$.

3) $\mathbb{F} \mid \mathbb{K}$ is finite, so $\mathbb{F} = \mathbb{K}(\gamma_1, ..., \gamma_s)$ with γ_j algebraic, and $\mathbb{L}$ is finite over $\mathbb{K}$, being generated by the finitely many roots of $\prod_j m_{\gamma_j, \mathbb{K}}$. We show that $\mathbb{L}$ is the compositum of the fields $\tau(\mathbb{F})$, where τ runs over the finitely many $\mathbb{K}$-monomorphisms $\mathbb{F} \to \Omega$. Each $\tau(\gamma_j)$ is a root of $m_{\gamma_j, \mathbb{K}}$ by Theorem 17.12, so $\tau(\mathbb{F}) \subseteq \mathbb{L}$. Conversely, for a root β of $m_{\gamma_j, \mathbb{K}}$ there is a $\mathbb{K}$-embedding $\mathbb{K}(\gamma_j) \to \Omega$ with $\gamma_j \mapsto \beta$, as in the proof of Theorem 17.13, and it extends to $\mathbb{F}$ by Lemma 17.19; so $\beta \in \tau(\mathbb{F})$ for some τ. If $\mathbb{K} = \mathbb{F}_0 \subseteq \cdots \subseteq \mathbb{F}_\ell = \mathbb{F}$ is a radical tower with $\mathbb{F}_i = \mathbb{F}_{i-1}(b_i)$, then $\tau(\mathbb{F}_i) = \tau(\mathbb{F}_{i-1})(\tau(b_i))$ and $\tau(b_i)^{n_i} = \tau(b_i^{n_i}) \in \tau(\mathbb{F}_{i-1})$, so each $\tau(\mathbb{F}) \mid \mathbb{K}$ is a root extension. By 2), applied finitely many times, $\mathbb{L} \mid \mathbb{K}$ is a root extension. ∎

Theorem 24.11 (Galois). Let $\operatorname{char} \mathbb{K} = 0$, $f \in \mathbb{K}[x]$, and let $\mathbb{F}$ be the splitting field of f over $\mathbb{K}$. Then f is solvable by radicals if and only if $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K})$ is solvable.

Proof. ($\Rightarrow$) Assume that f is solvable by radicals. Each root of f lies in a root extension of $\mathbb{K}$; by Lemma 24.10 2), their compositum is a root extension containing all roots of f. So there is a radical tower $\mathbb{K} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \cdots \subseteq \mathbb{K}_n$ with $\mathbb{K}_{i+1} = \mathbb{K}_i(b_i)$, $b_i^{m_i} = a_i \in \mathbb{K}_i$, and $\mathbb{F} \subseteq \mathbb{K}_n$. Let $m = \operatorname{lcm}(m_0, ..., m_{n-1})$, ξ a primitive m-th root of unity, $\mathbb{E}_0 = \mathbb{K}(\xi)$ and $\mathbb{E}_{i+1} = \mathbb{E}_i(b_i)$. Then $\mathbb{E}_0 \mid \mathbb{K}$ is abelian Galois by Proposition 20.4, and each $\mathbb{E}_{i+1} \mid \mathbb{E}_i$ is cyclic by Theorem 24.7 1), since $m_i \mid m$ and $\mathbb{E}_i$ contains the m_i-th roots of unity $\xi^{m/m_i \cdot j}$.

Let $\mathbb{L}$ be the normal closure of $\mathbb{E}_n \mid \mathbb{K}$. By the proof of Lemma 24.10 3), $\mathbb{L}$ is the compositum of the fields $\tau(\mathbb{E}_n)$ for the $\mathbb{K}$-monomorphisms $\tau : \mathbb{E}_n \to \Omega$. Since $\tau(\xi)$ is a primitive m-th root of unity, $\mathbb{K}(\tau(\xi)) = \mathbb{K}(\xi)$, both being the splitting field of $x^m - 1$. So $\tau(\mathbb{E}_n) = \mathbb{K}(\xi)(\tau(b_0), ..., \tau(b_{n-1}))$ with $\tau(b_i)^{m_i} = \tau(a_i) \in \tau(\mathbb{E}_i)$. Concatenating these towers for the finitely many τ gives

$$\mathbb{K} \subseteq \mathbb{K}(\xi) = \mathbb{L}_0 \subseteq \mathbb{L}_1 \subseteq \cdots \subseteq \mathbb{L}_N = \mathbb{L}, \quad \mathbb{L}_{j+1} = \mathbb{L}_j(c_j), \quad c_j^{e_j} \in \mathbb{L}_j, \quad e_j \mid m.$$

As above, each $\mathbb{L}_{j+1} \mid \mathbb{L}_j$ is cyclic by Theorem 24.7 1), and $\mathbb{L}_0 \mid \mathbb{K}$ is abelian Galois. $\mathbb{L} \mid \mathbb{K}$ is Galois, being normal (Theorem 17.21) in characteristic 0, and $\mathbb{F} \subseteq \mathbb{L}$, as $\mathbb{L}$ contains all roots of f.

Let $G = \mathrm{Gal}(\mathbb{L} \mid \mathbb{K})$ and $G_j = \mathrm{Gal}(\mathbb{L} \mid \mathbb{L}_j)$, so $G \geq G_0 \geq G_1 \geq \cdots \geq G_N = 1$. Since $\mathbb{L}_{j+1} \mid \mathbb{L}_j$ is normal, Theorem 18.12 applied to the Galois extension $\mathbb{L} \mid \mathbb{L}_j$ (Corollary 17.22) gives $G_{j+1} \trianglelefteq G_j$ and

$$G_j/G_{j+1} = \mathrm{Gal}(\mathbb{L} \mid \mathbb{L}_j)/\mathrm{Gal}(\mathbb{L} \mid \mathbb{L}_{j+1}) \cong \mathrm{Gal}(\mathbb{L}_{j+1} \mid \mathbb{L}_j),$$

which is abelian. Likewise $G_0 \trianglelefteq G$ with $G/G_0 \cong \mathrm{Gal}(\mathbb{L}_0 \mid \mathbb{K})$ abelian. By Proposition 23.4, G is solvable. Finally $\mathbb{F} \mid \mathbb{K}$ is normal and $\mathbb{K} \subseteq \mathbb{F} \subseteq \mathbb{L}$, so by Theorem 18.12,

$$\mathrm{Gal}(\mathbb{F} \mid \mathbb{K}) \cong \mathrm{Gal}(\mathbb{L} \mid \mathbb{K})/\mathrm{Gal}(\mathbb{L} \mid \mathbb{F}),$$

The projection $\pi : \mathrm{Gal}(\mathbb{L} \mid \mathbb{K}) \to \mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$ is surjective, so $\pi(H') = \pi(H)'$ for every subgroup H, and by induction $\pi\big(G^{(i)}\big) = \mathrm{Gal}(\mathbb{F} \mid \mathbb{K})^{(i)}$. As $G^{(k)} = 1$ for some k, $\mathrm{Gal}(\mathbb{F} \mid \mathbb{K})^{(k)} = 1$, and $\mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$ is solvable.

($\Leftarrow$) Assume $\mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$ is solvable. By Proposition 23.6 there is a normal series

$$1 = G_n \trianglelefteq G_{n-1} \trianglelefteq \cdots \trianglelefteq G_0 = \mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$$

with G_{i-1}/G_i cyclic of order m_i. Let $\mathbb{K}_i = \mathcal{F}(G_i)$. Then $\mathbb{K} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \cdots \subseteq \mathbb{K}_n = \mathbb{F}$, by Theorem 18.8 and Theorem 18.11. By Theorem 18.11, $G_{i-1} = \mathrm{Gal}(\mathbb{F} \mid \mathbb{K}_{i-1})$, and since $G_i \trianglelefteq G_{i-1}$, Theorem 18.12 applied to $\mathbb{F} \mid \mathbb{K}_{i-1}$ shows that $\mathbb{K}_i \mid \mathbb{K}_{i-1}$ is Galois with $\mathrm{Gal}(\mathbb{K}_i \mid \mathbb{K}_{i-1}) \cong G_{i-1}/G_i$, cyclic of order m_i.

Let $m = \mathrm{lcm}(m_1, \ldots, m_n)$, $\mathbb{E}_0 = \mathbb{K}(\xi_m)$ and $\mathbb{E}_i = \mathbb{E}_0\mathbb{K}_i$, so $\mathbb{E}_i = \mathbb{E}_{i-1}\mathbb{K}_i$. By Theorem 17.21, $\mathbb{K}_i$ is the splitting field over $\mathbb{K}_{i-1}$ of some $g \in \mathbb{K}_{i-1}[x]$, so $\mathbb{E}_i = \mathbb{E}_{i-1}(\text{roots of } g)$ is the splitting field of g over $\mathbb{E}_{i-1}$, and $\mathbb{E}_i \mid \mathbb{E}_{i-1}$ is Galois. For $\sigma \in \mathrm{Gal}(\mathbb{E}_i \mid \mathbb{E}_{i-1})$, σ permutes the roots of g (Theorem 17.12), so $\sigma(\mathbb{K}_i) = \mathbb{K}_i$, and $\sigma|_{\mathbb{K}_i}$ fixes $\mathbb{K}_{i-1} \subseteq \mathbb{E}_{i-1}$. The restriction $\mathrm{Gal}(\mathbb{E}_i \mid \mathbb{E}_{i-1}) \to \mathrm{Gal}(\mathbb{K}_i \mid \mathbb{K}_{i-1})$ is an injective homomorphism, since $\mathbb{E}_i$ is generated over $\mathbb{E}_{i-1}$ by the roots of g. Hence $\mathrm{Gal}(\mathbb{E}_i \mid \mathbb{E}_{i-1})$ is isomorphic to a subgroup of a cyclic group of order m_i, so it is cyclic of some order $d_i \mid m_i$. The field $\mathbb{E}_{i-1} \supseteq \mathbb{E}_0$ contains the primitive d_i-th root of unity ξ_m^{m/d_i}, so by Theorem 24.7 2), $\mathbb{E}_i = \mathbb{E}_{i-1}\big(\sqrt[d_i]{c_i}\big)$ with $c_i \in \mathbb{E}_{i-1}$. Together with $\mathbb{E}_0 = \mathbb{K}\big(\sqrt[m]{1}\big)$, this shows that $\mathbb{E}_n \mid \mathbb{K}$ is a root extension. It contains $\mathbb{K}_n = \mathbb{F}$, hence all roots of f, so f is solvable by radicals. ■

Remark 24.12 (Recall). *Cauchy's theorem.* If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p.

Lemma 24.13. Let p be a prime and $G \leq S_p$. If G contains a p-cycle and a transposition, then $G = S_p$.

Proof. Let $(a\ b) \in G$ be a transposition and $c \in G$ a p-cycle. Some power c^k with $1 \leq k < p$ maps a to b; since p is prime, c^k is again a p-cycle. Relabelling the points, $a = 1$, $b = 2$ and $c^k = (1\ 2\ \ldots\ p)$. Conjugating $(1\ 2)$ by $(c^k)^j$ gives $(j+1\ j+2)$, so G contains all transpositions $(i\ i+1)$, $1 \leq i \leq p-1$, which generate S_p. ■

Example 24.14. Show that $f(x) = x^5 - 20x + 6 \in \mathbb{Q}[x]$ is not solvable by radicals.

Solution. By Remark 22.1 with $p = 2$, f is irreducible over $\mathbb{Q}$. Let $\mathbb{E} \subseteq \mathbb{C}$ be its splitting field. Since $\mathrm{char}\ \mathbb{Q} = 0$, f is separable (Lemma 17.7) and has five distinct roots $\alpha_1, \ldots, \alpha_5$. $G = \mathrm{Gal}(\mathbb{E} \mid \mathbb{Q})$

permutes the roots and is determined by this action, so $G \leq S_5$. As $[\mathbb{Q}(\alpha_i) : \mathbb{Q}] = 5$, the tower law gives $5 \mid [\mathbb{E} : \mathbb{Q}] = |G|$. By Cauchy's theorem G has an element of order 5; an element of S_5 of order 5 has cycle lengths with least common multiple 5, so it is a 5-cycle.

Now $f'(x) = 5x^4 - 20$ and $f''(x) = 20x^3$. As f'' has a single real zero, f' has at most 2 real zeros by Rolle's theorem, and so f has at most 3 real zeros. From the values

x	-3	0	1	3
$f(x)$	-177	6	-13	189

and the intermediate value theorem, f has at least 3 real zeros. So f has exactly three real roots $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ and two nonreal roots $\alpha_4, \alpha_5 \in \mathbb{C}$ with $\overline{\alpha_4} = \alpha_5$. Complex conjugation maps $\mathbb{E}$ onto $\mathbb{E}$ by Proposition 17.20, so its restriction lies in G; it fixes $\alpha_1, \alpha_2, \alpha_3$ and swaps α_4, α_5, so $(4\ 5) \in G$. By Lemma 24.13, $G = S_5$, which is not solvable by Theorem 23.9. By Theorem 24.11, f is not solvable by radicals. ◊